INFORMATICS INSTITUTE OF TECHNOLOGY

INFORMATICS INSTITUTE OF TECHNOLOGY DEPARTMENT OF BUSINESS MANAGEMENT

DOC327 – WORKING WITH DATA

COURSEWORK – 2022/23 (Foundation Jan 2023)

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20222330	Akash Sathiyanathan

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Without Mr. Chathura W. Wickramasinghe, we would not have acquired the knowledge necessary to complete this paper; as a result, we would like to dedicate a little portion of this book to him. We also want to express our gratitude to all the other IIT professors who inspired and helped us along the road. Finally, we want to express our gratitude to our parents and friends for supporting us in our undertakings.

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Part A

Question 01

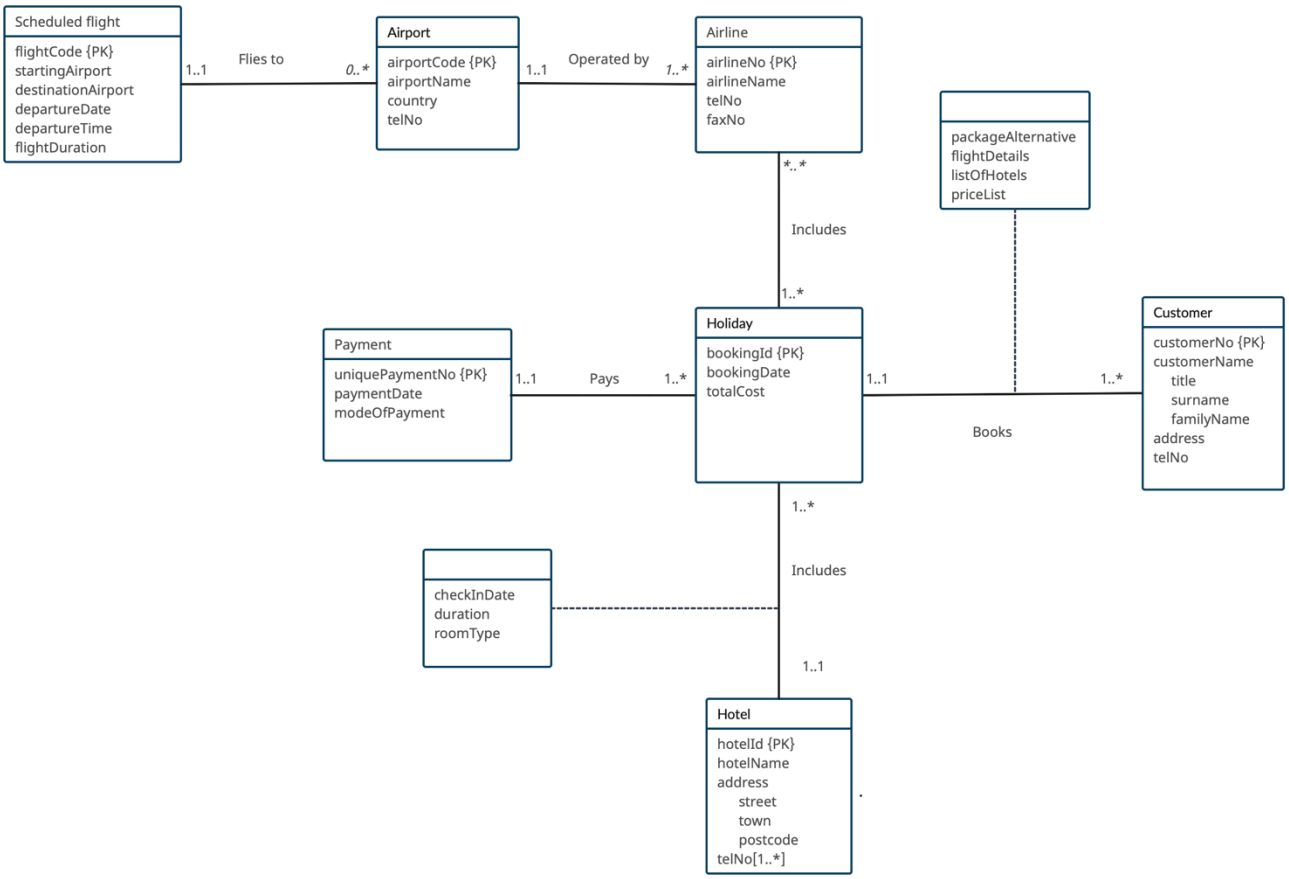

Table 1 Entity Relational Diagram

Question 02

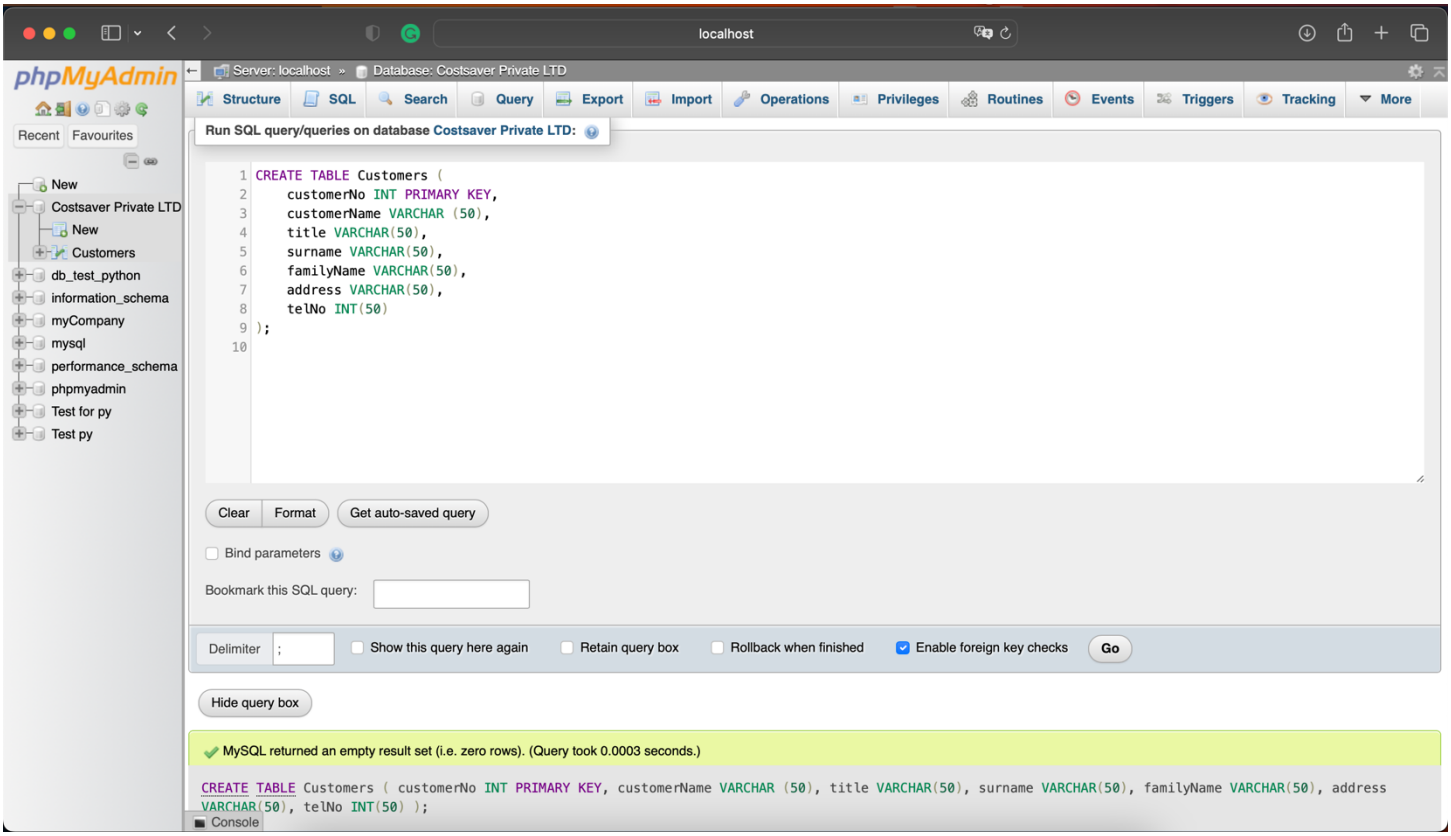

Figure 1 Costsaver Private LTD Customers

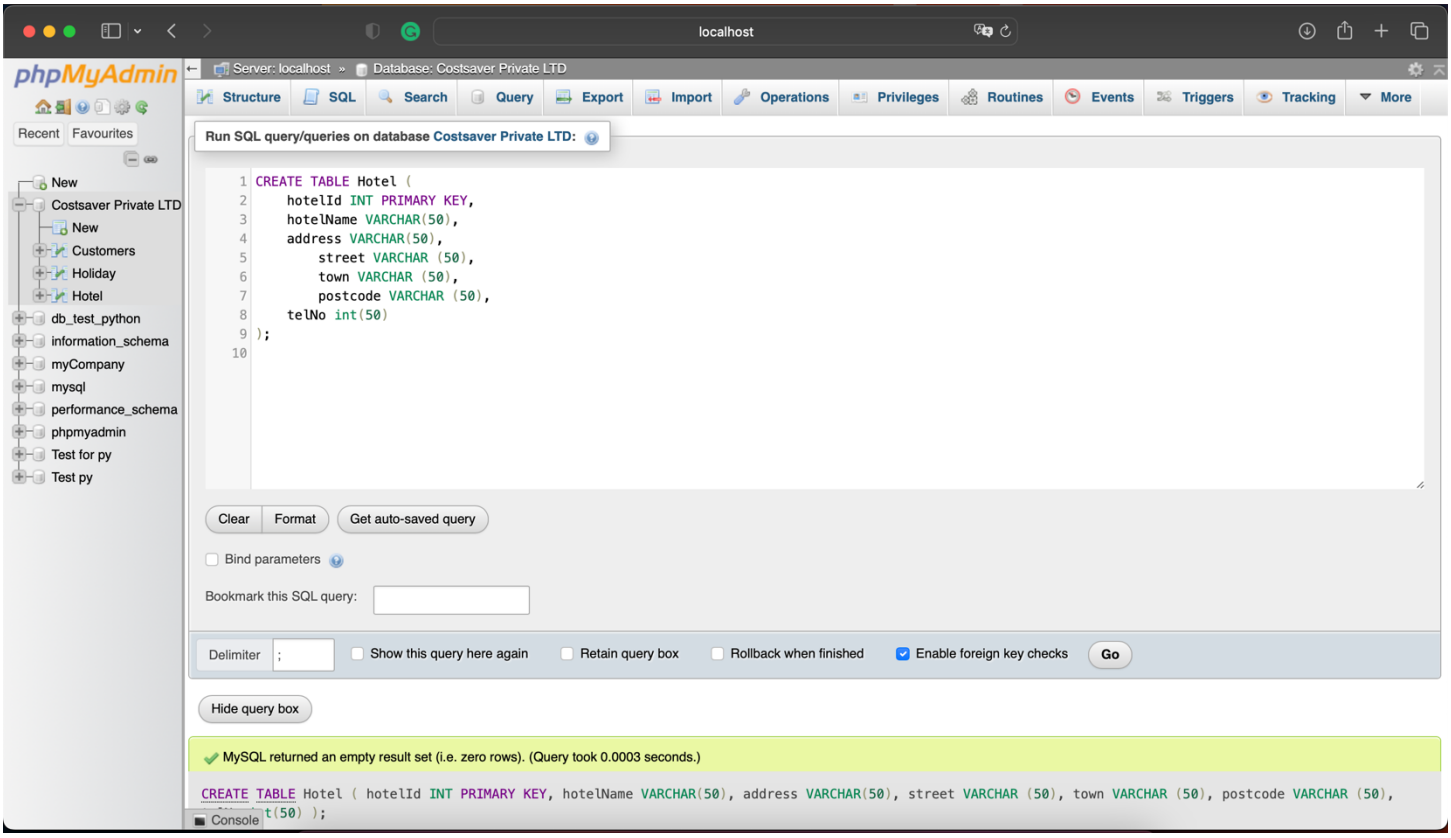

Figure 2 Hotel table

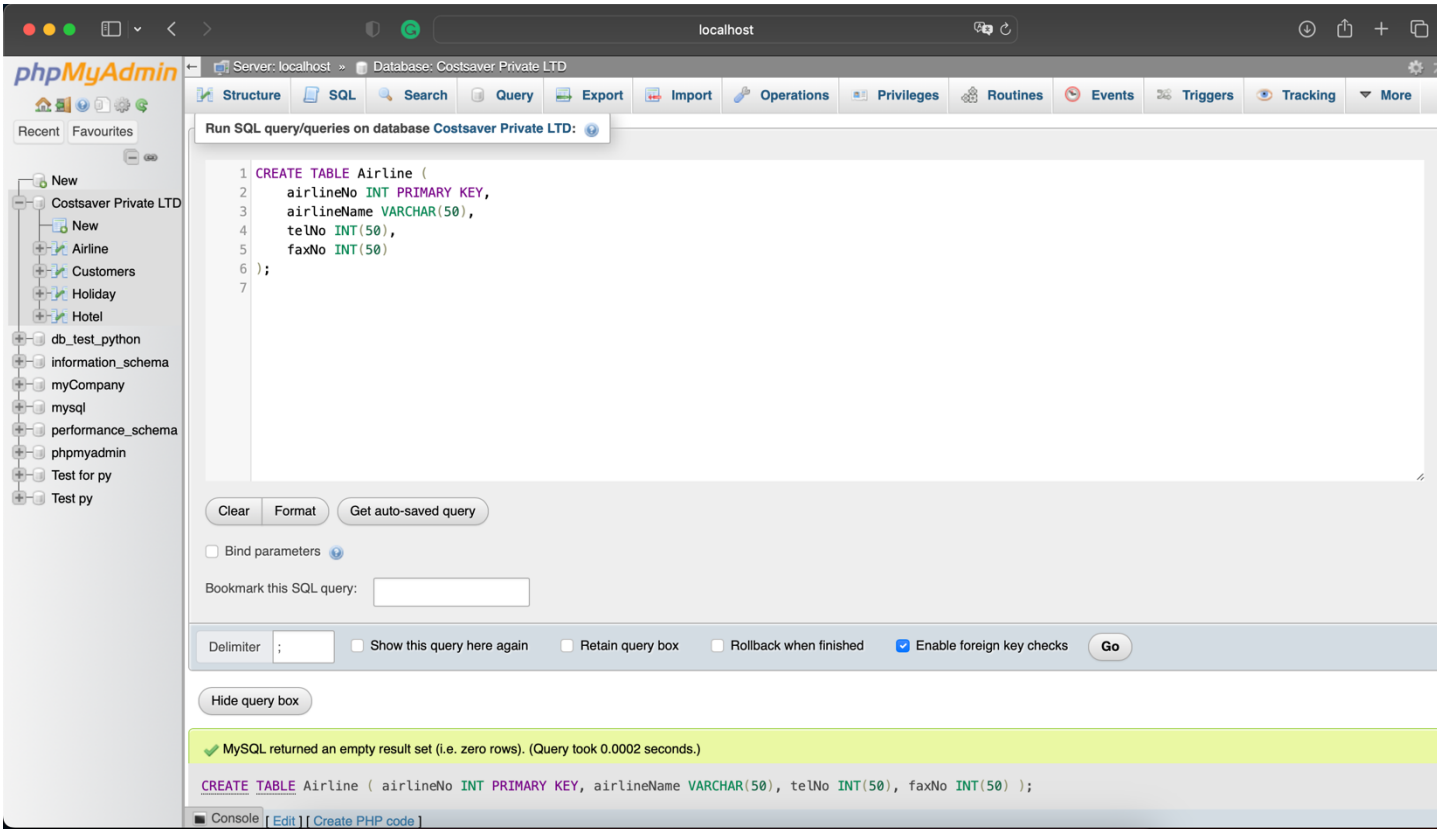

Figure 3 Airline table

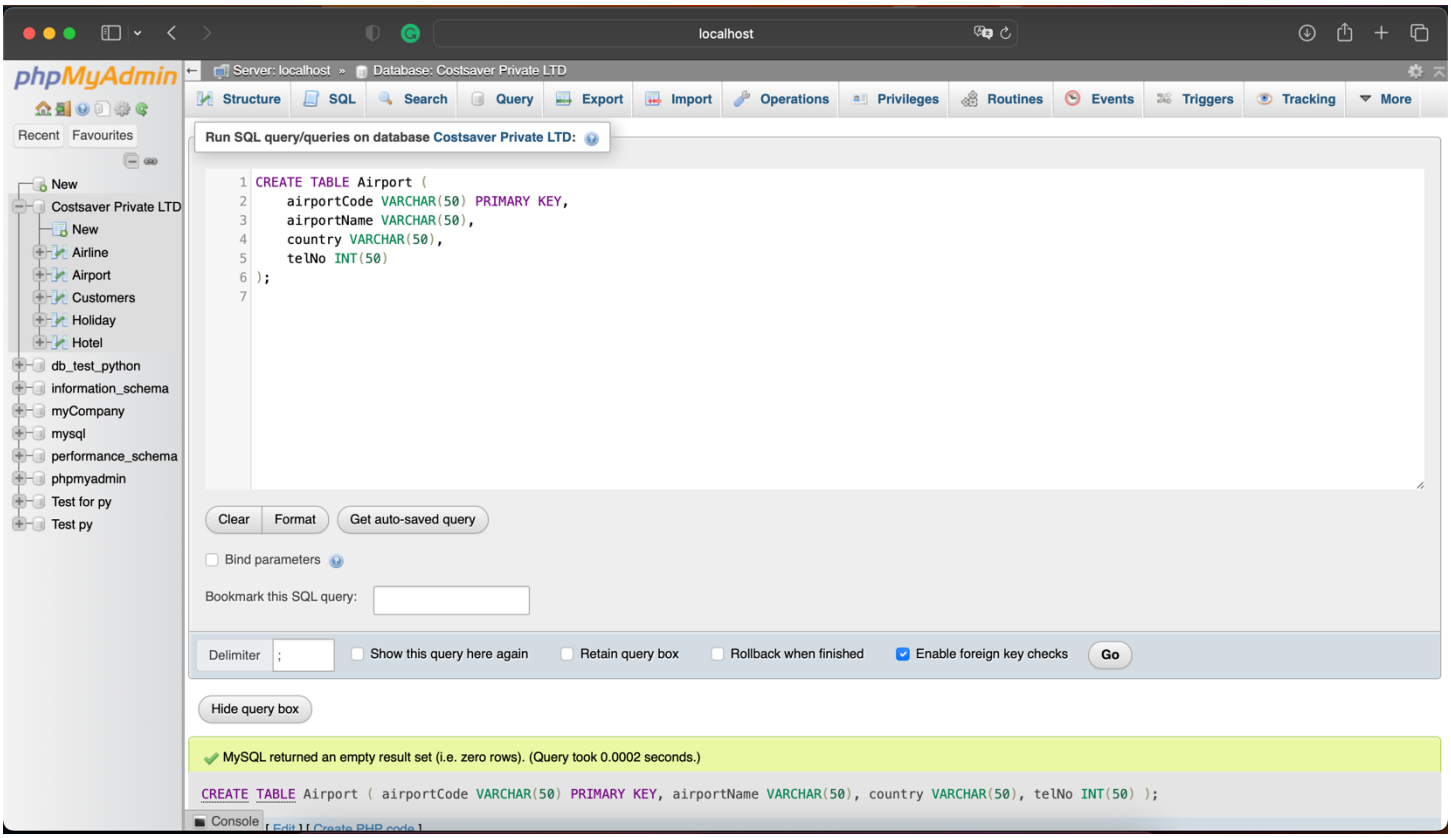

Figure 4 Airport table

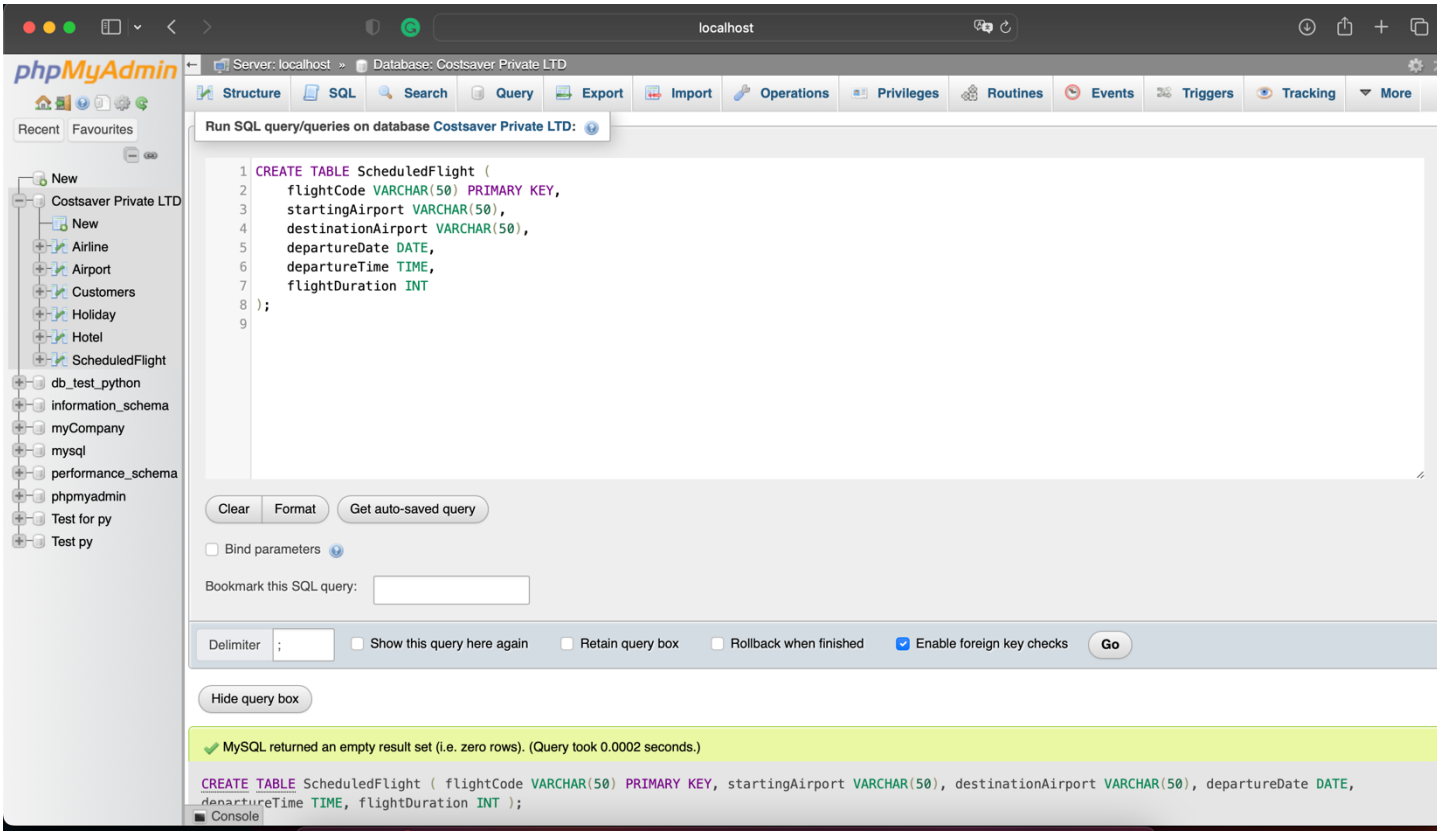

Figure 5 SchediledFlight table

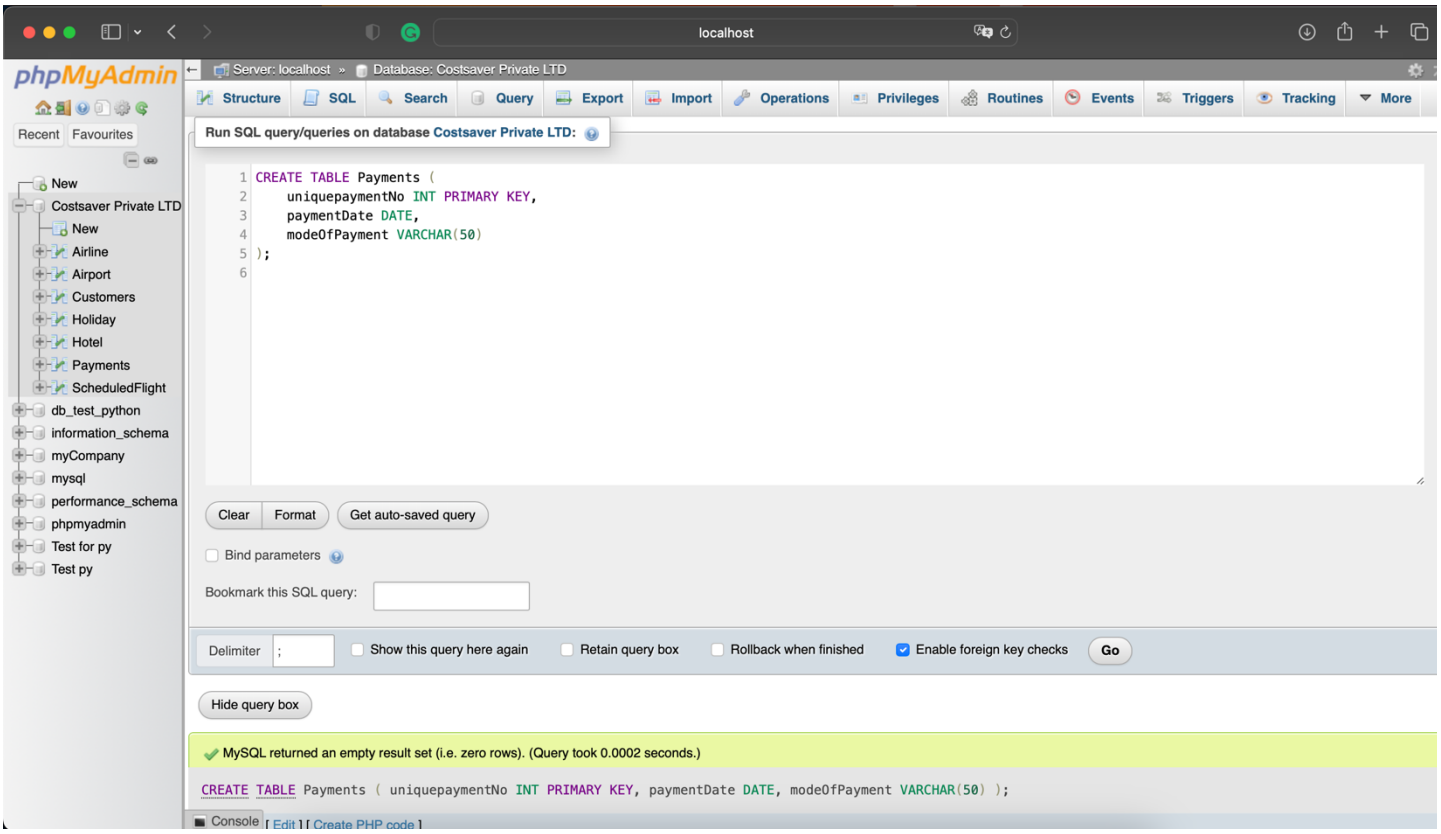

Figure 6 Payments

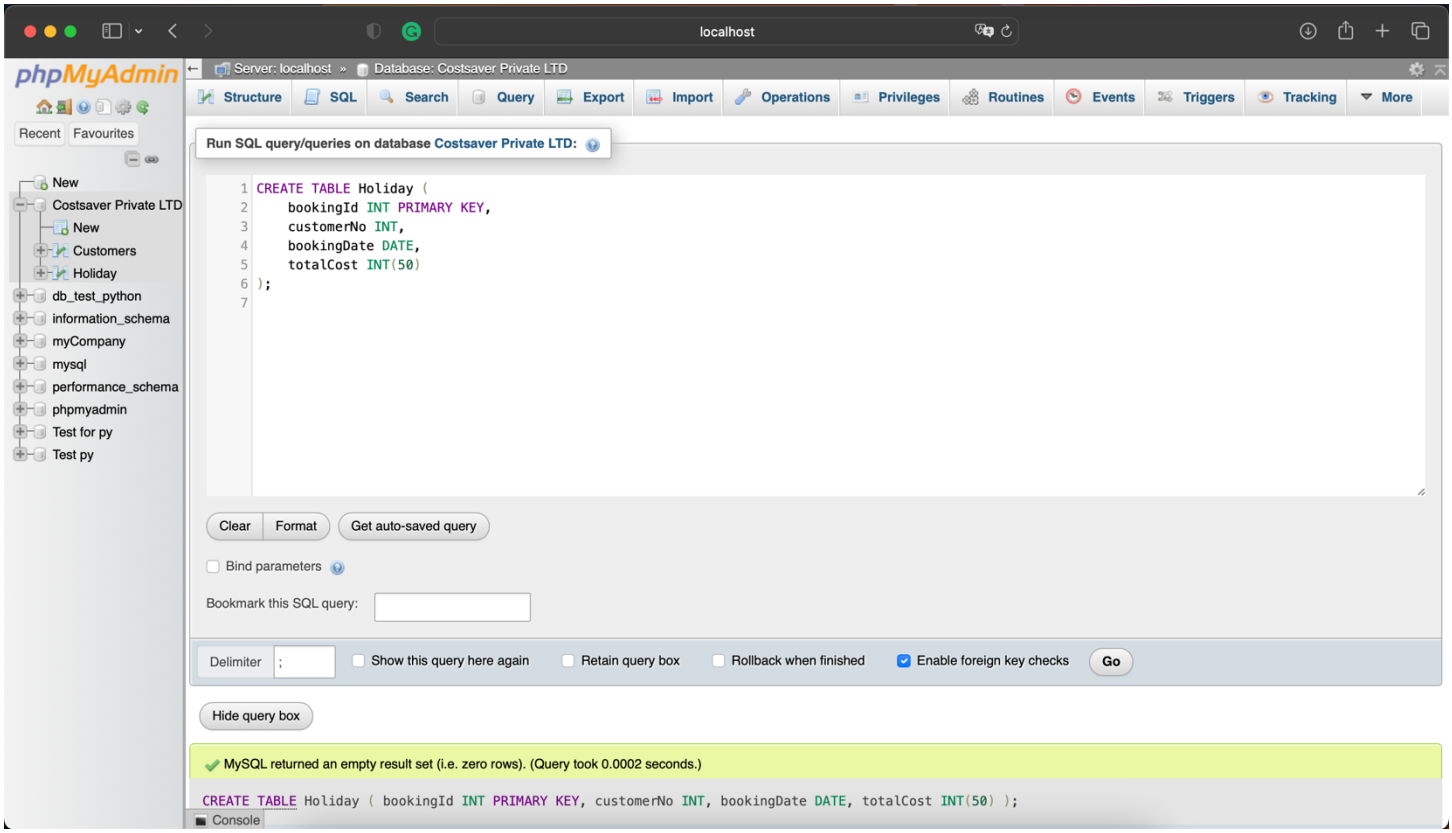

Figure 7 Holiday

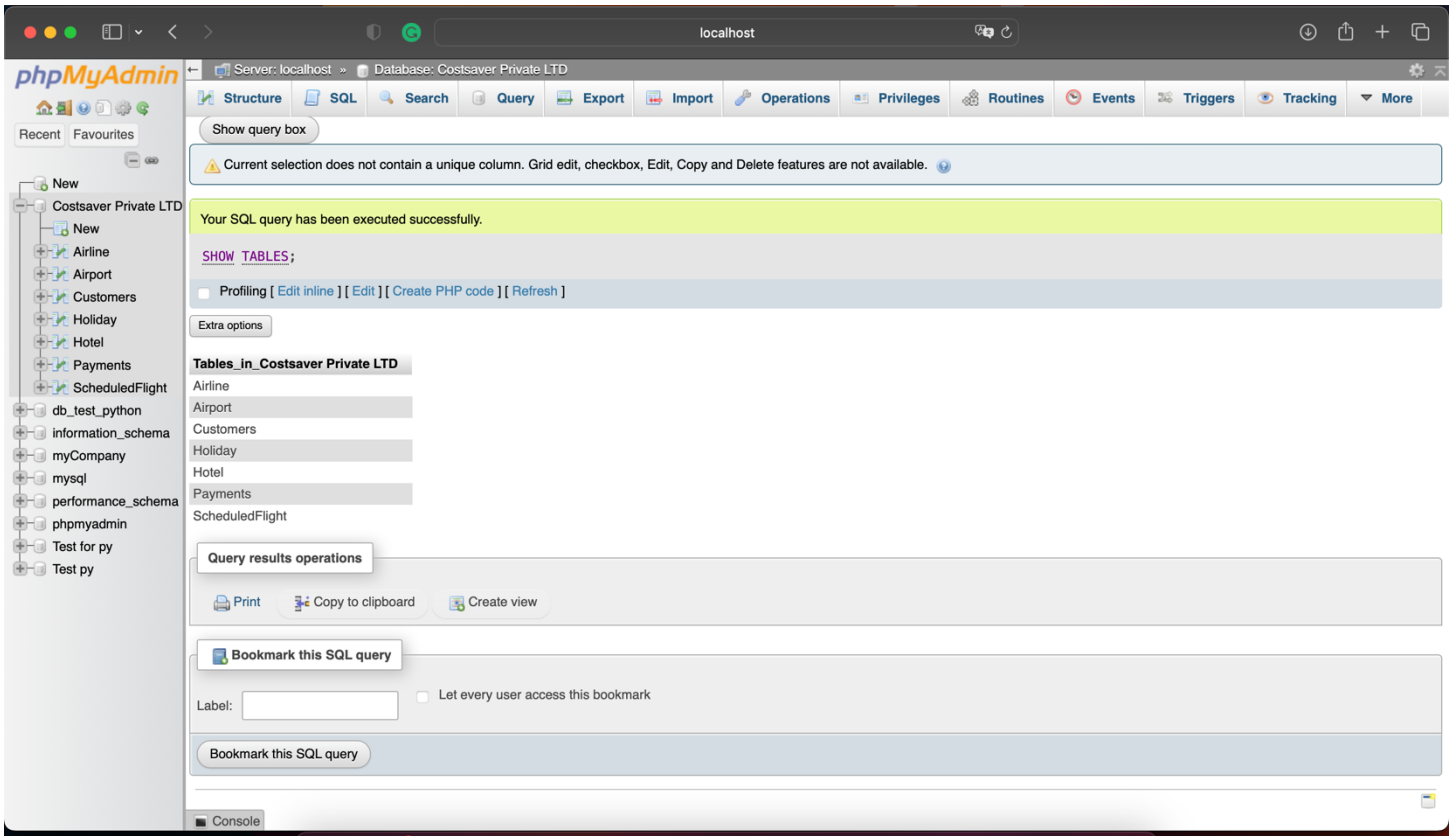

Figure 8 Tables in Costsaver Private LTD

Question 03

Candidate Keys, Primary Keys, Composite Keys, Surrogate Keys, and Foreign Keys

1) Customer:
   - Candidate Key : customerNo
   - Primary Key : customerNo
   - Surrogate Key : -
   - Composite Key : address
   - Foreign Key : -

2) Holiday:
   - Candidate Key : bookingId
   - Primary Key : bookingId
   - Surrogate Key : -
   - Composite Key : -
   - Foreign Key : customerNo

3) Flight:
   - Candidate Key : flightId
   - Primary Key : flightId
   - Surrogate Key : -
   - Composite Key : -
   - Foreign Key : bookingId , airlineNo

4) Hotels:
   - Candidate Key : hotelId
   - Primary Key : hotelId
   - Surrogate Key : -
   - Composite Key : address
   - Foreign Key : -

5) Airline:
   - Candidate Key : airlineNo
   - Primary Key : airlineNo
   - Surrogate Key : -
   - Composite Key : -
   - Foreign Key : -

6) ScheduledFlights:
   - Candidate Key : flightCode
   - Primary Key : flightCode
   - Surrogate Key : -
   - Composite Key : -
   - Foreign Key : flight_id

7) Airports:
   - Candidate Key : airportCode
   - Primary Key : airportCode
   - Surrogate Key : -
   - Composite Key : -
   - Foreign Key : -

8) Payments:
   - Candidate Key : paymentNo
   - Primary Key : paymentNo
   - Surrogate Key : -
   - Composite Key : -
   - Foreign Key : bookingId

Part B

Question 01

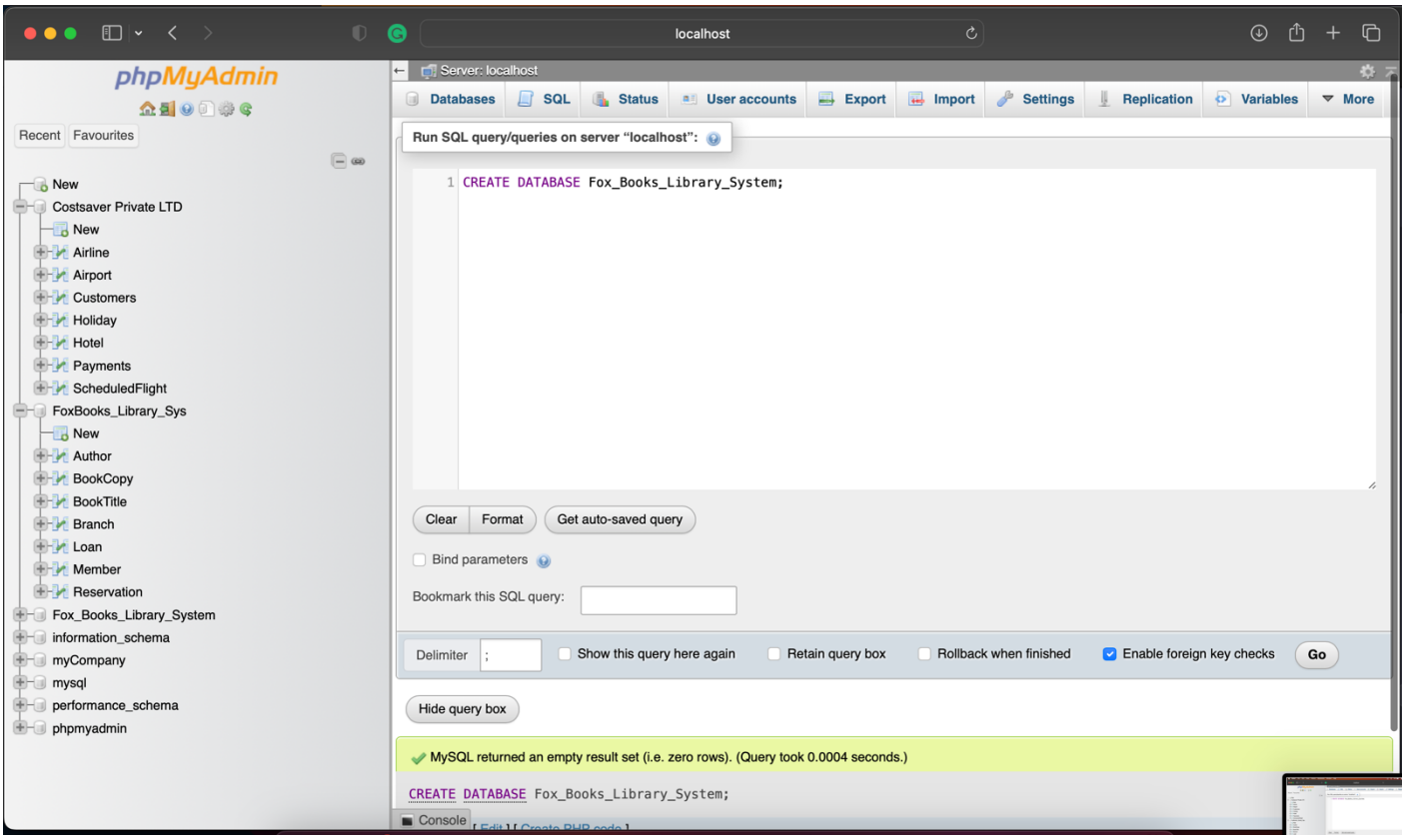

Figure 9 Fox Books Library System

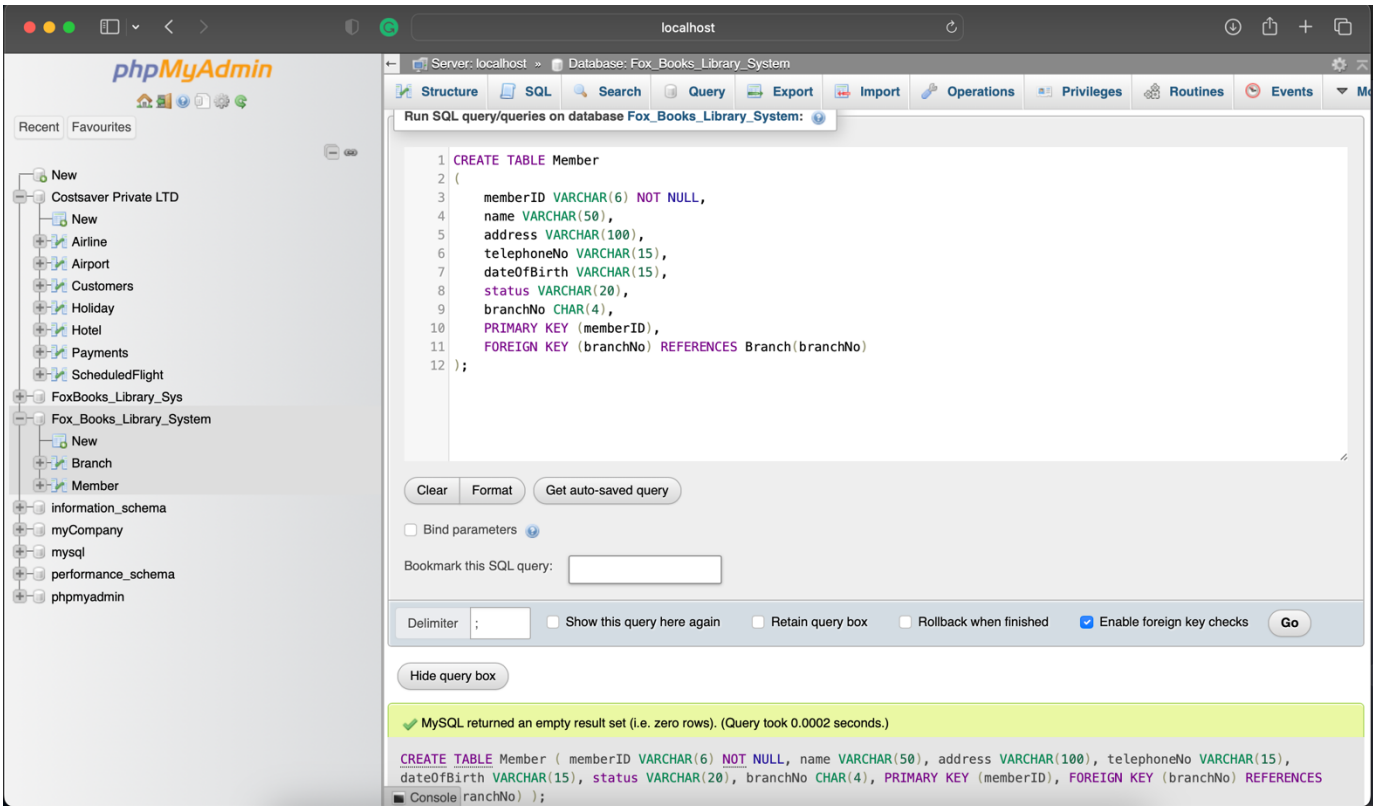

Figure 10 Create Member table

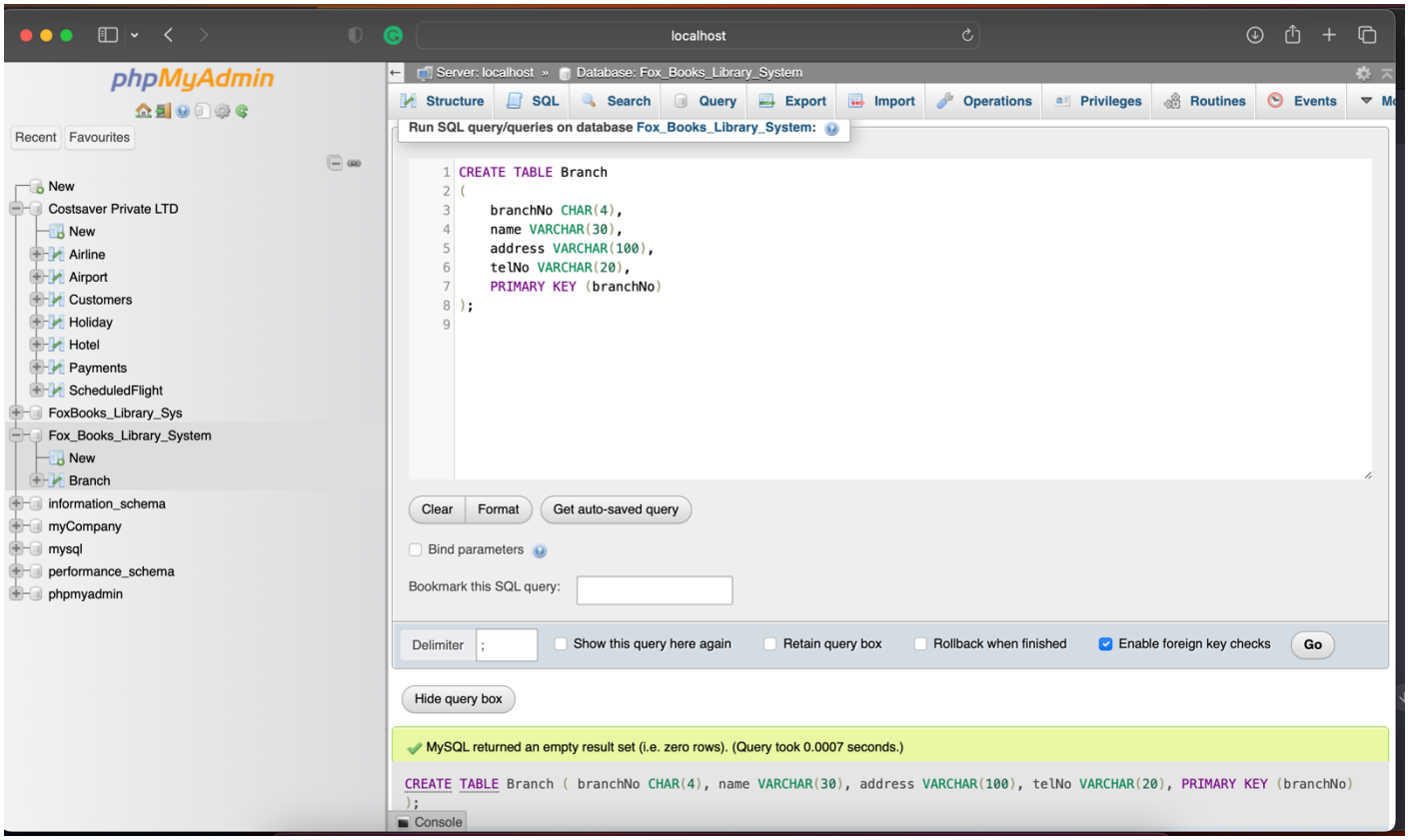

Figure 11 Create Branch table

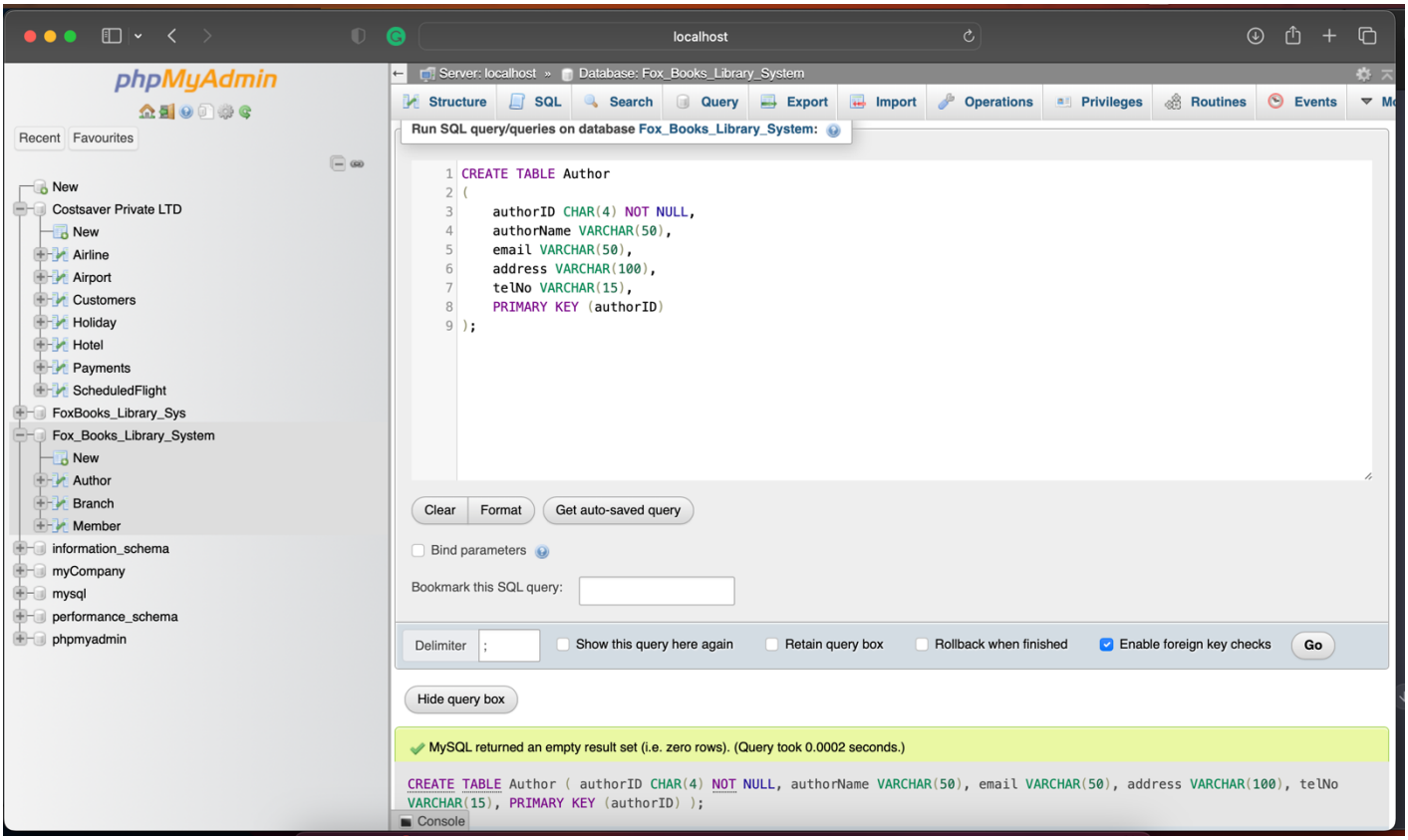

Figure 12 Create Author table

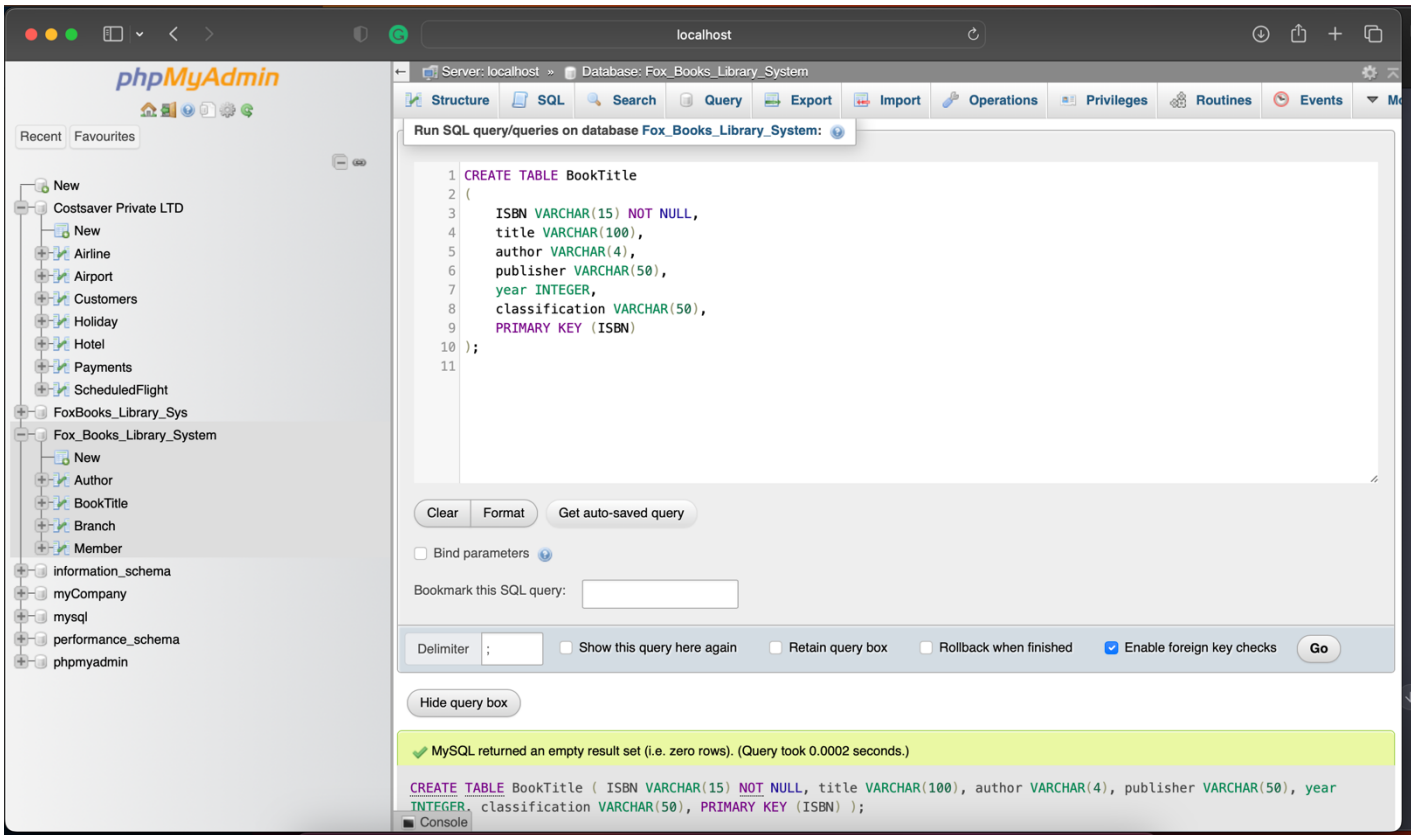

Figure 13 Create Booktitle table

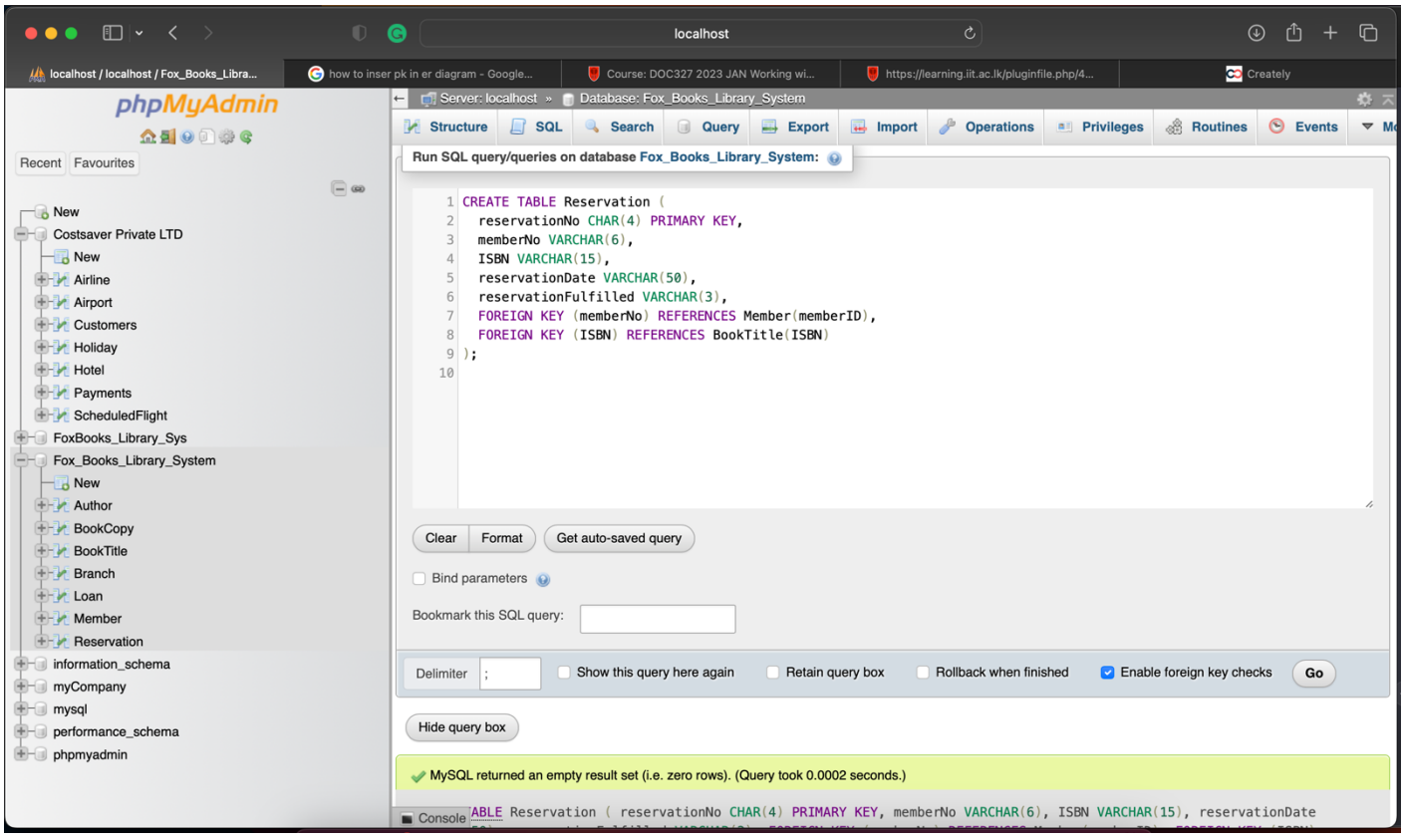

Figure 14 Create Reservation table

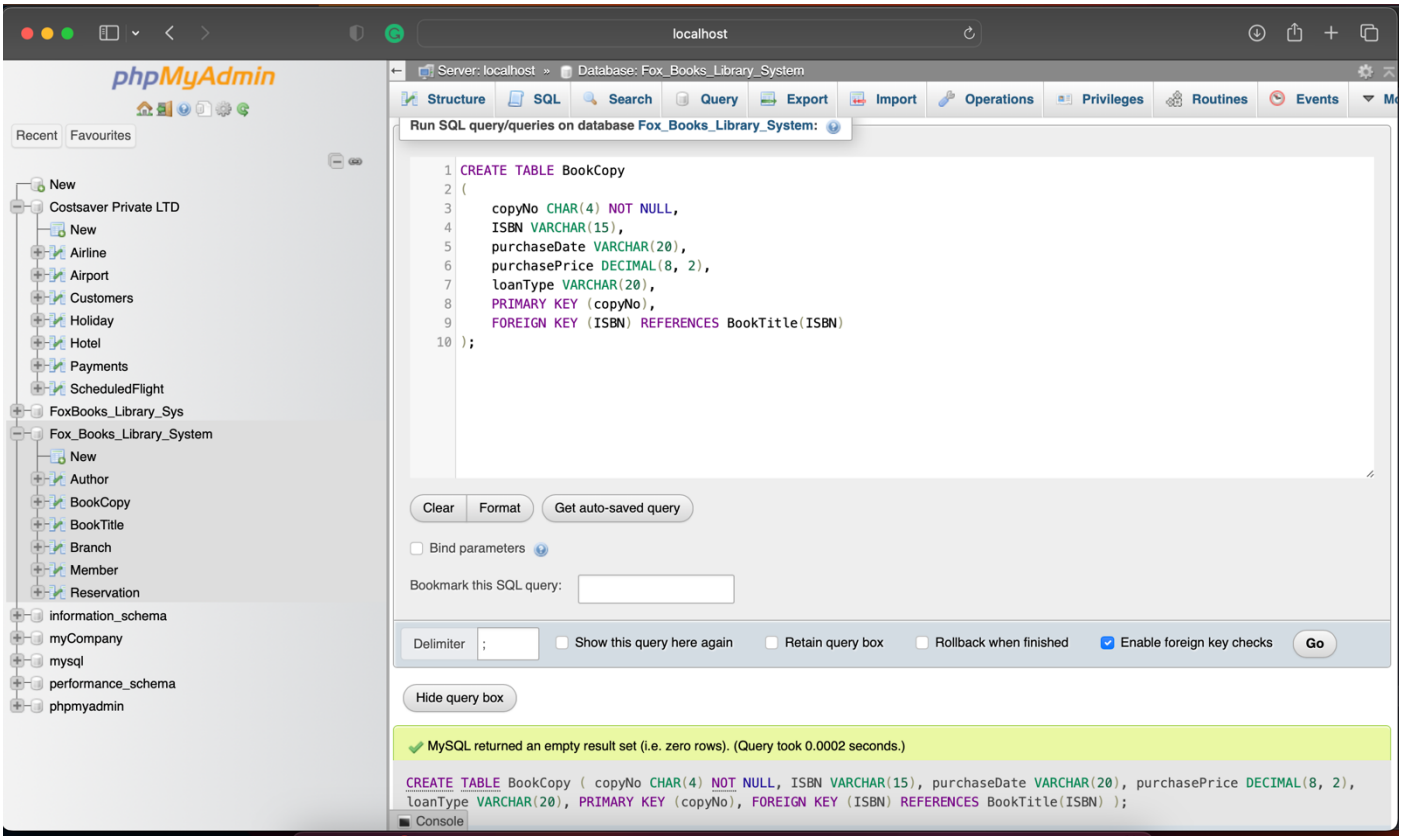

Figure 15 Create Bookcopy table

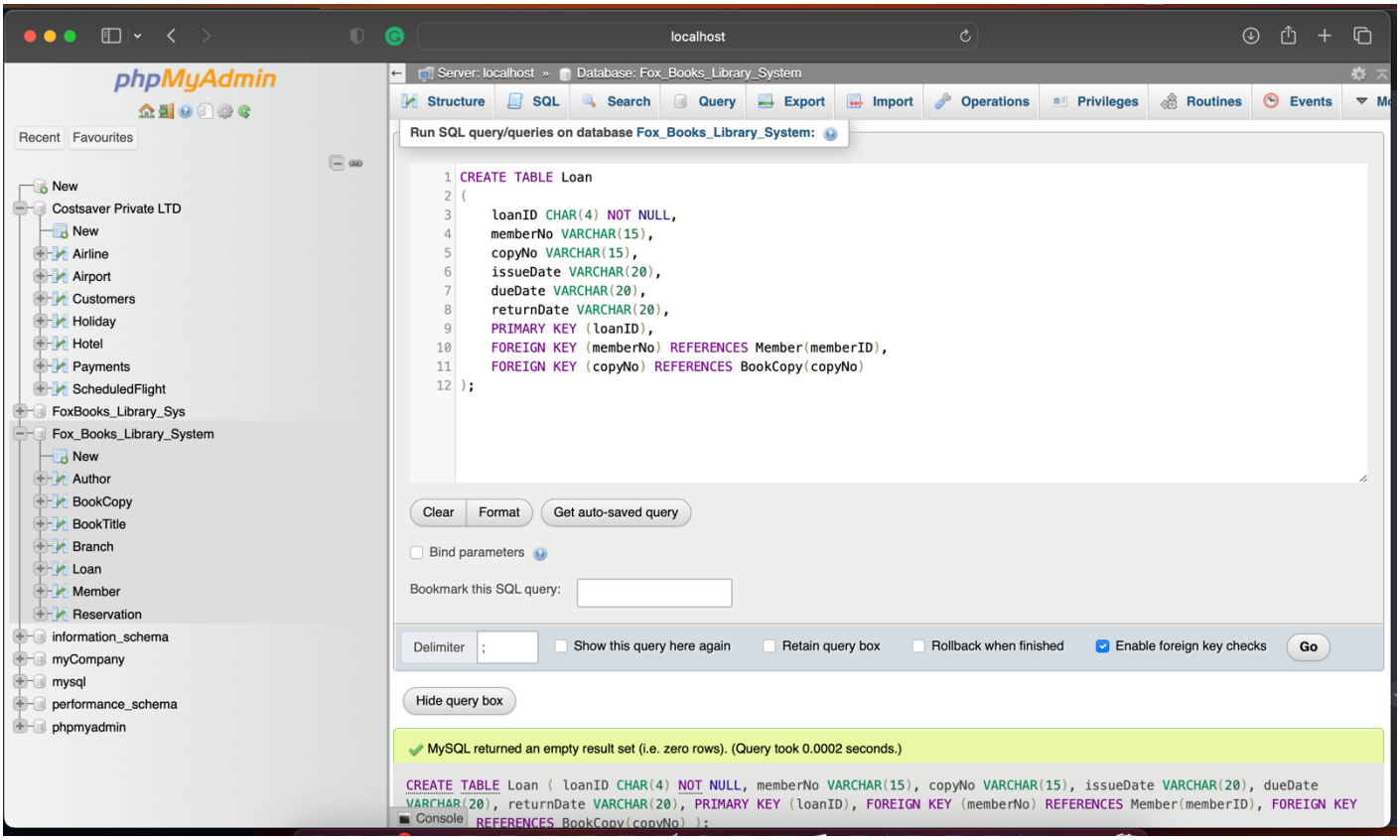

Figure 16 Create Loan table

Question 02

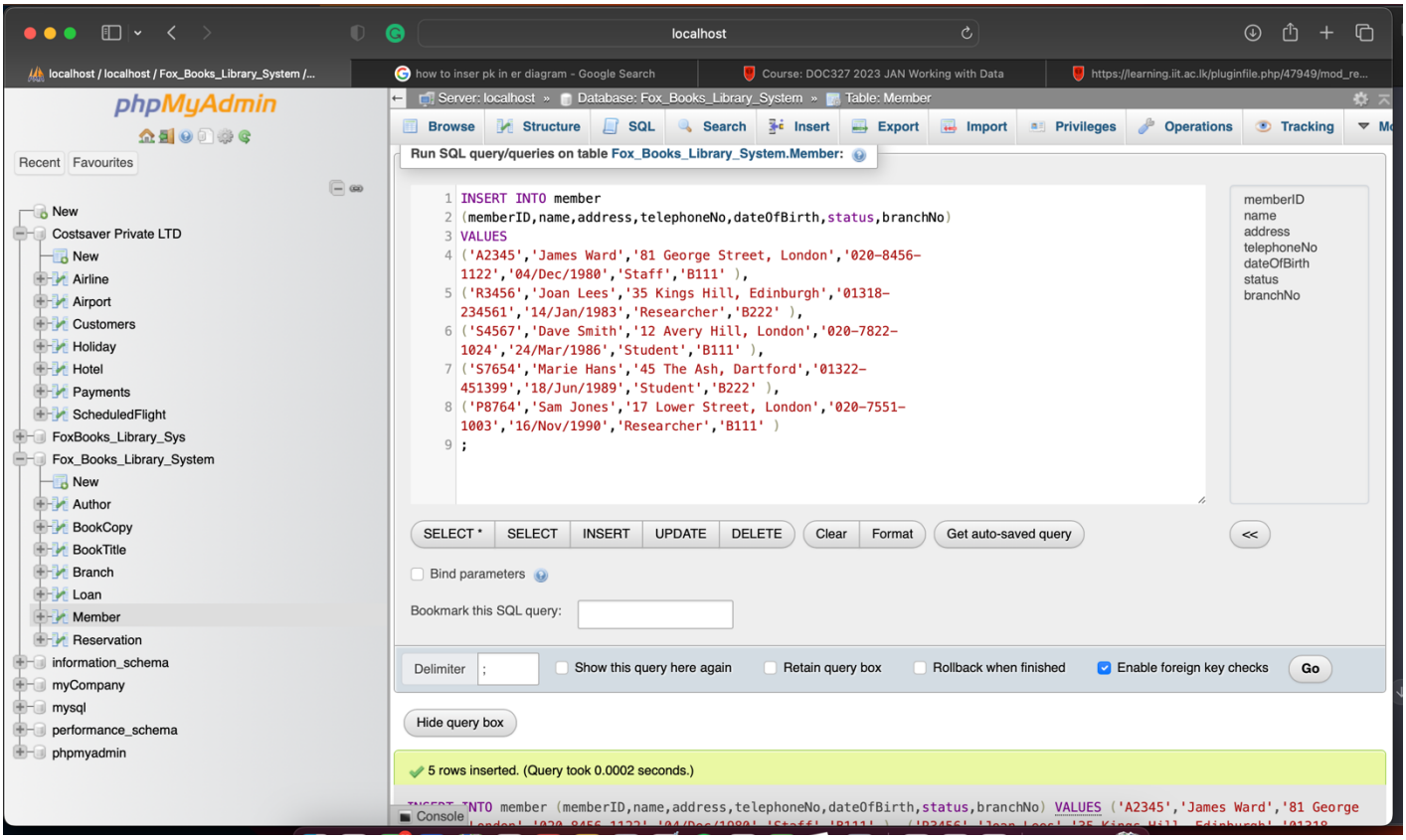

Figure 17 Member table data

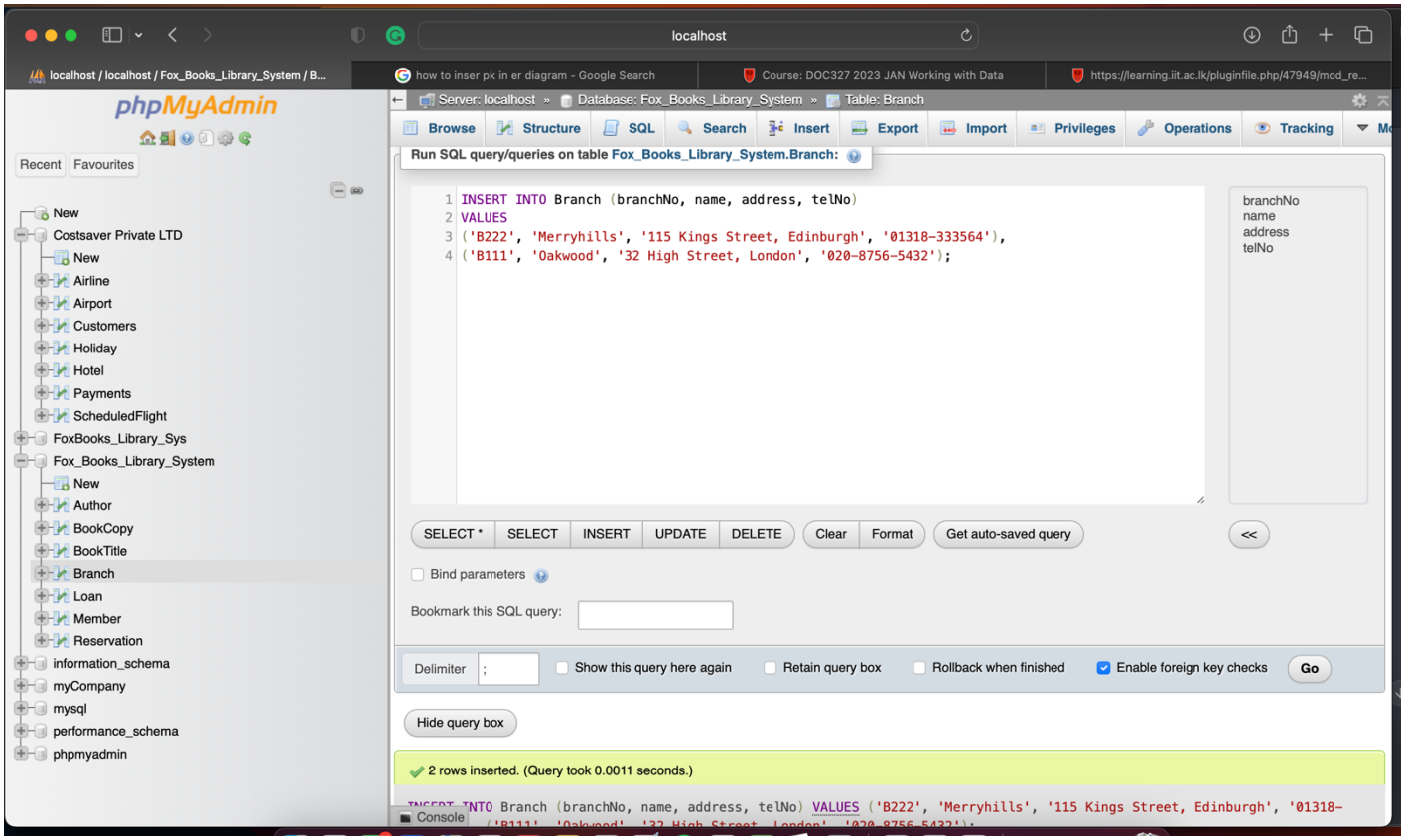

Figure 18 Branch table data

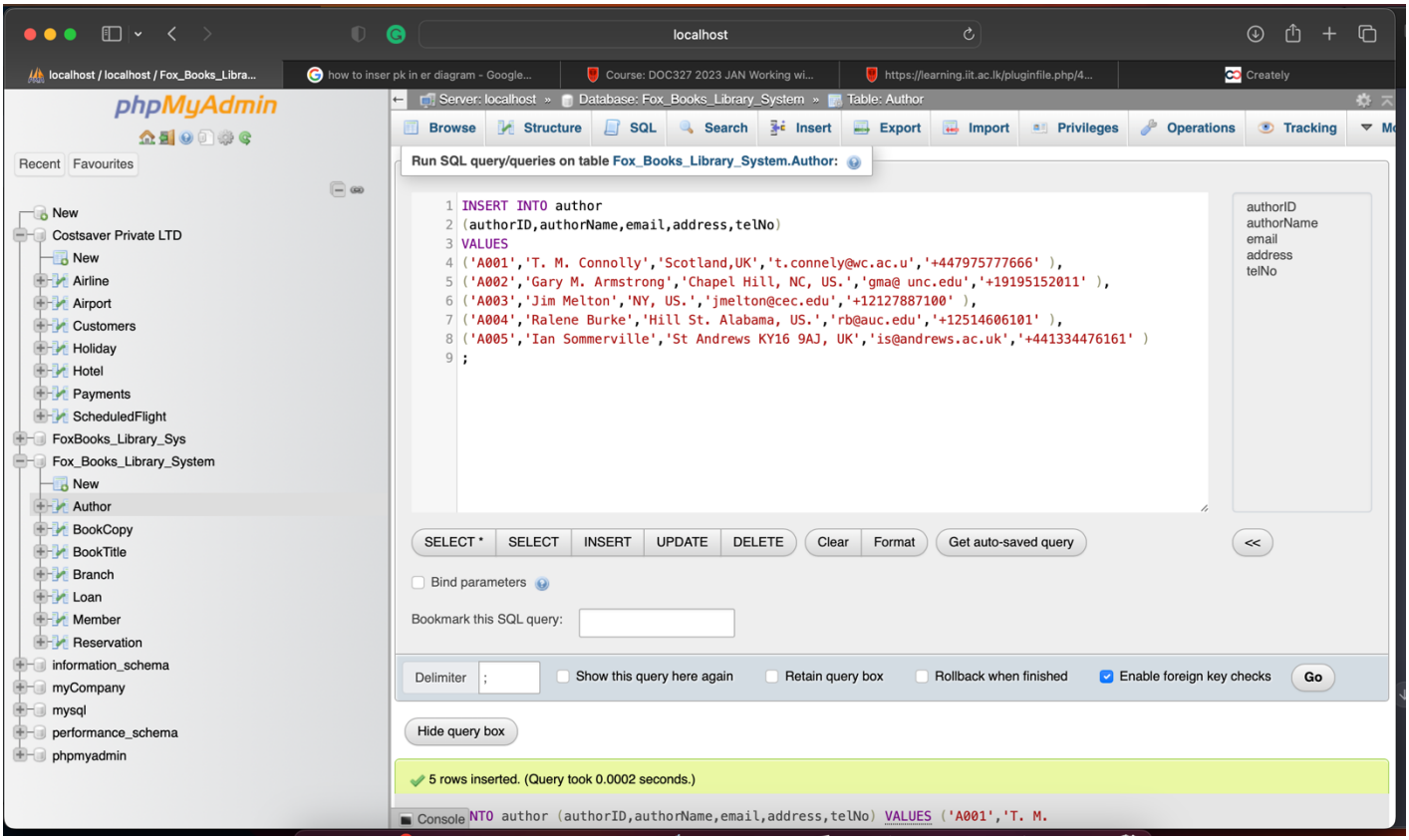

Figure 19 Author table data

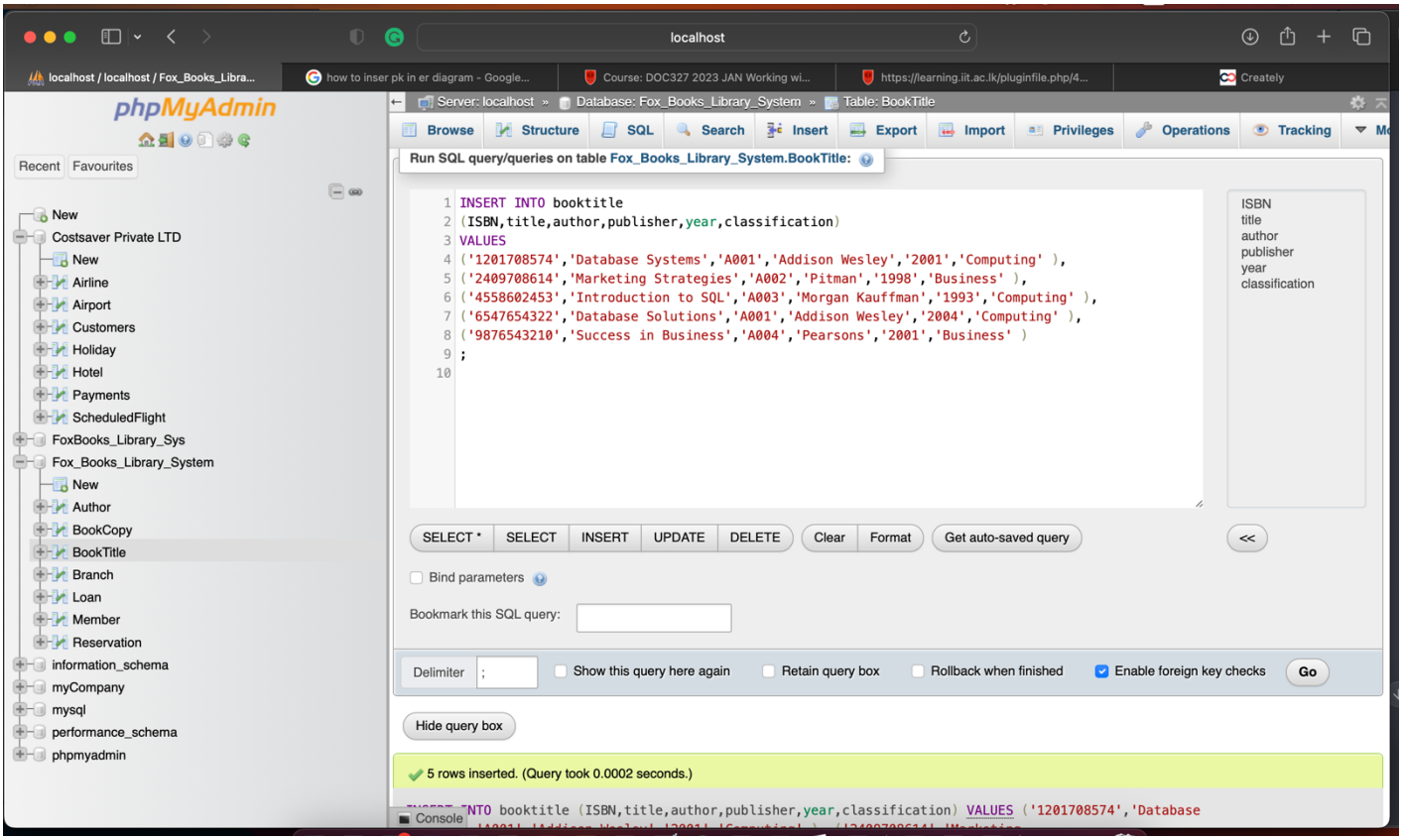

Figure 20 Booktitle data

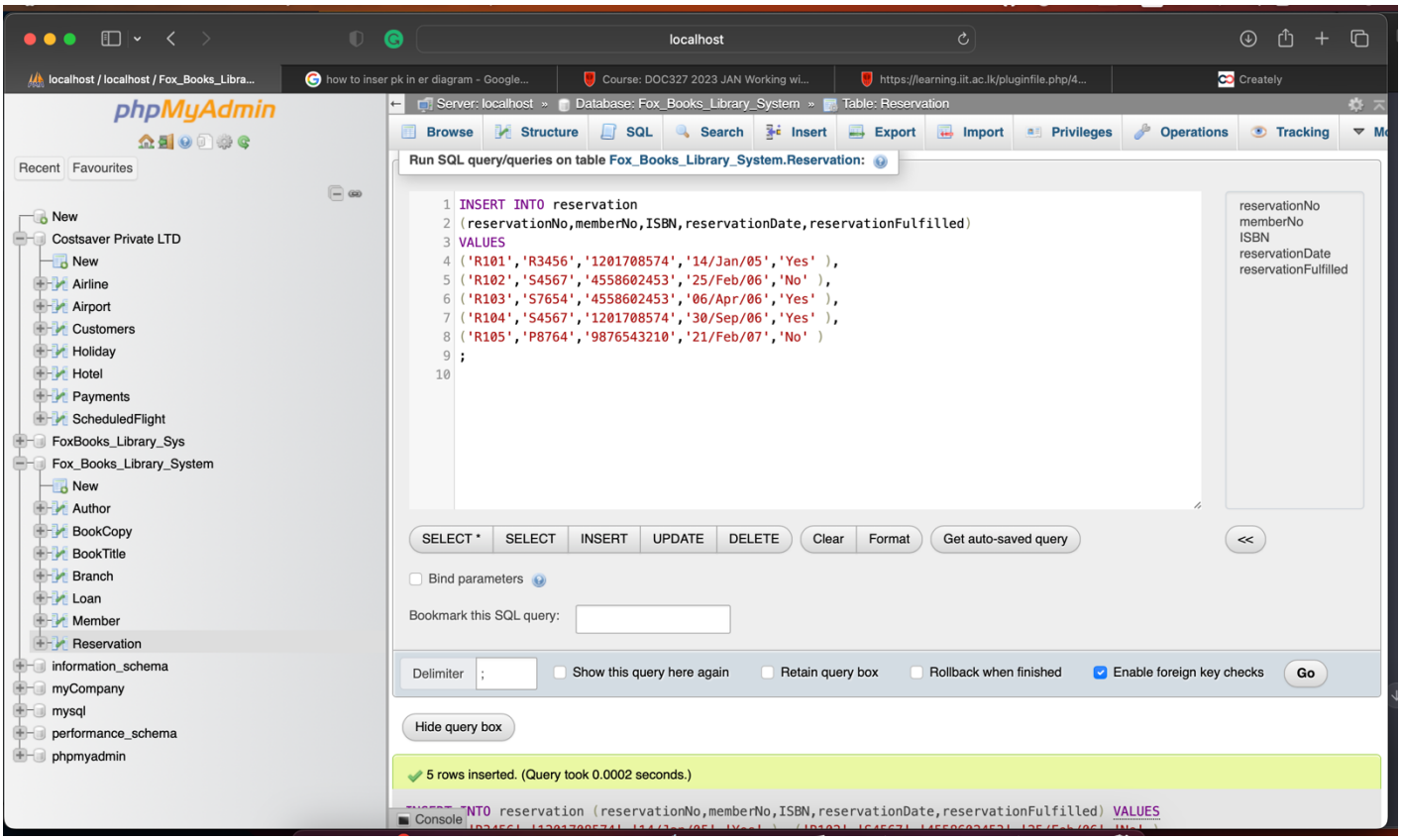

Figure 21 Reservation table data

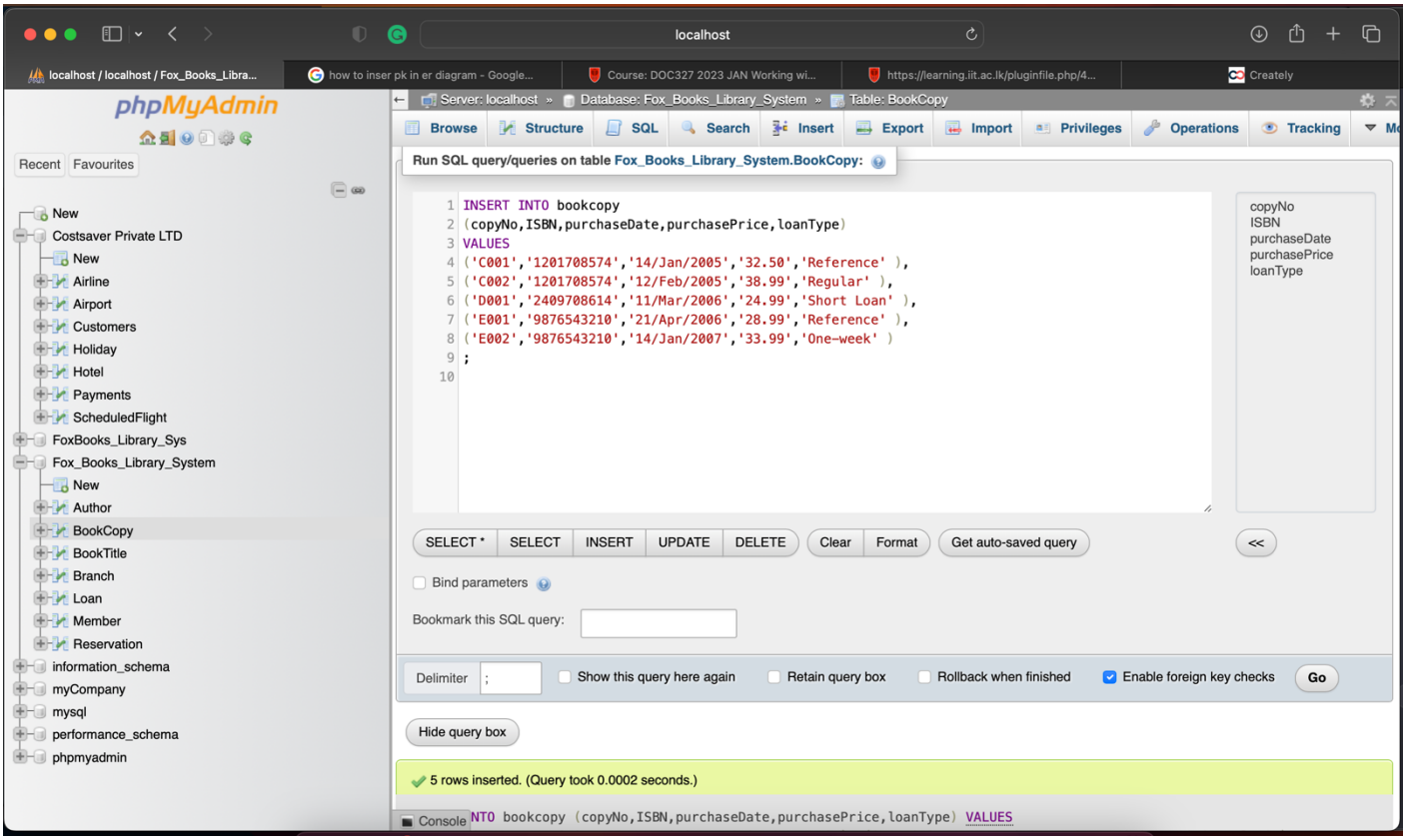

Figure 22 Bookcopy table data

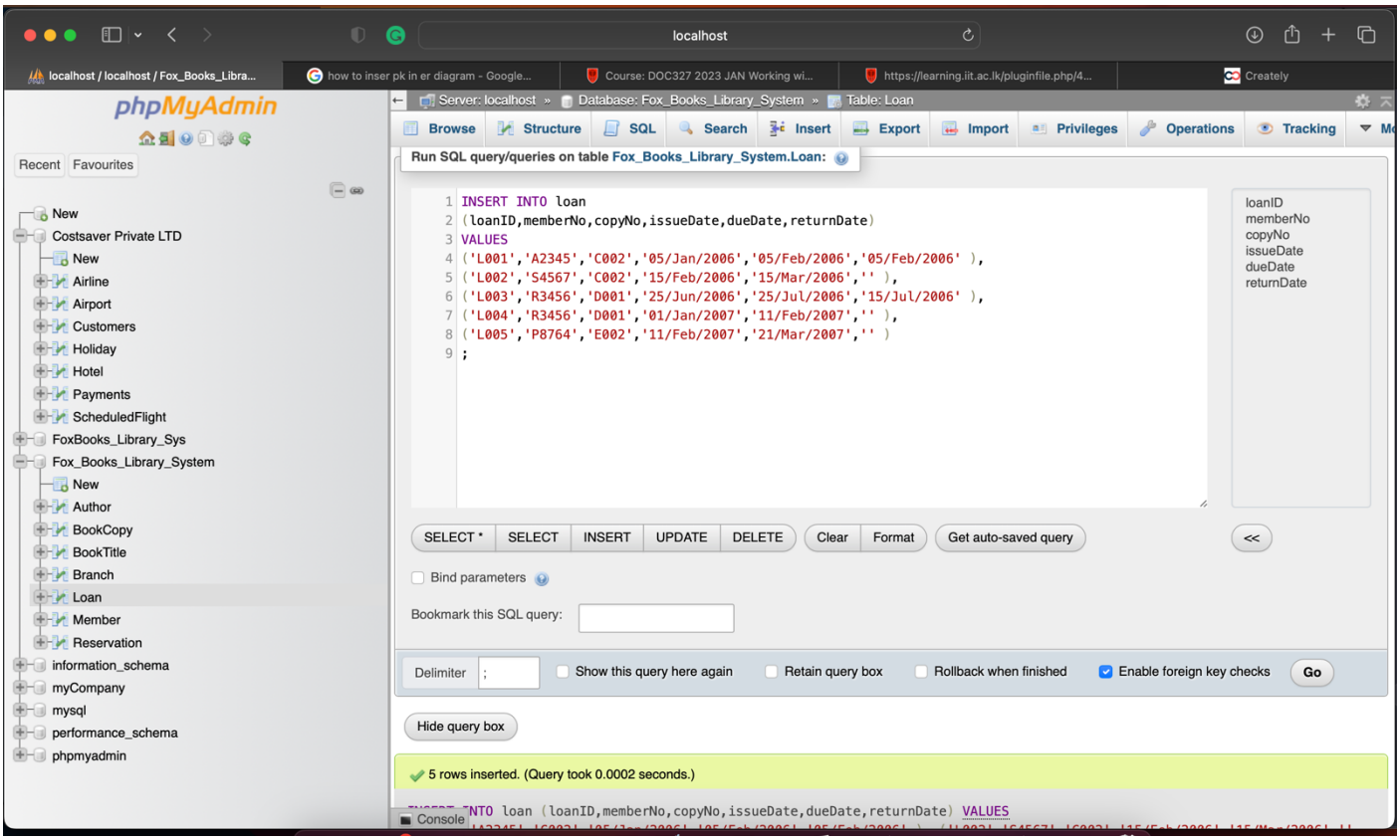

Figure 23 Loan table data

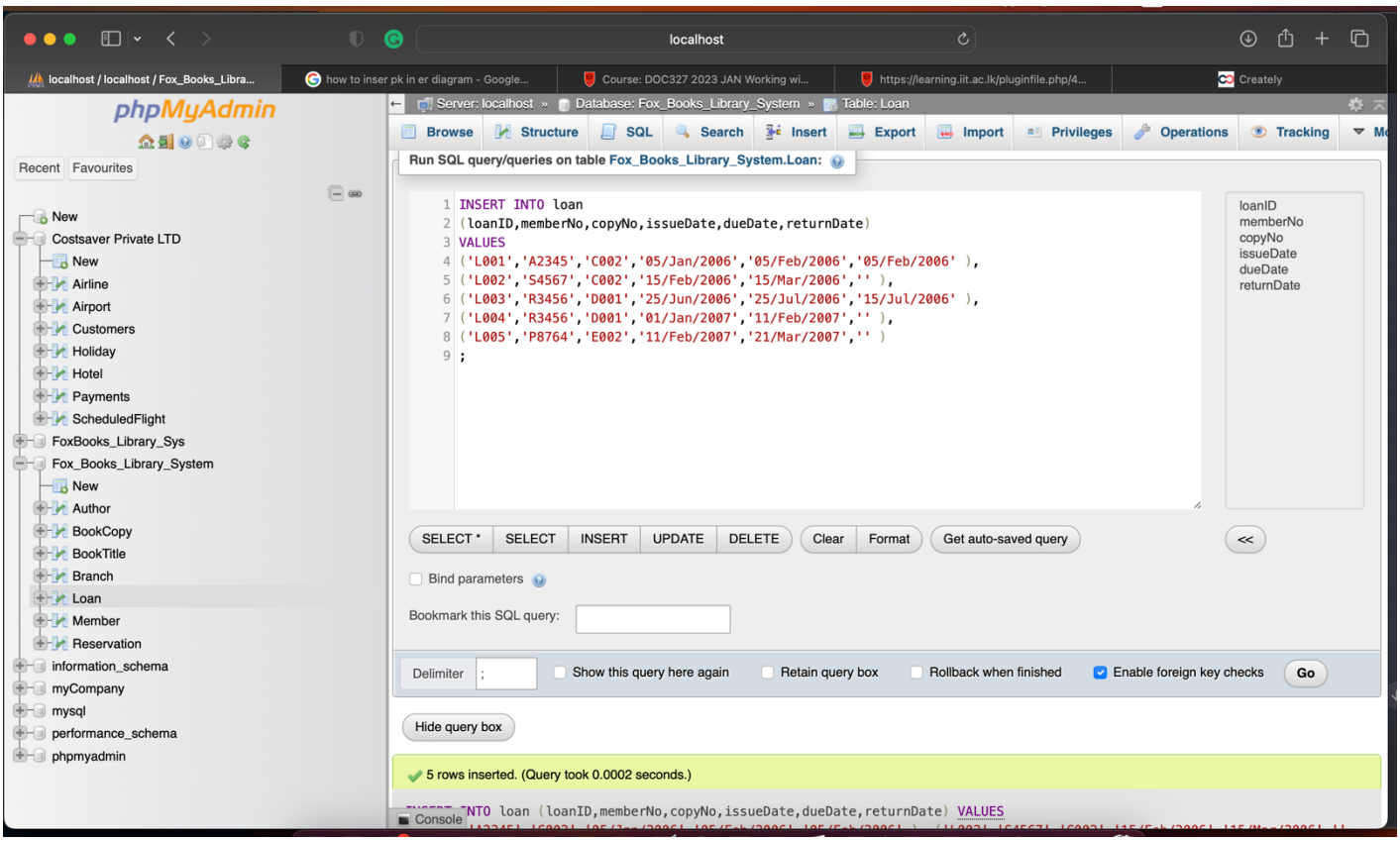

Figure 24 Loan table data

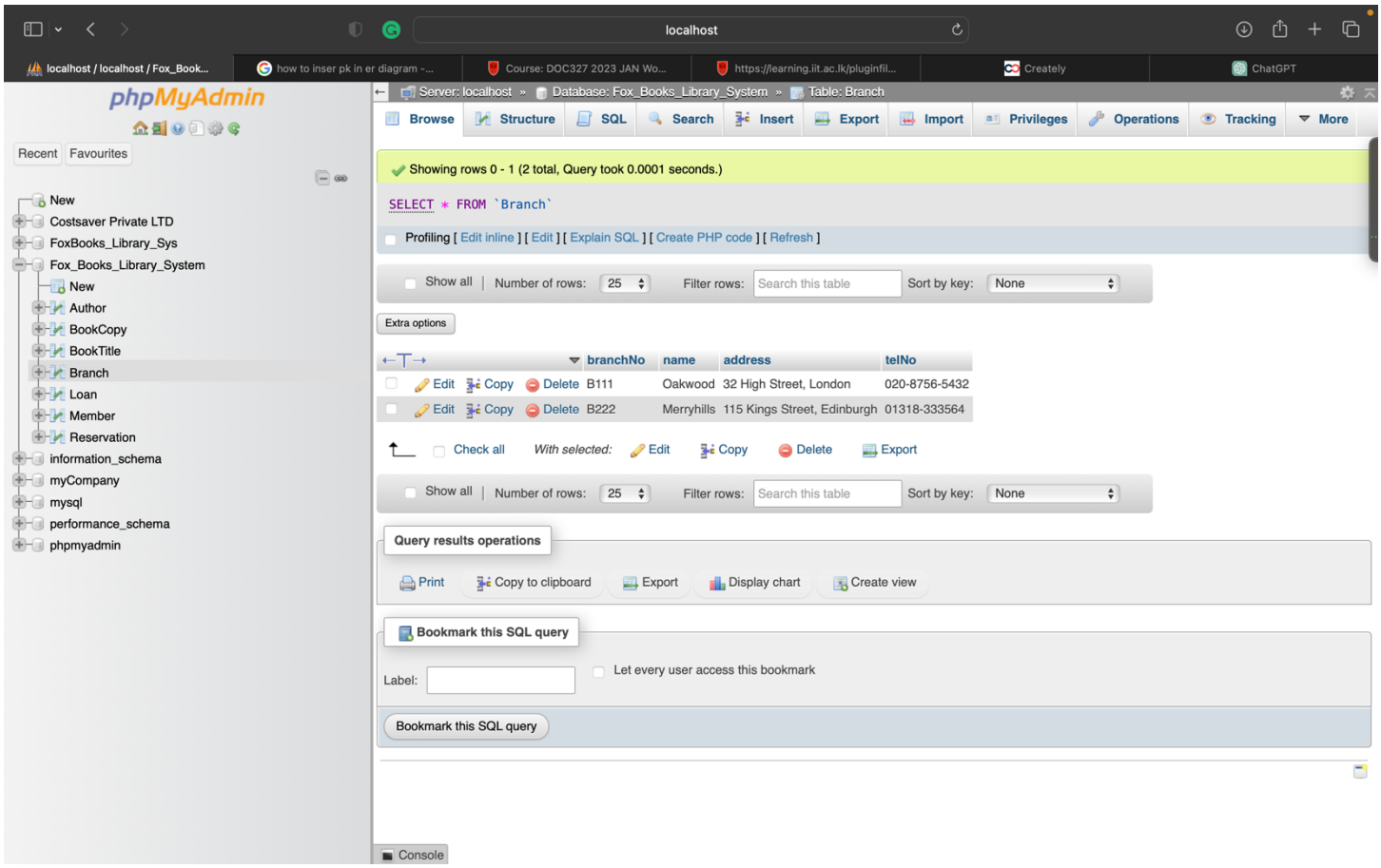

Figure 25 Branch table

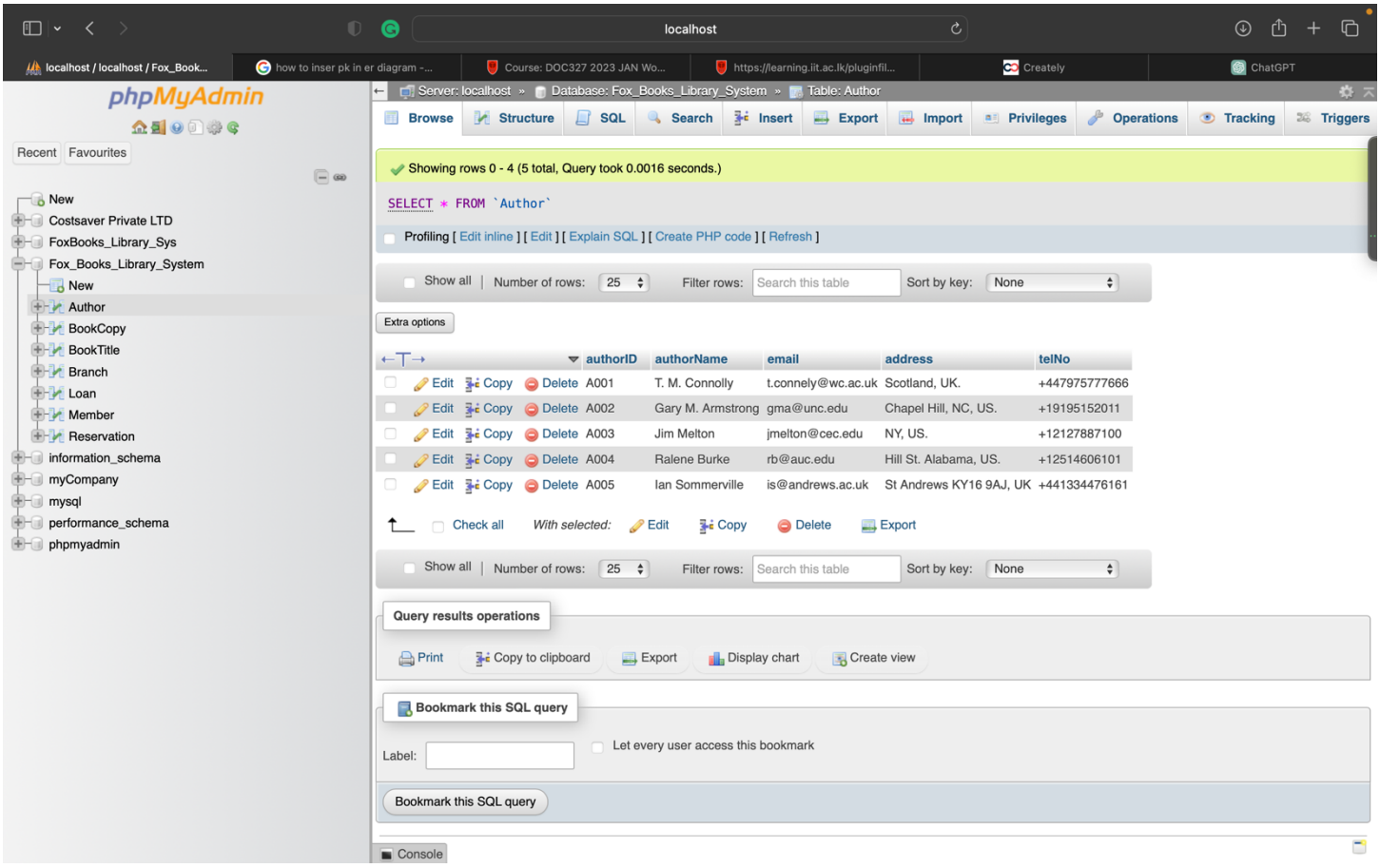

Figure 26 Author table

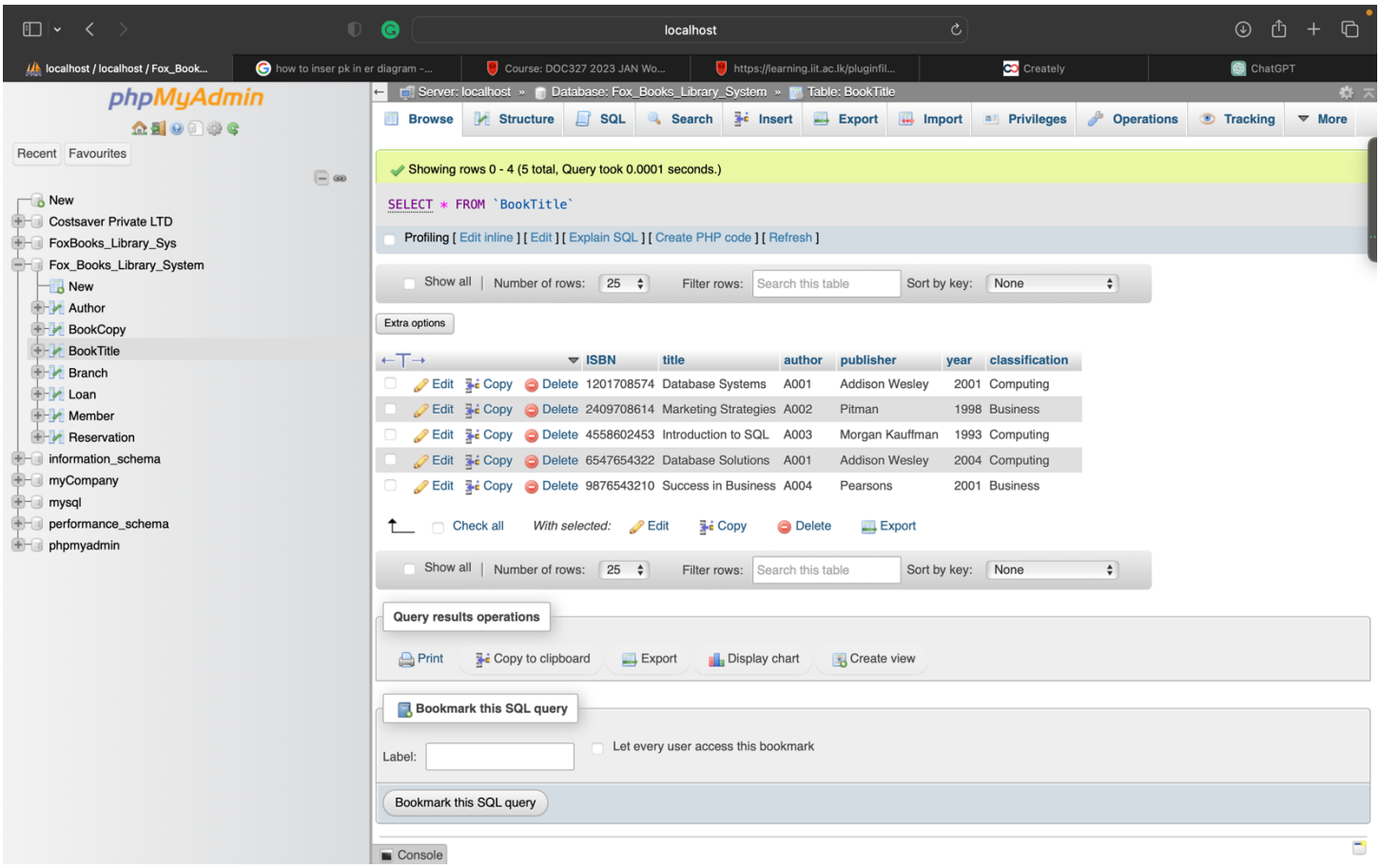

Figure 27 Booktitle table

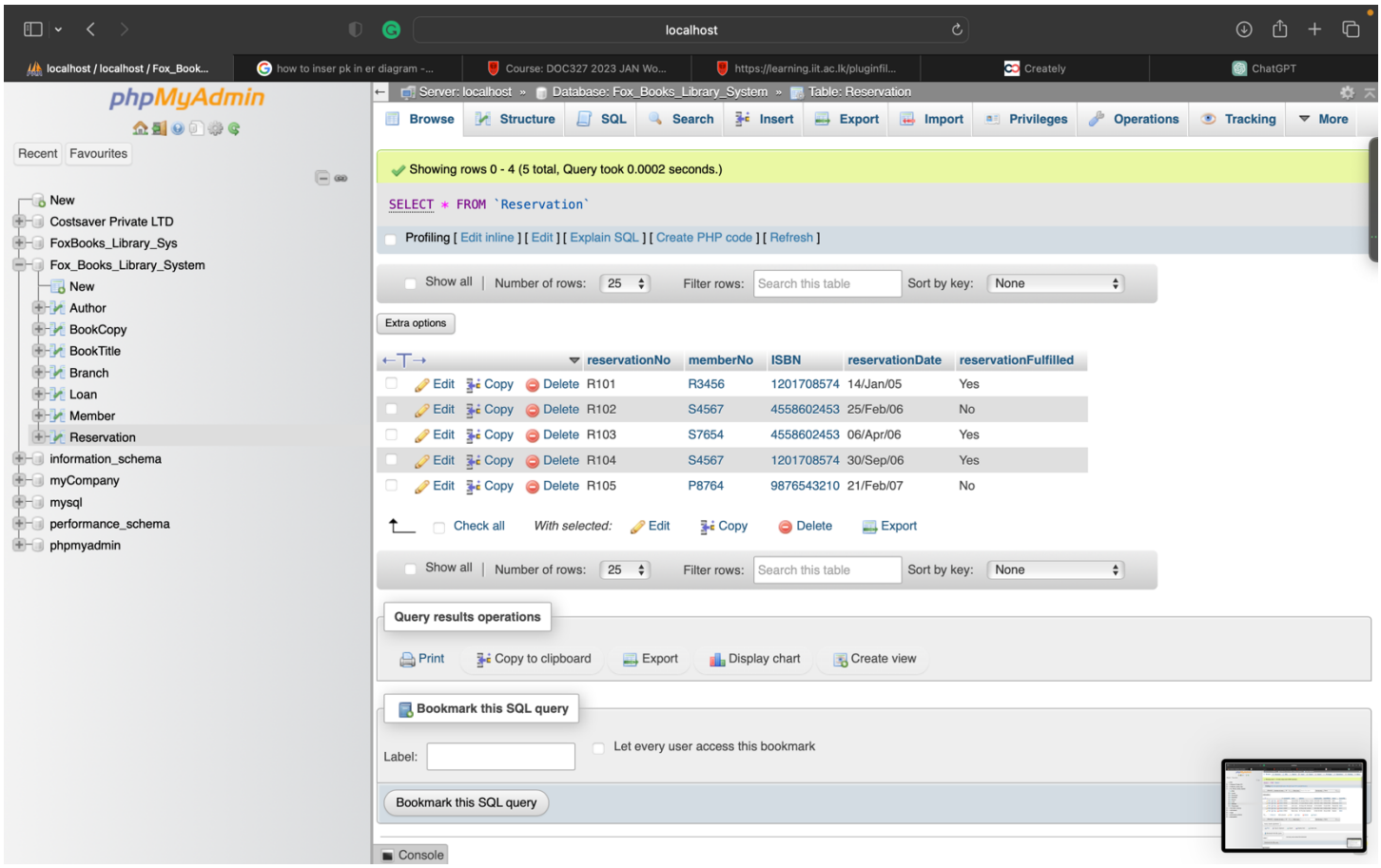

Figure 28 Reservation table

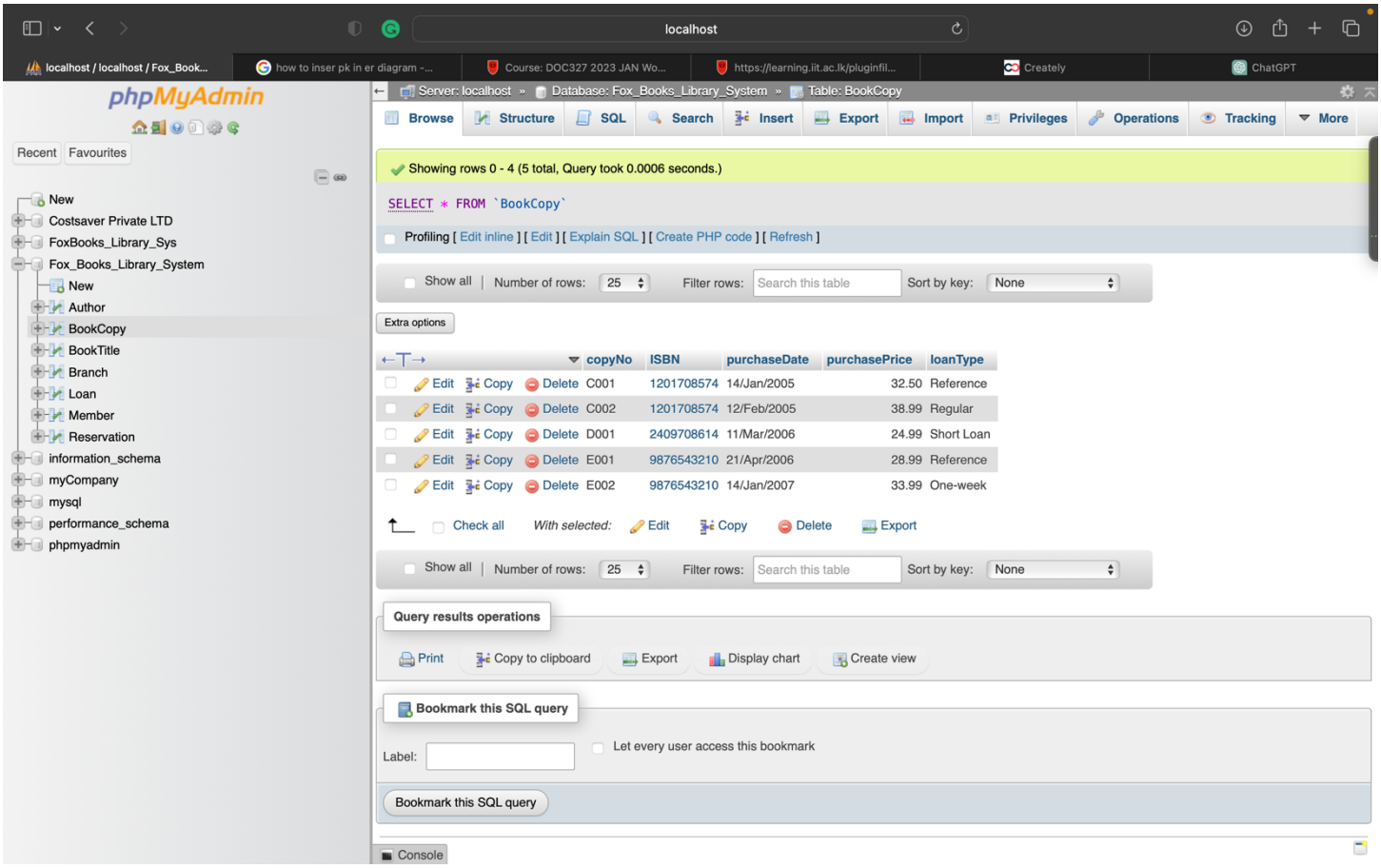

Figure 29 Bookcopy table

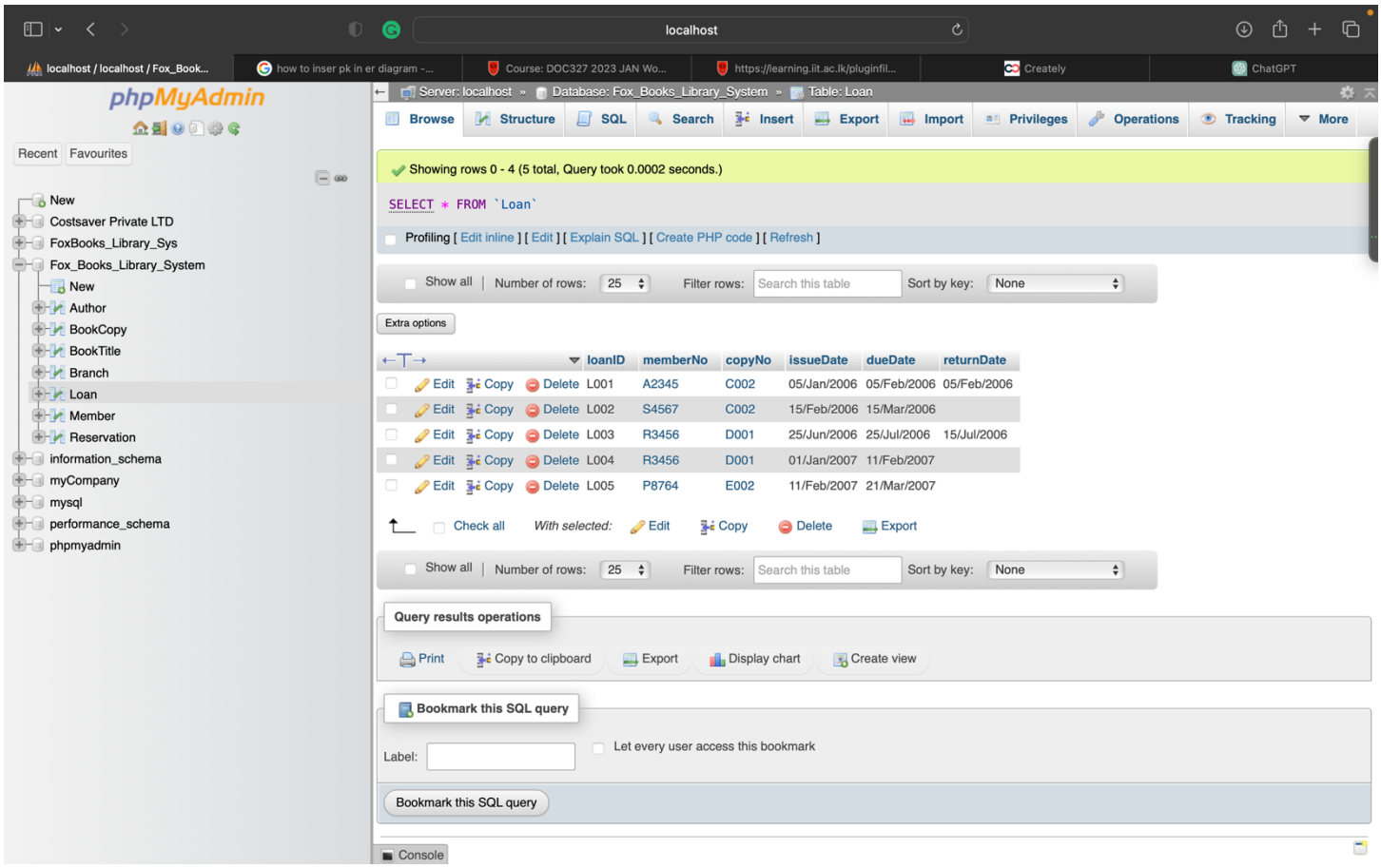

Figure 30 Loan table

Question 03

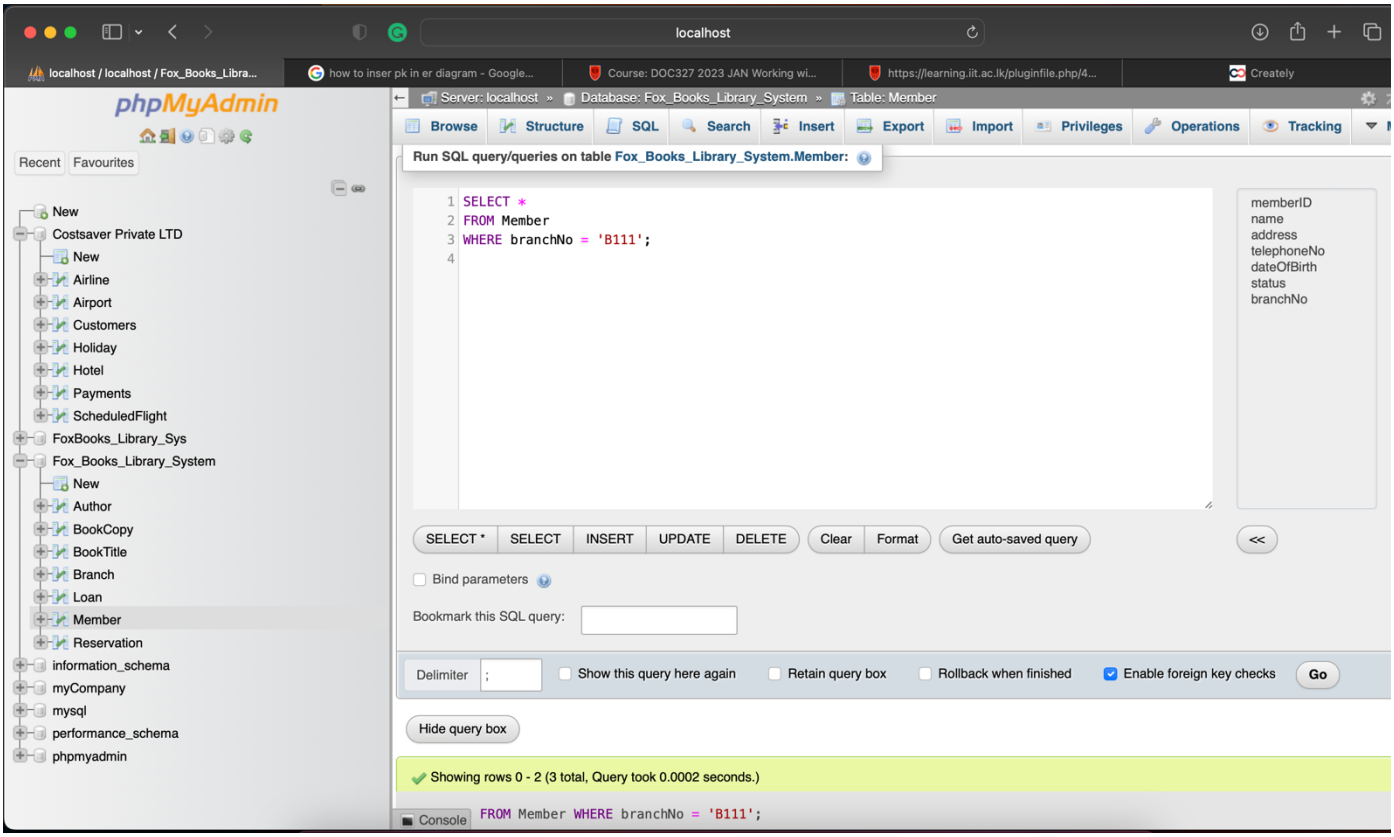

Figure 32 Q3

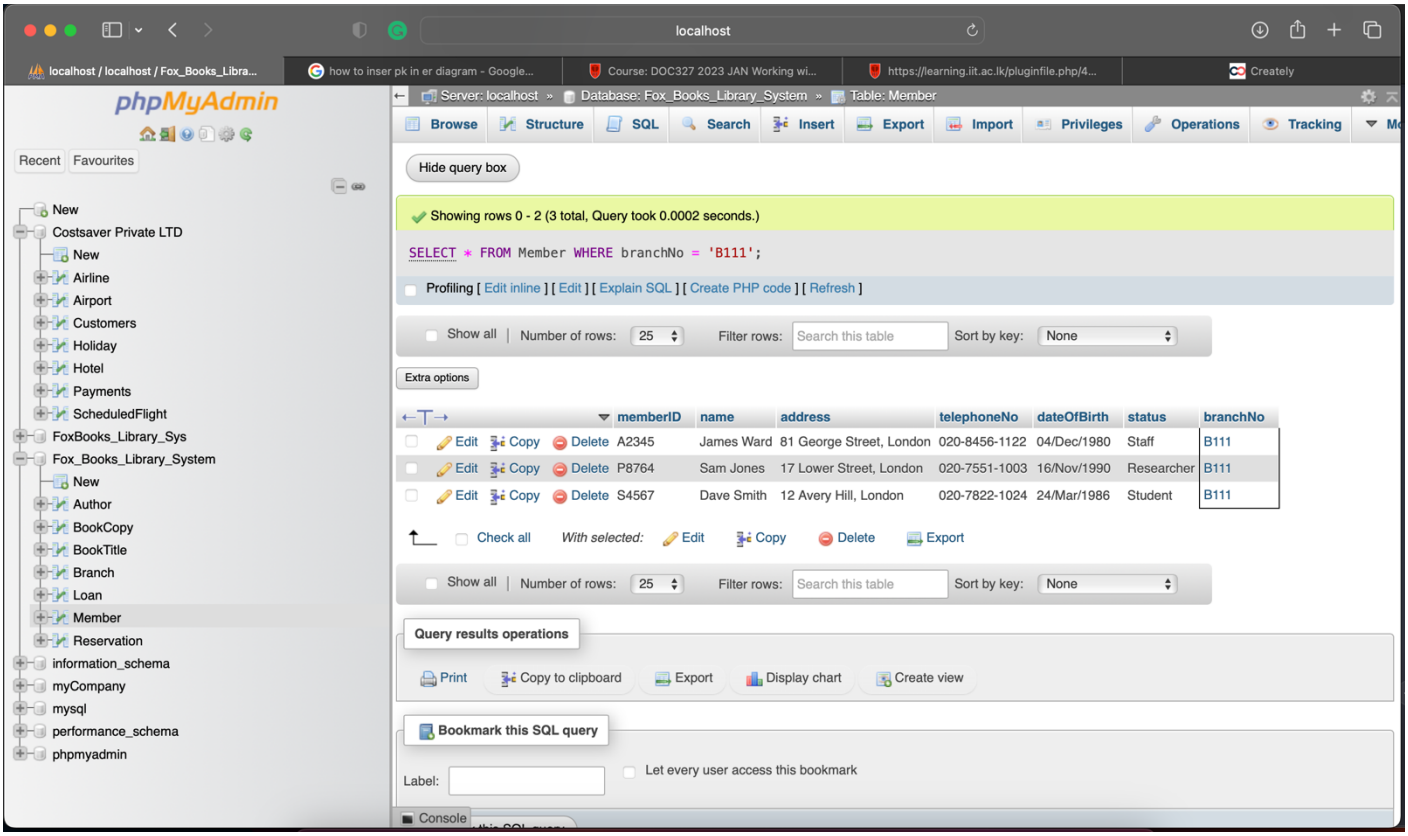

Figure 31 Q3

b)

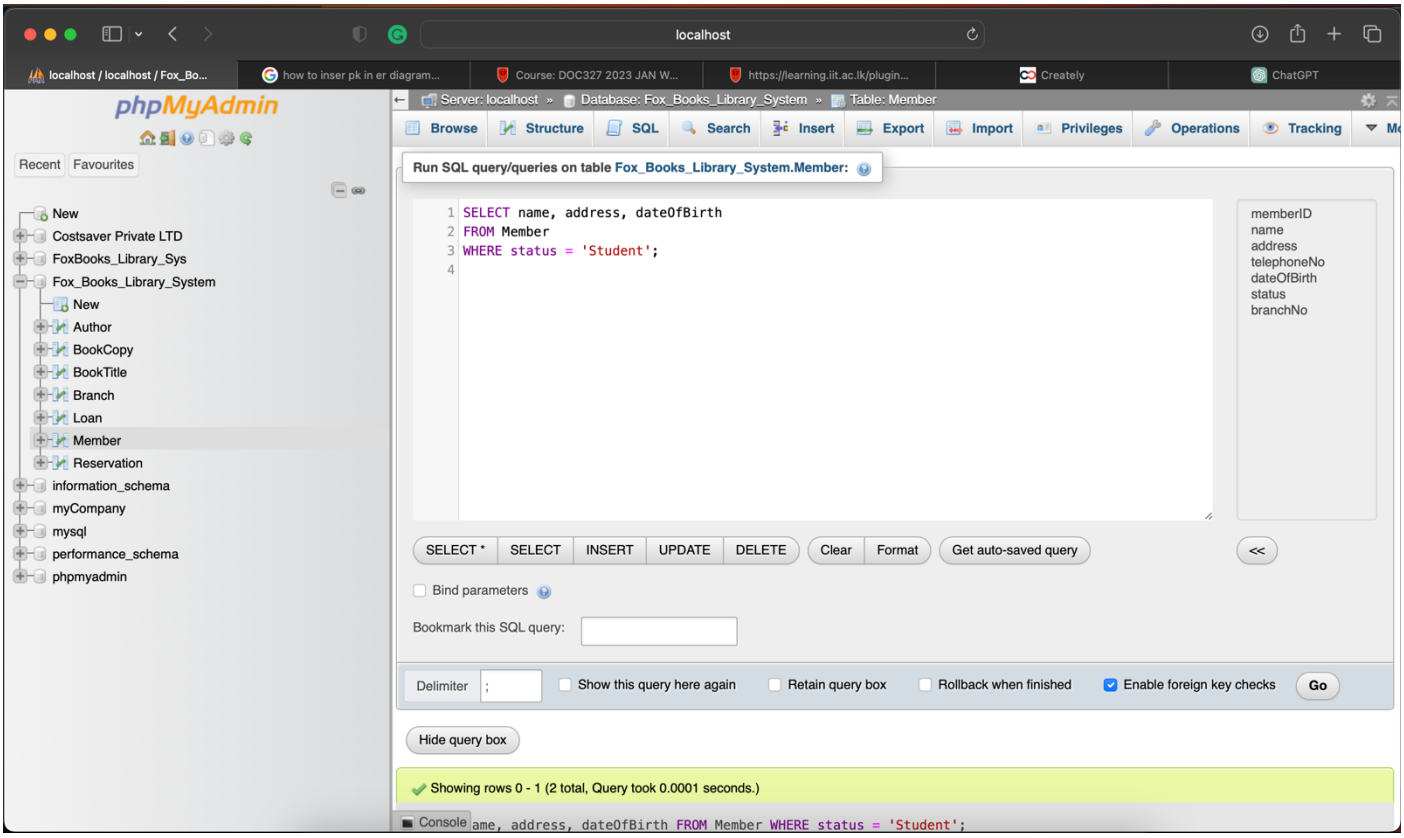

Figure 33 Q3

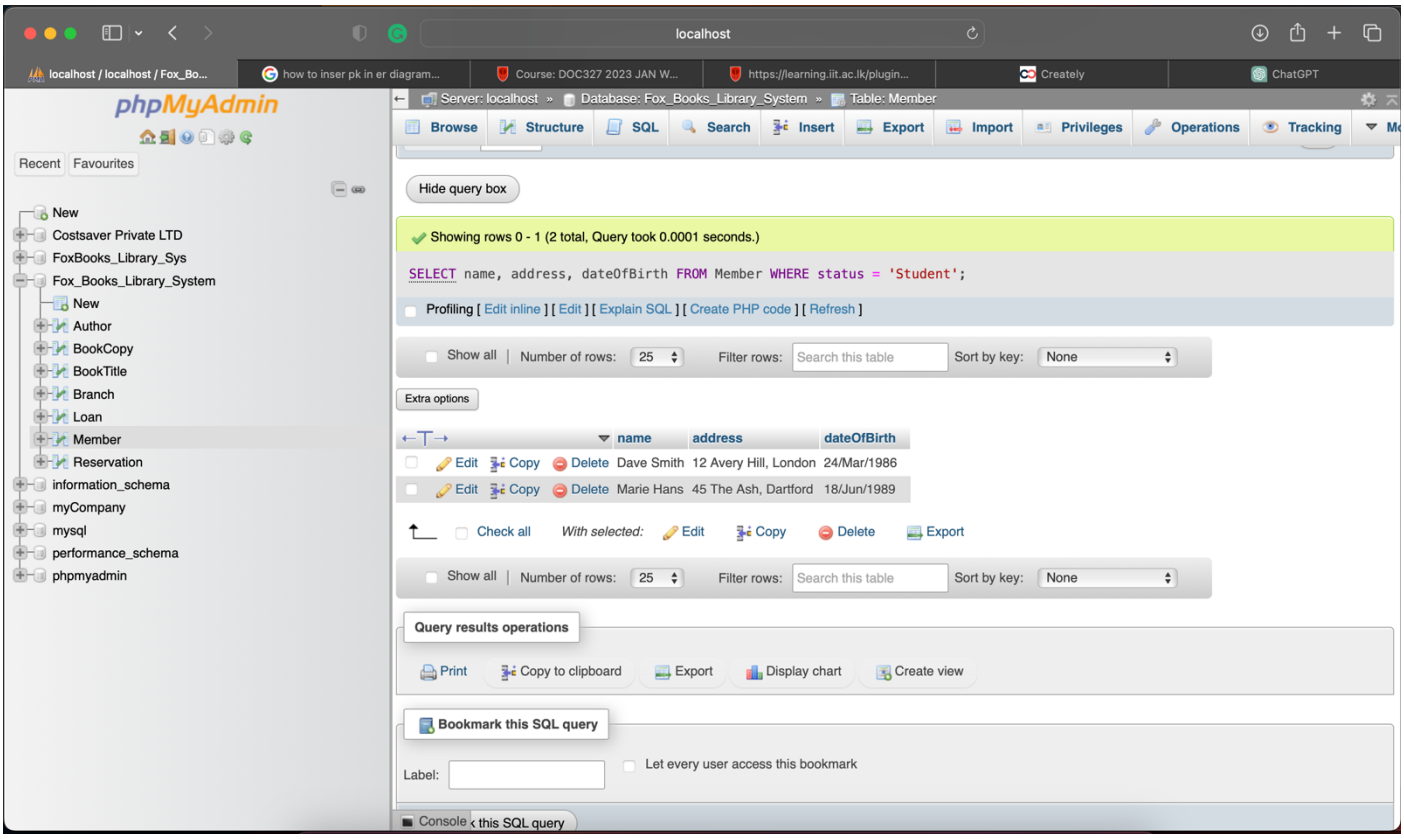

Figure 34 Q3

c)

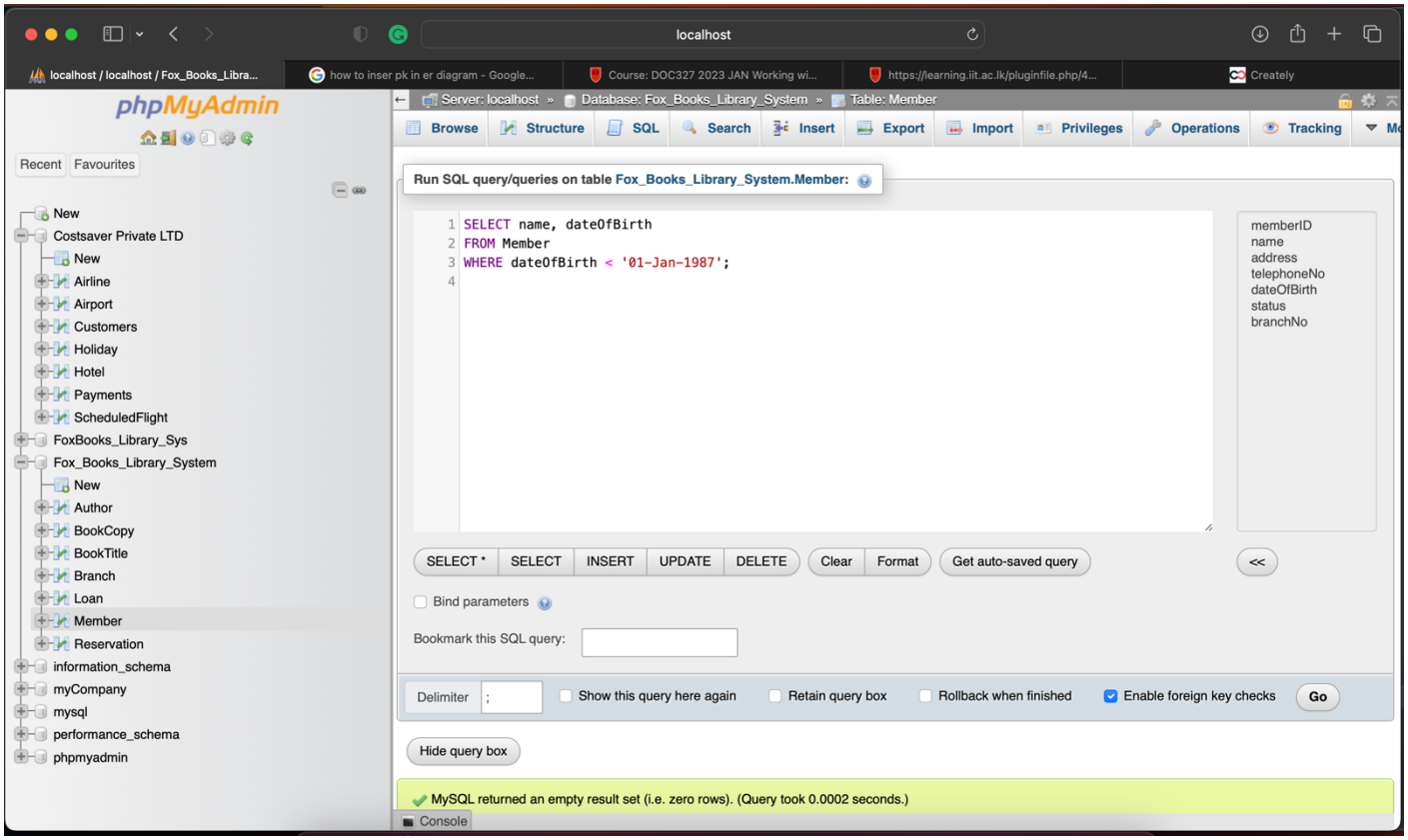

Figure 35 Q3

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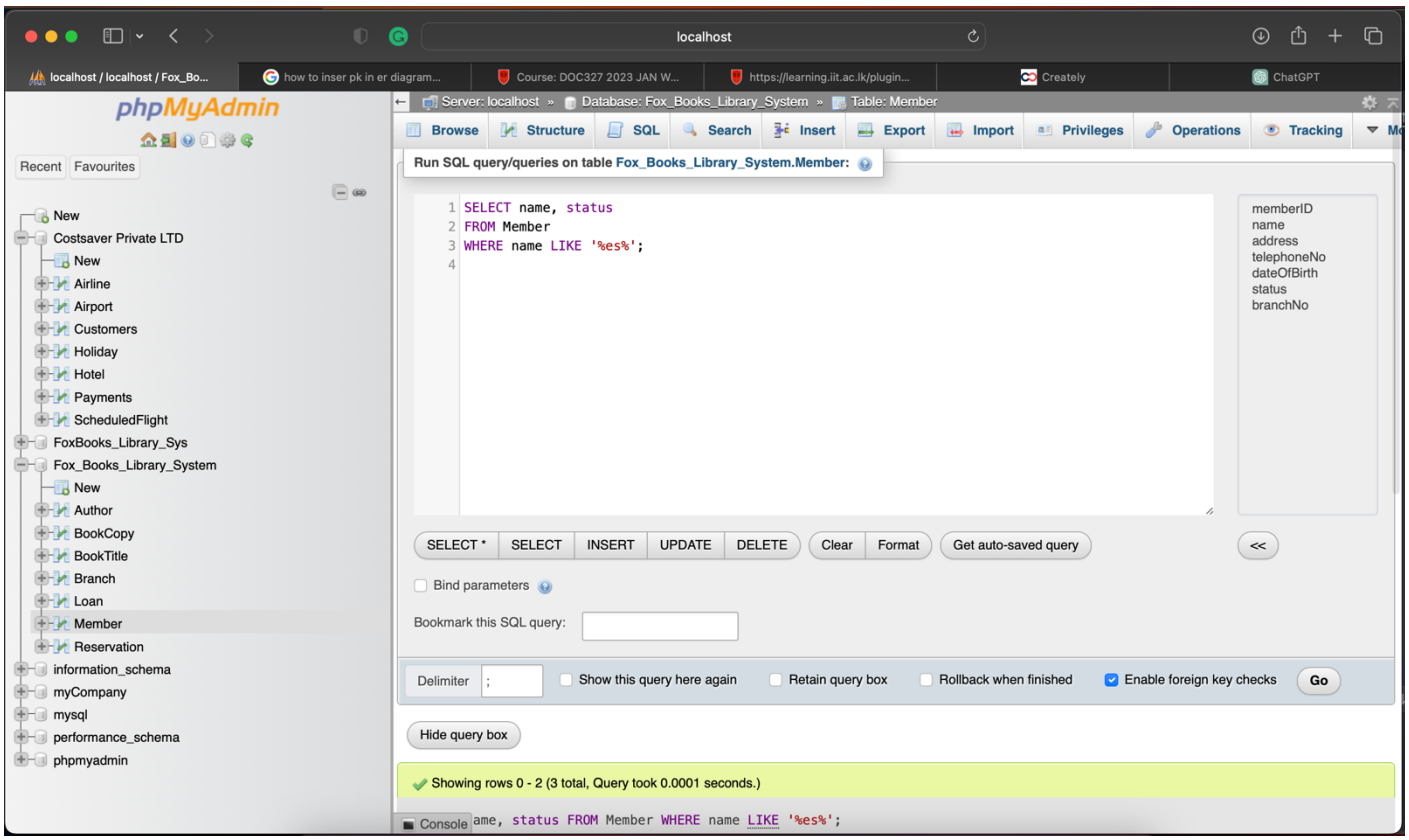

Figure 36 Q3

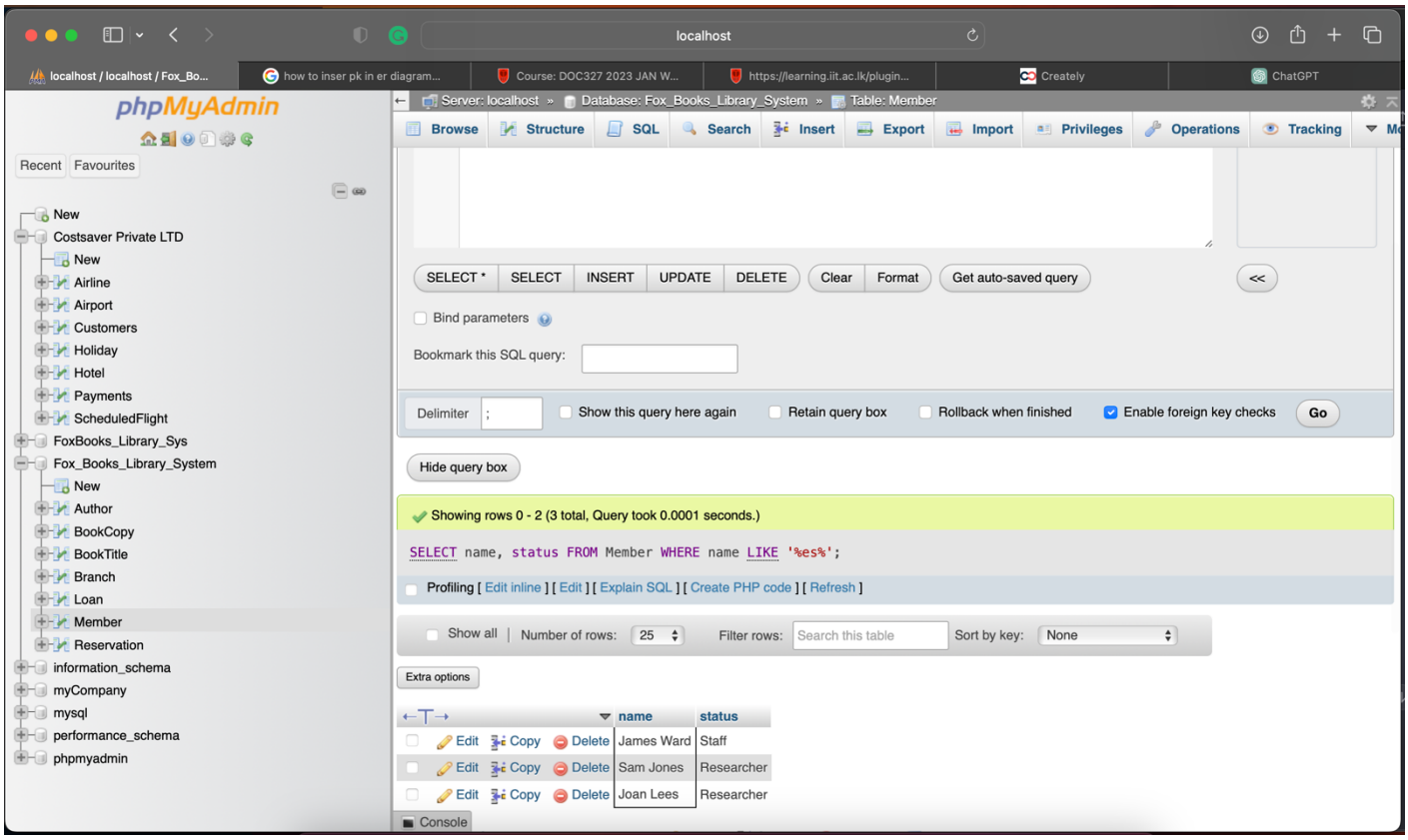

Figure 37 Q3

e)

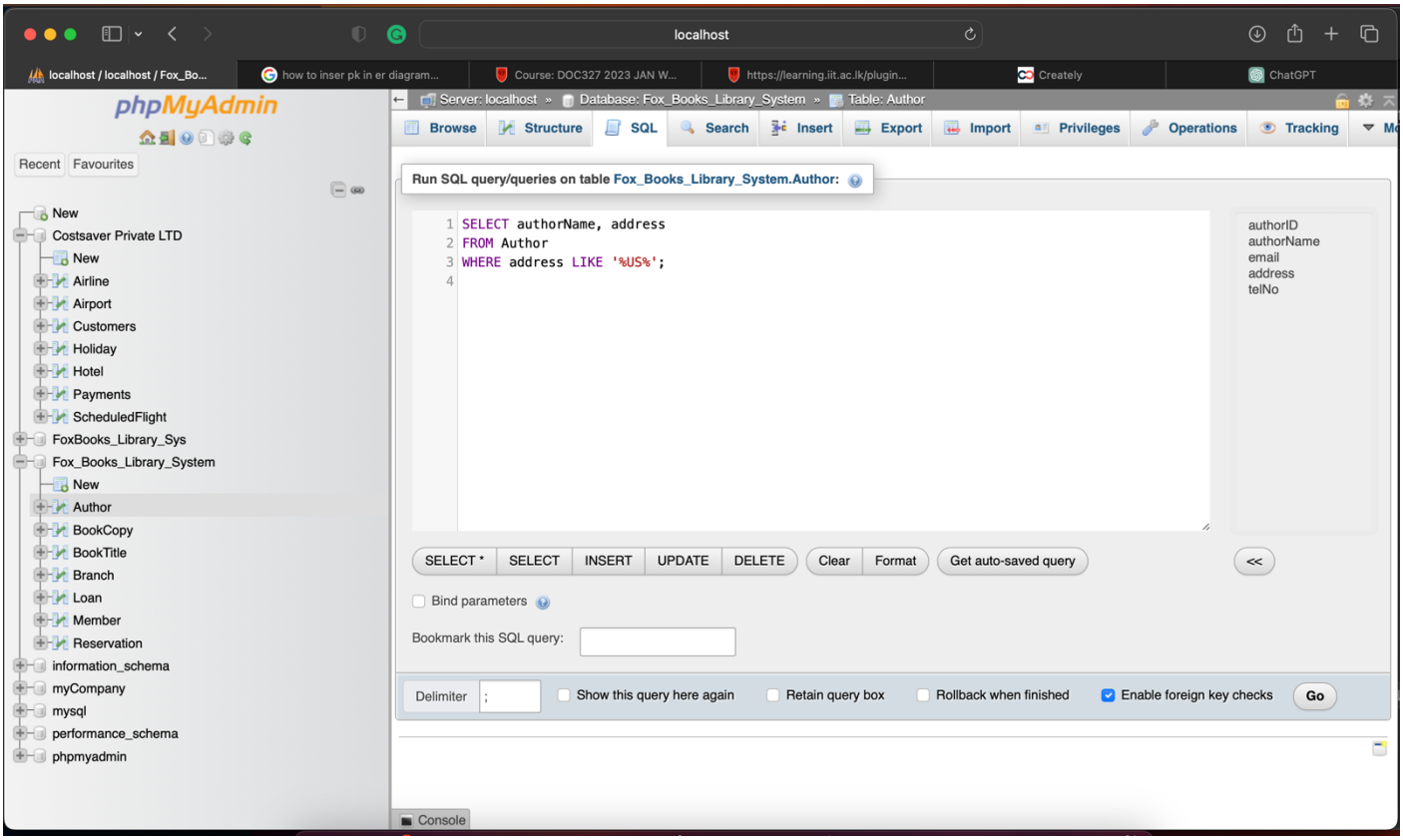

Figure 38 Q3

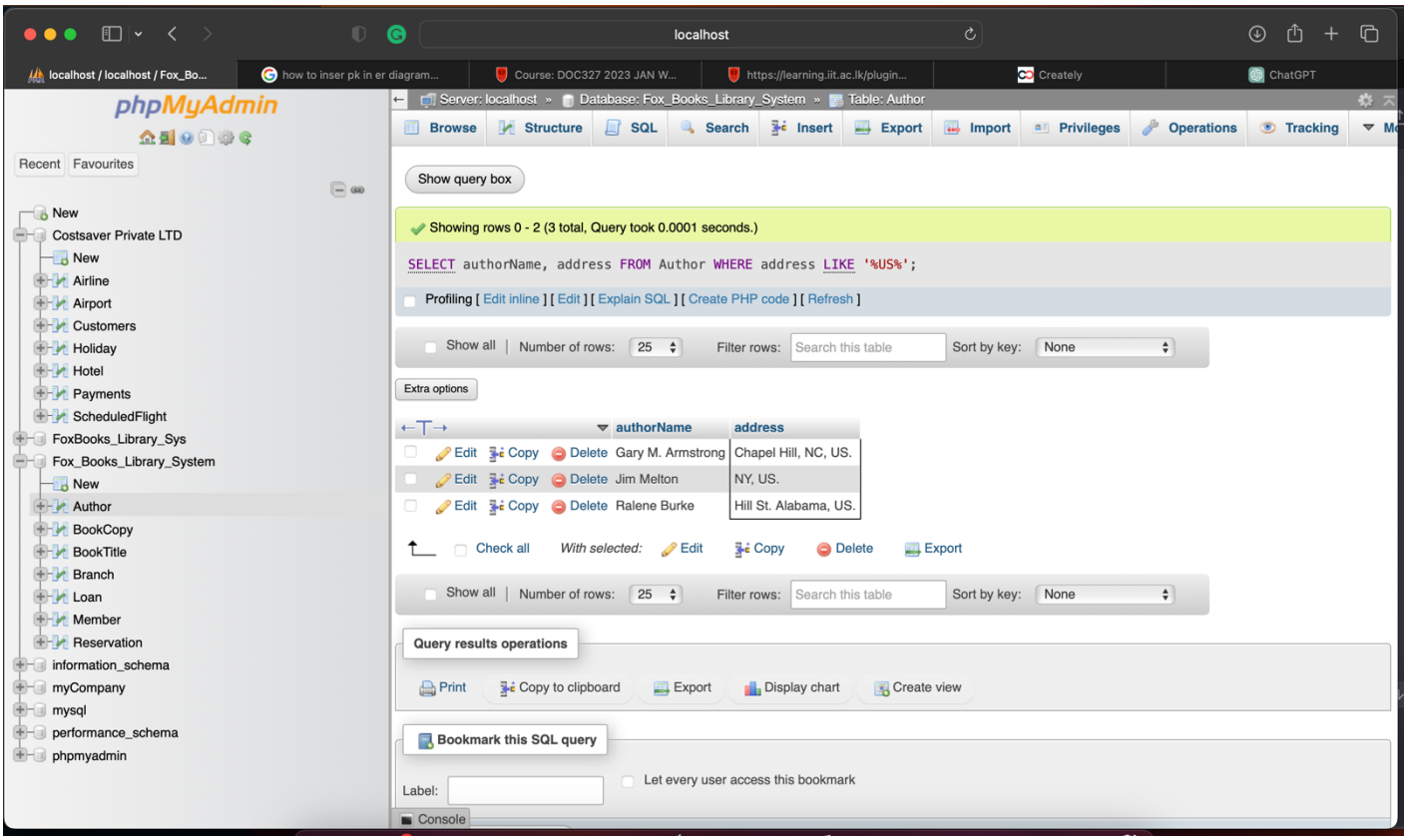

Figure 39 Q3

f)

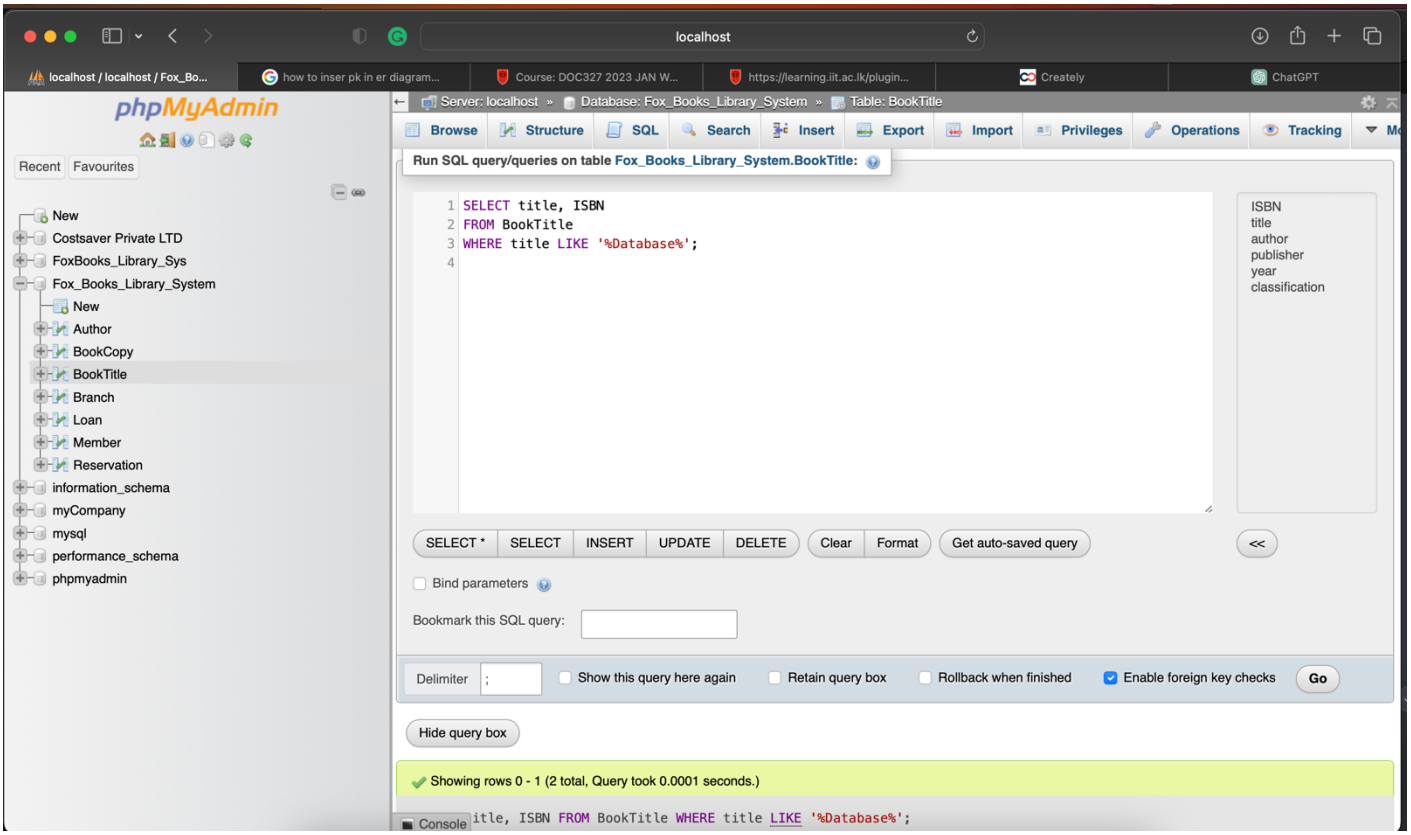

Figure 40 Q3

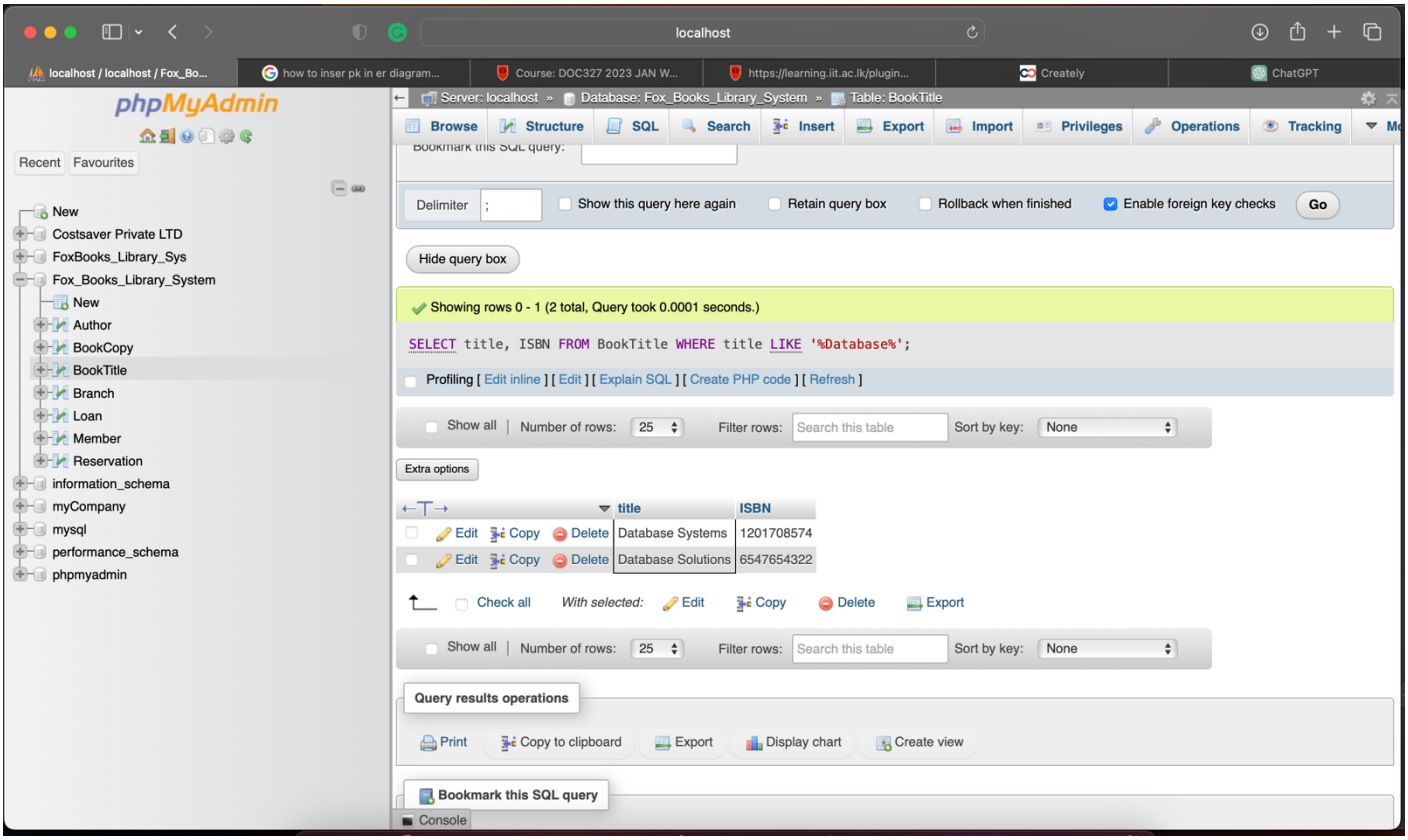

Figure 42 Q3

g)

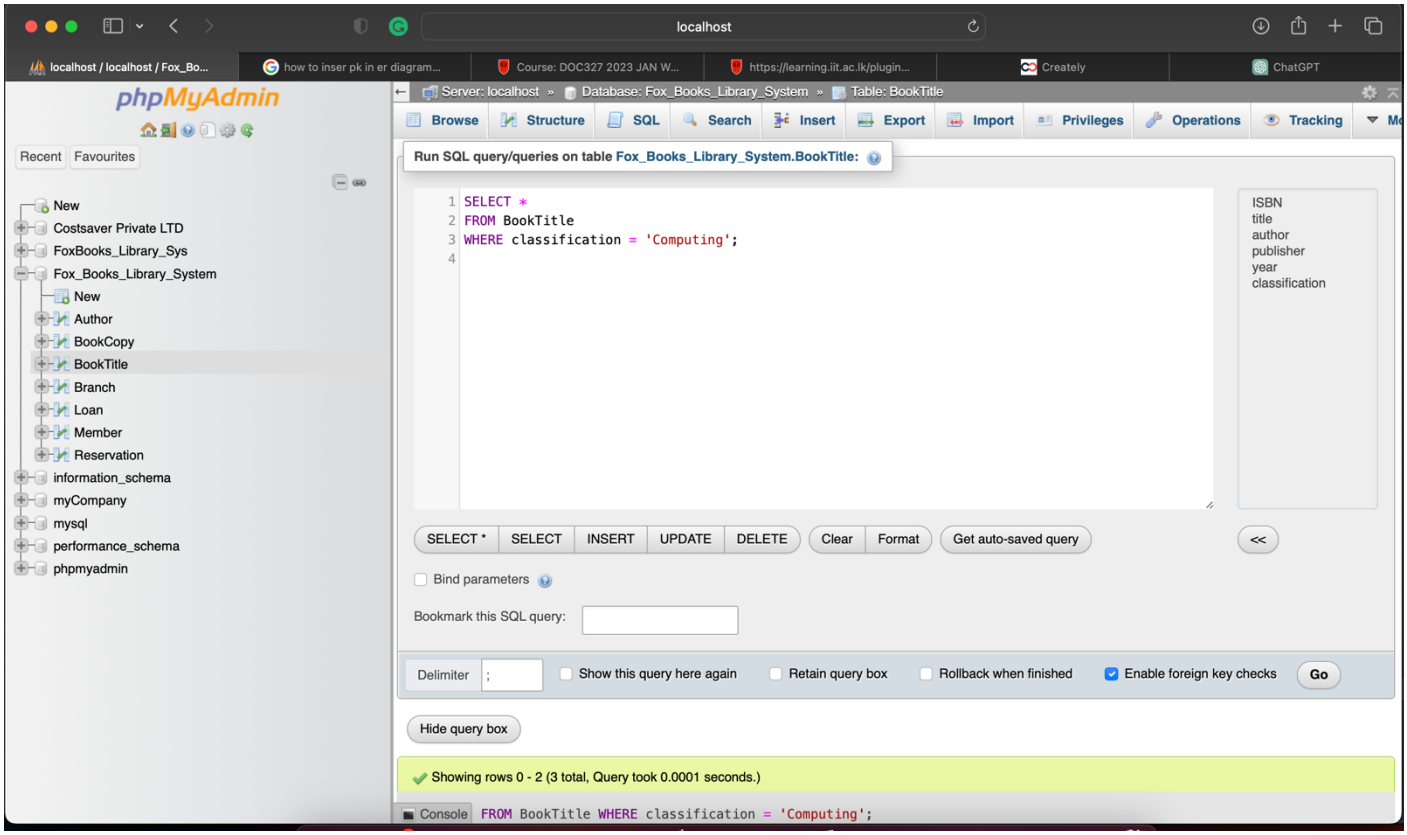

Figure 41 Q3

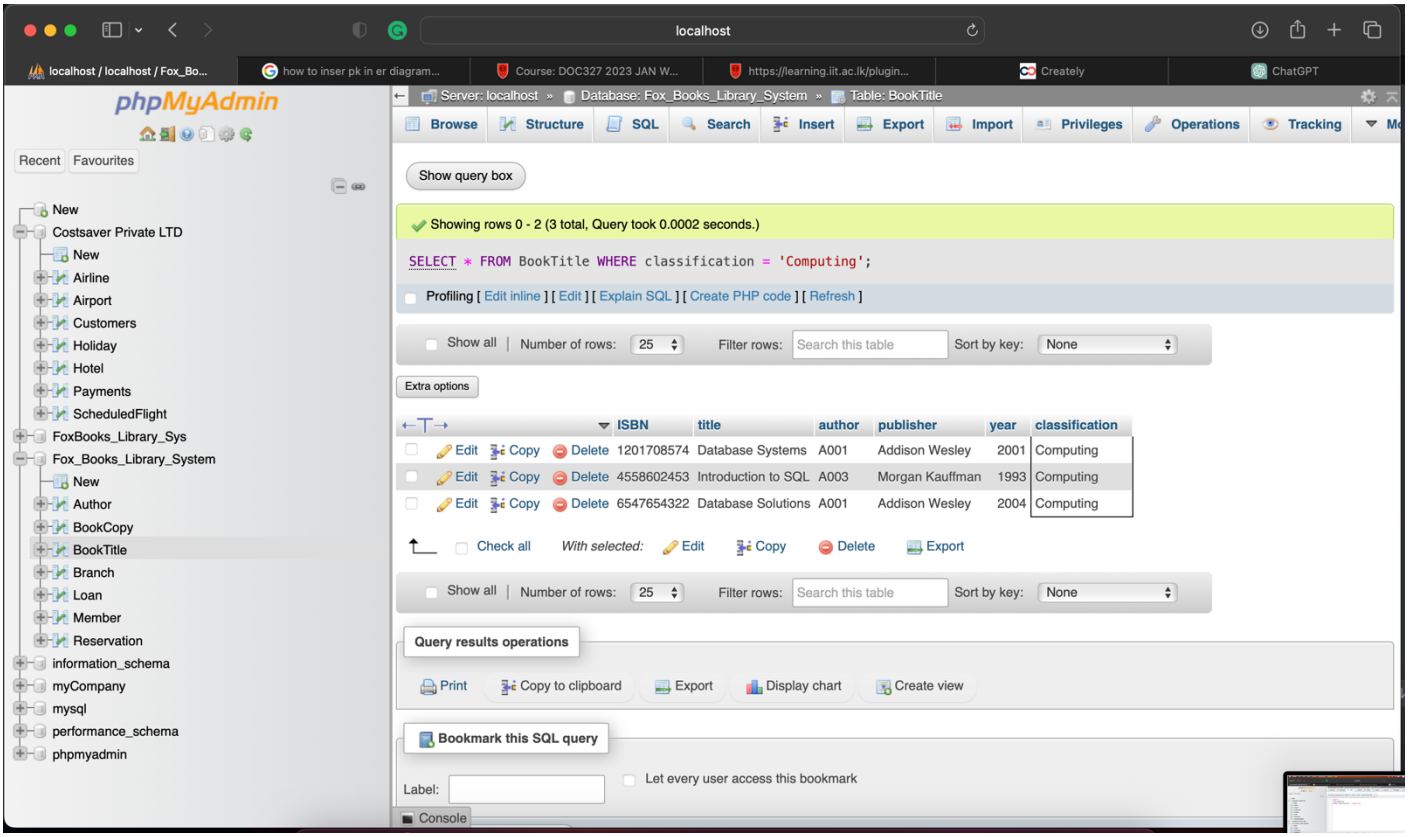

Figure 43 Q3

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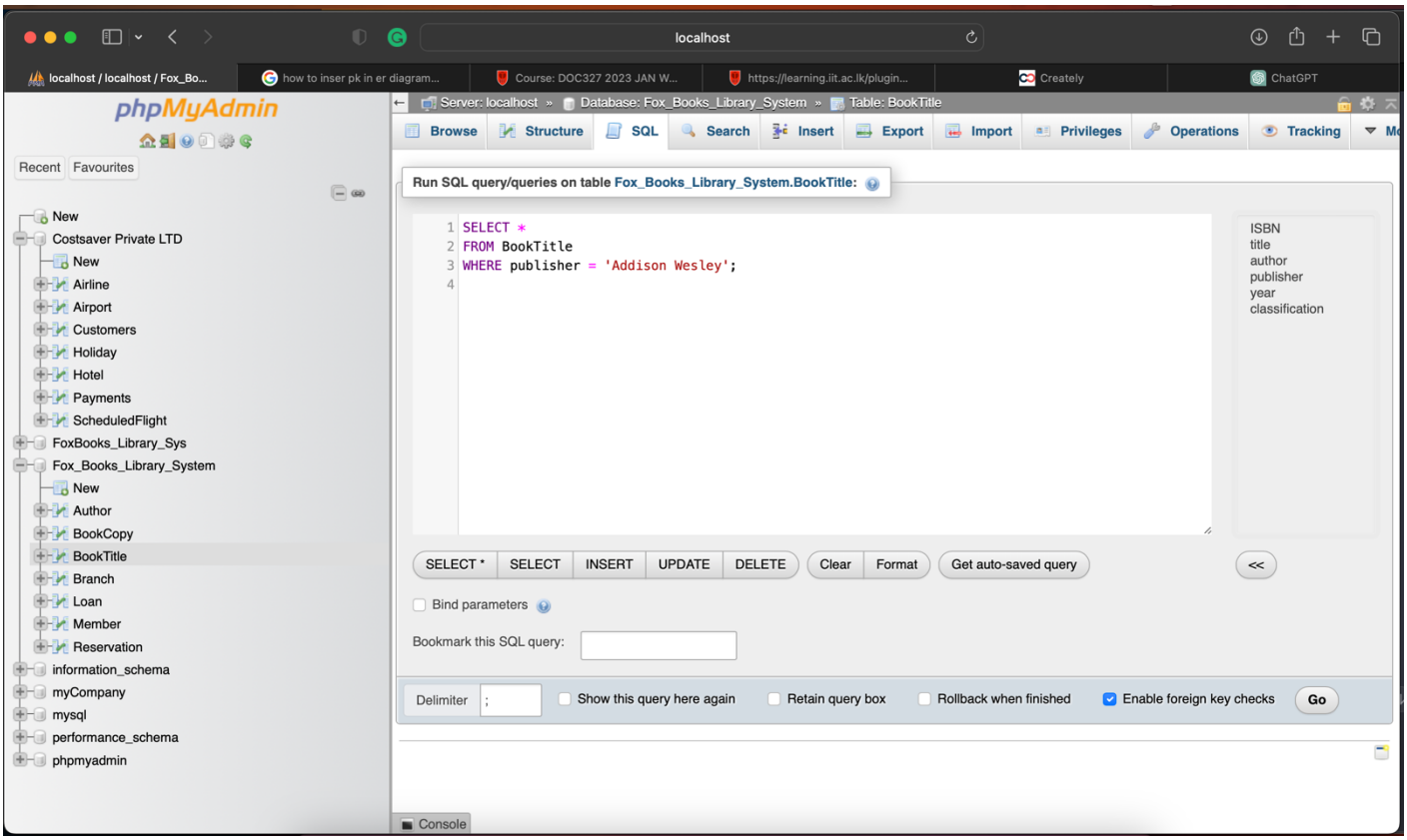

Figure 44 Q3

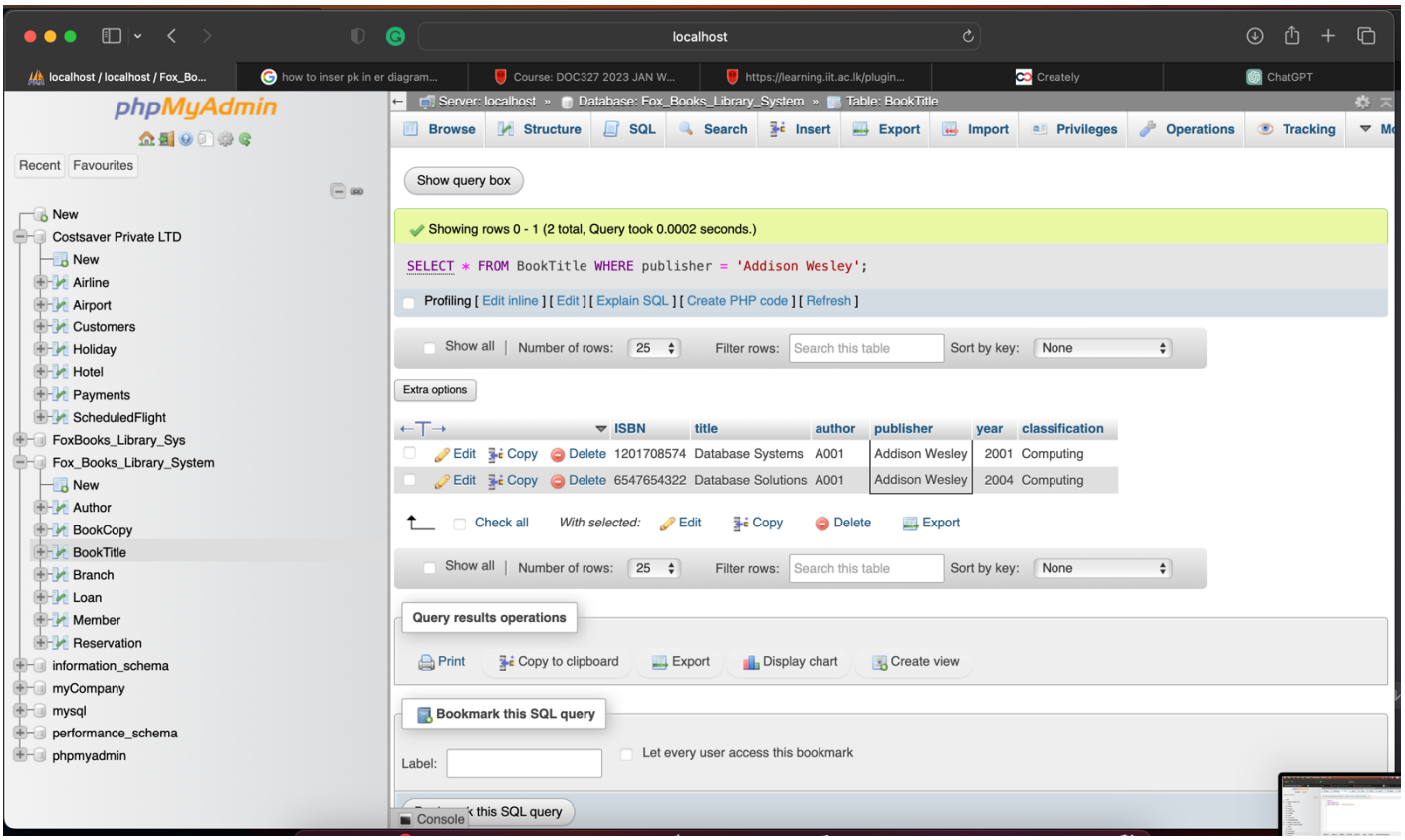

Figure 45 Q3

i)

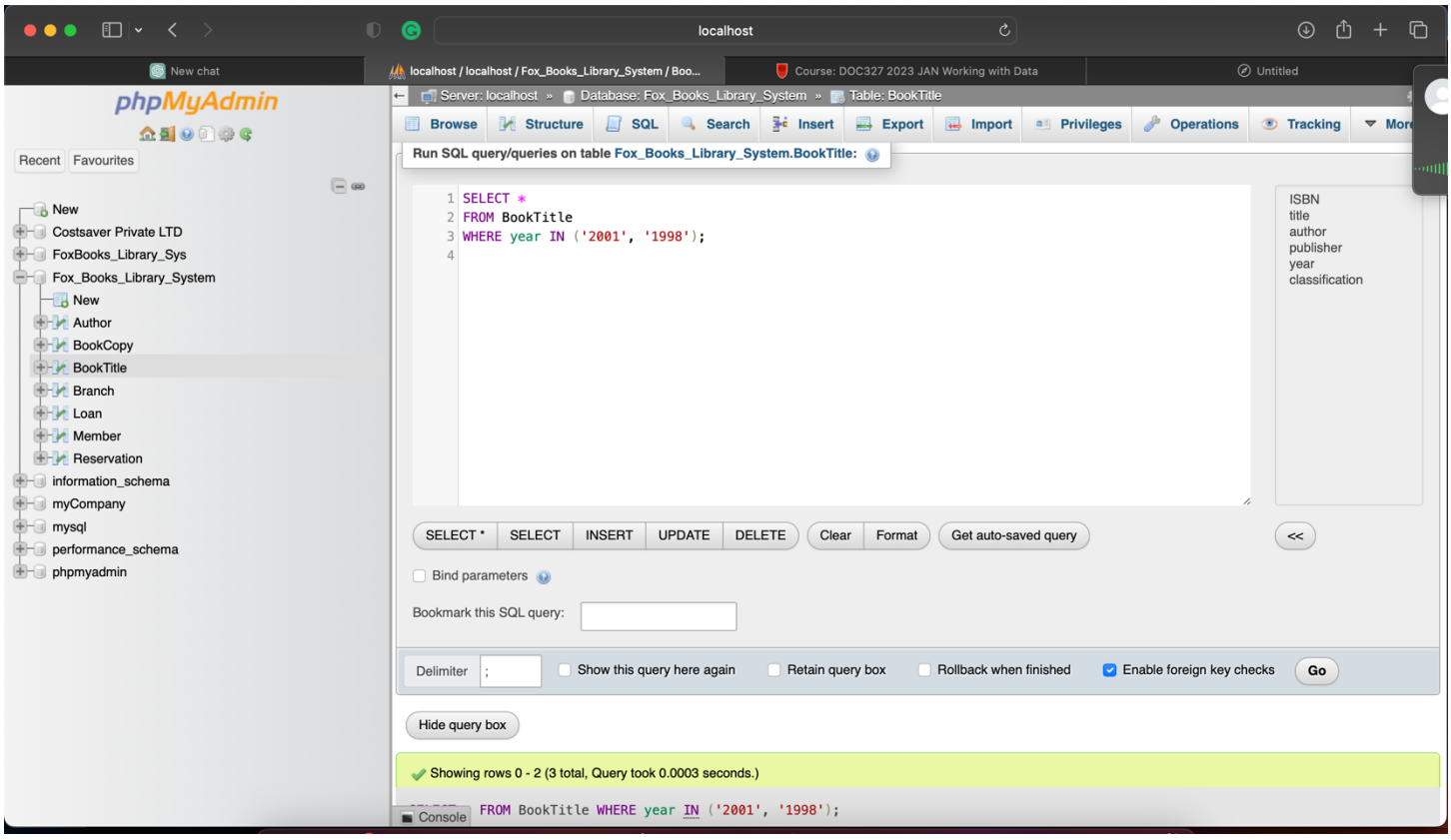

Figure 46 Q3

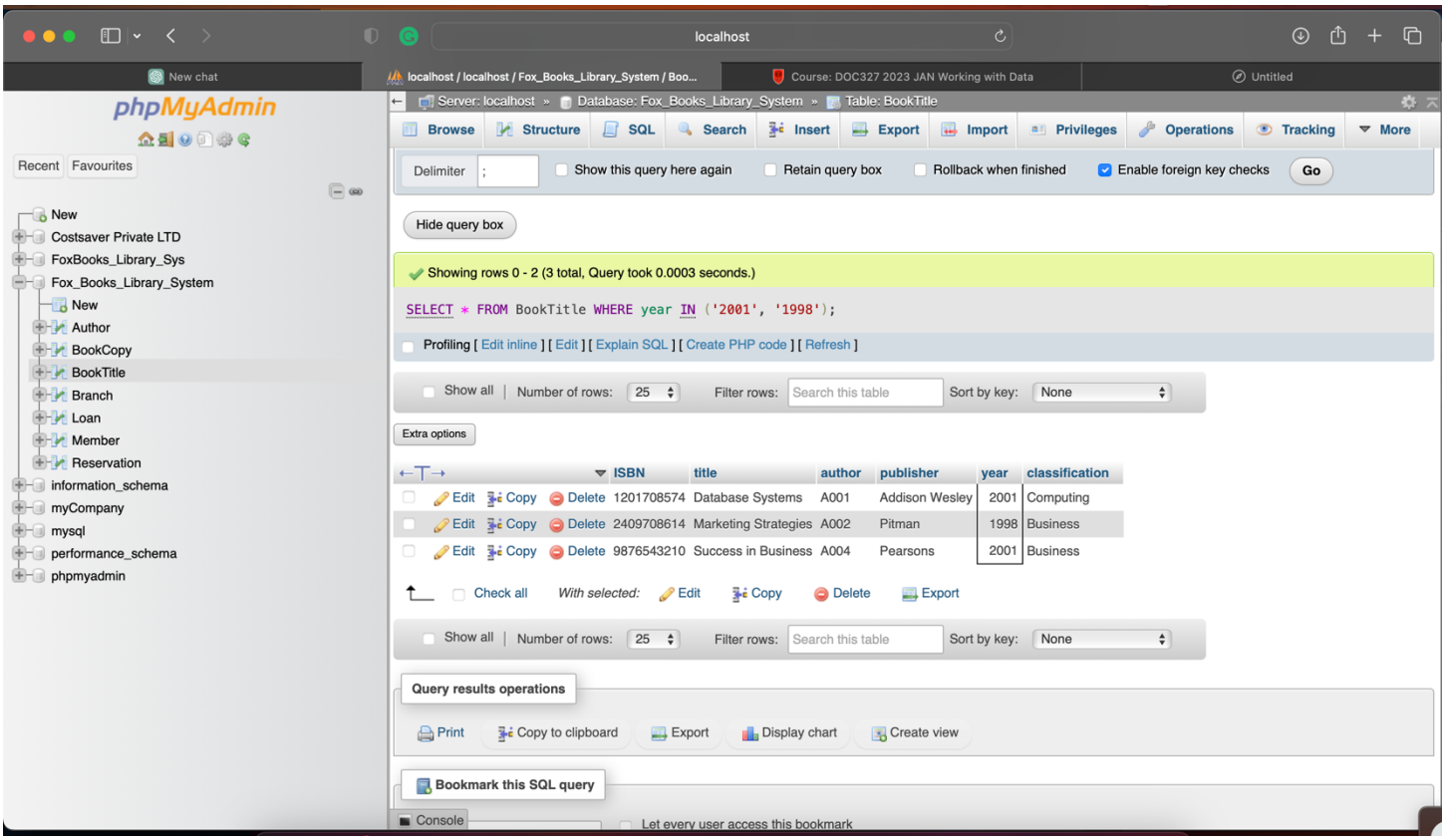

Figure 47 Q3

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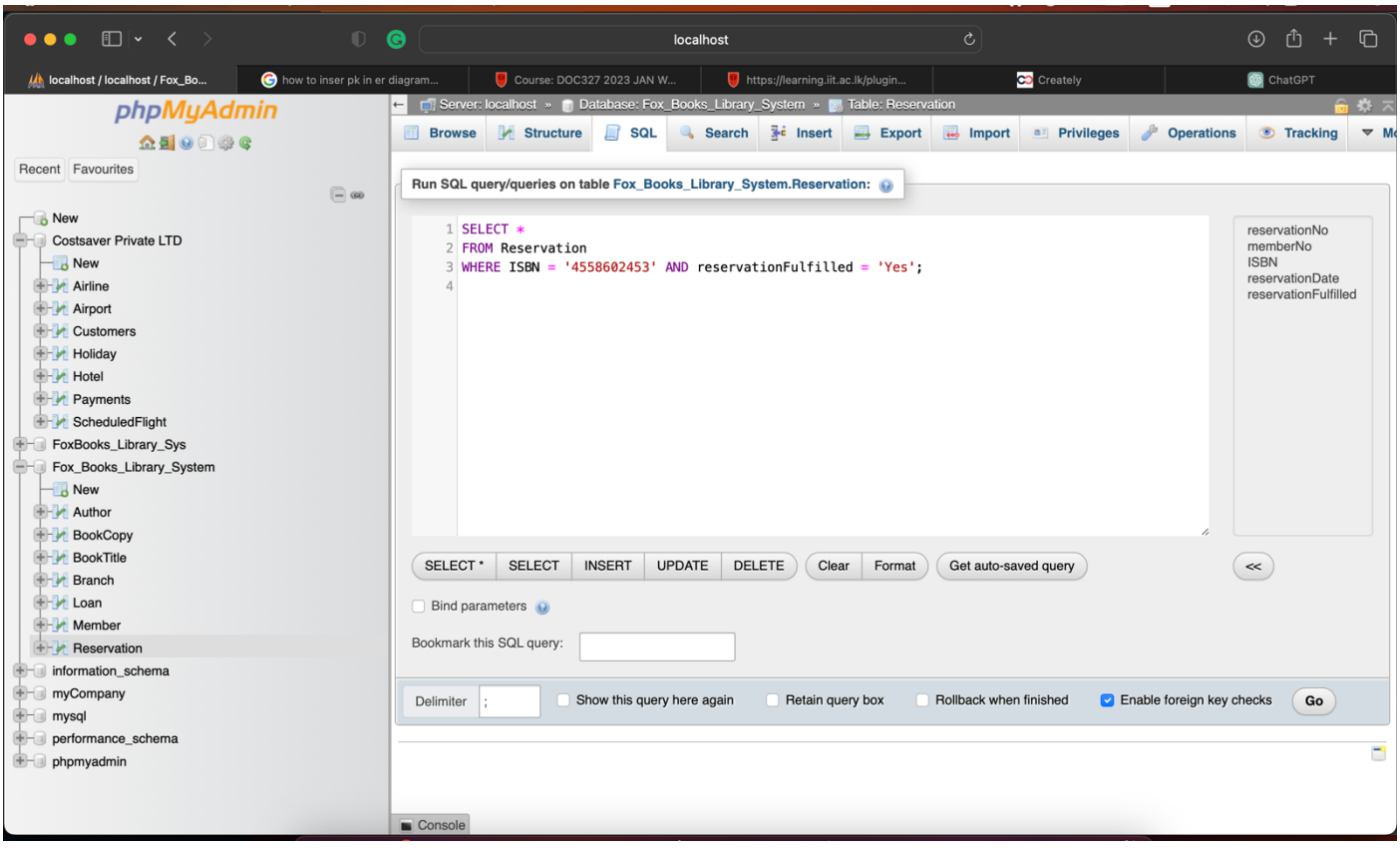

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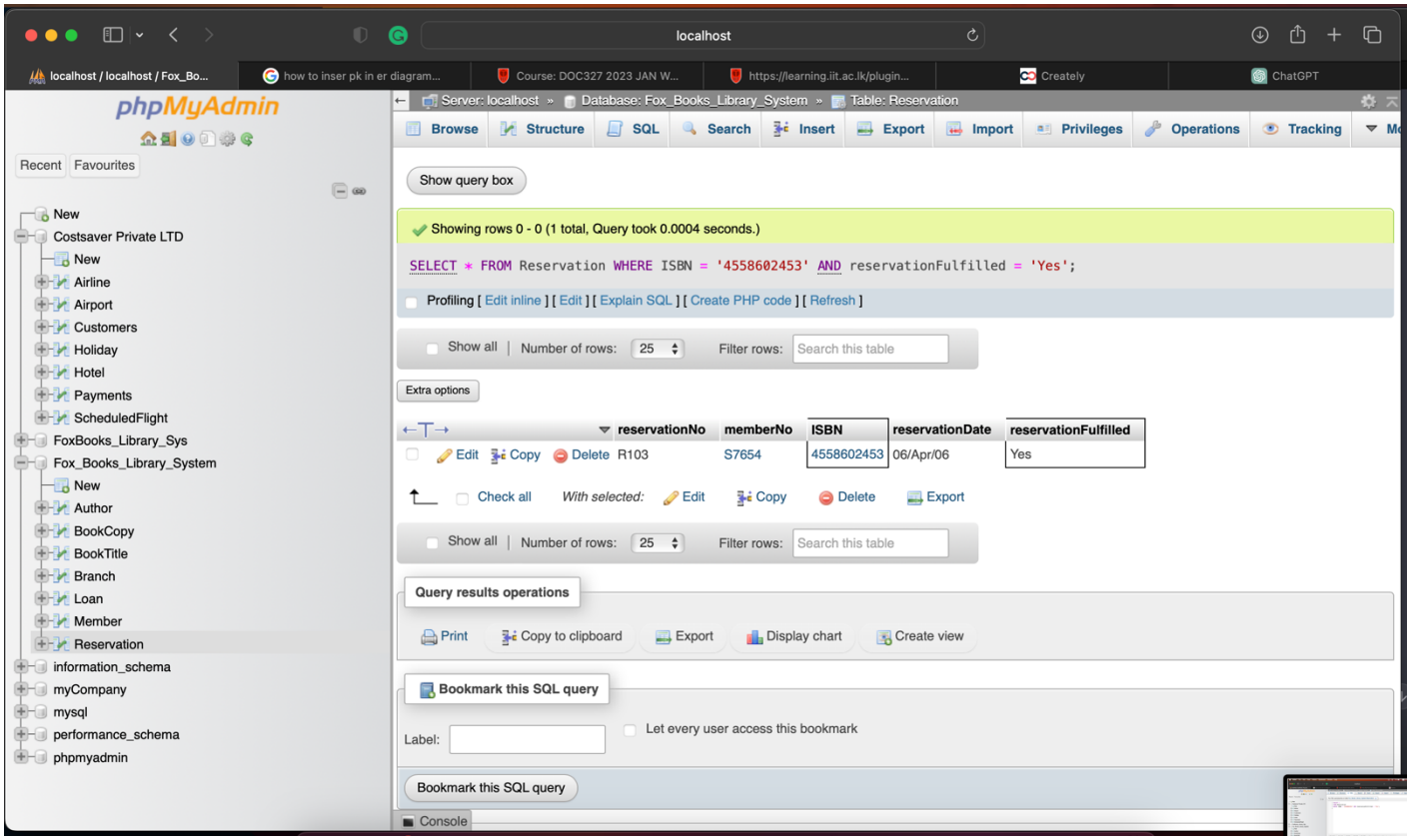

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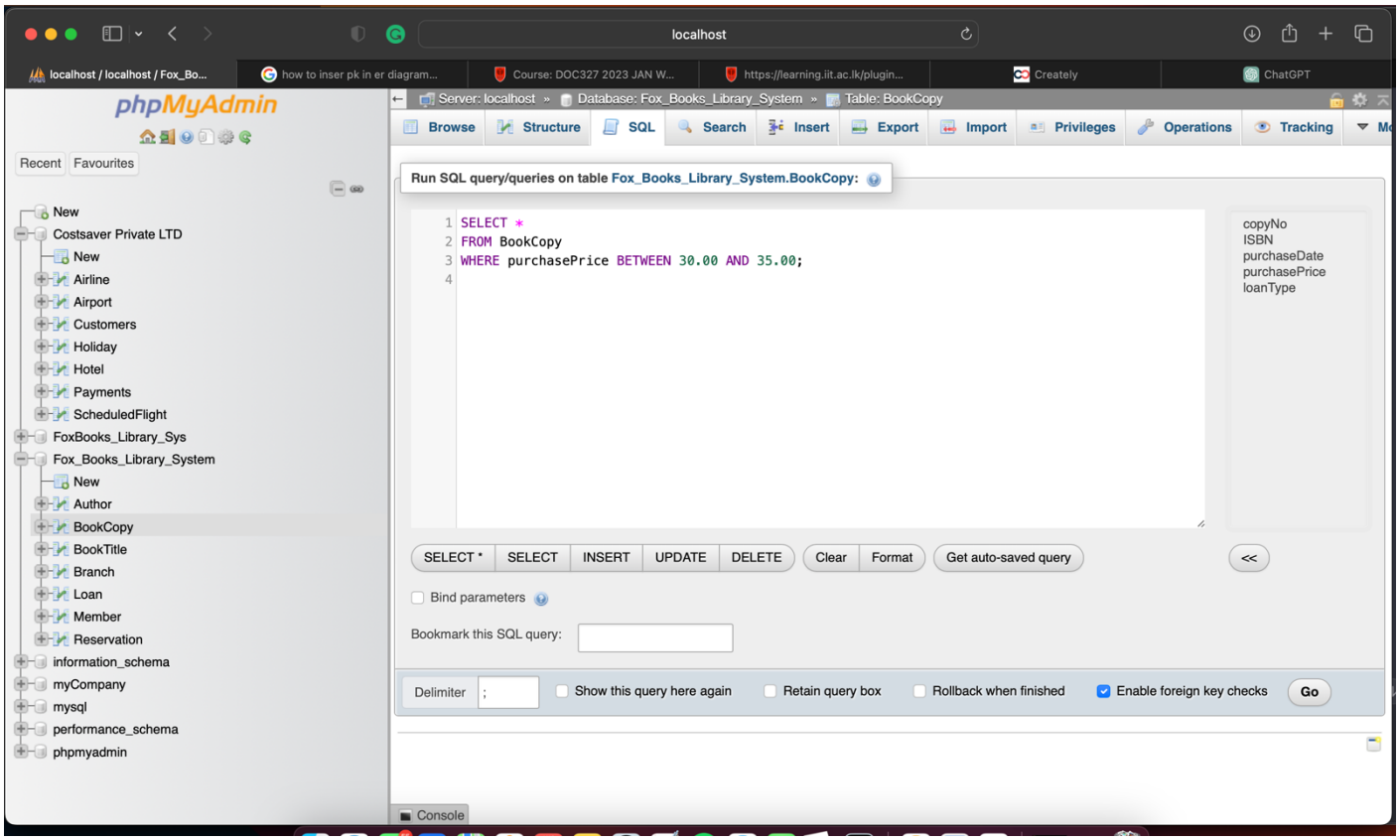

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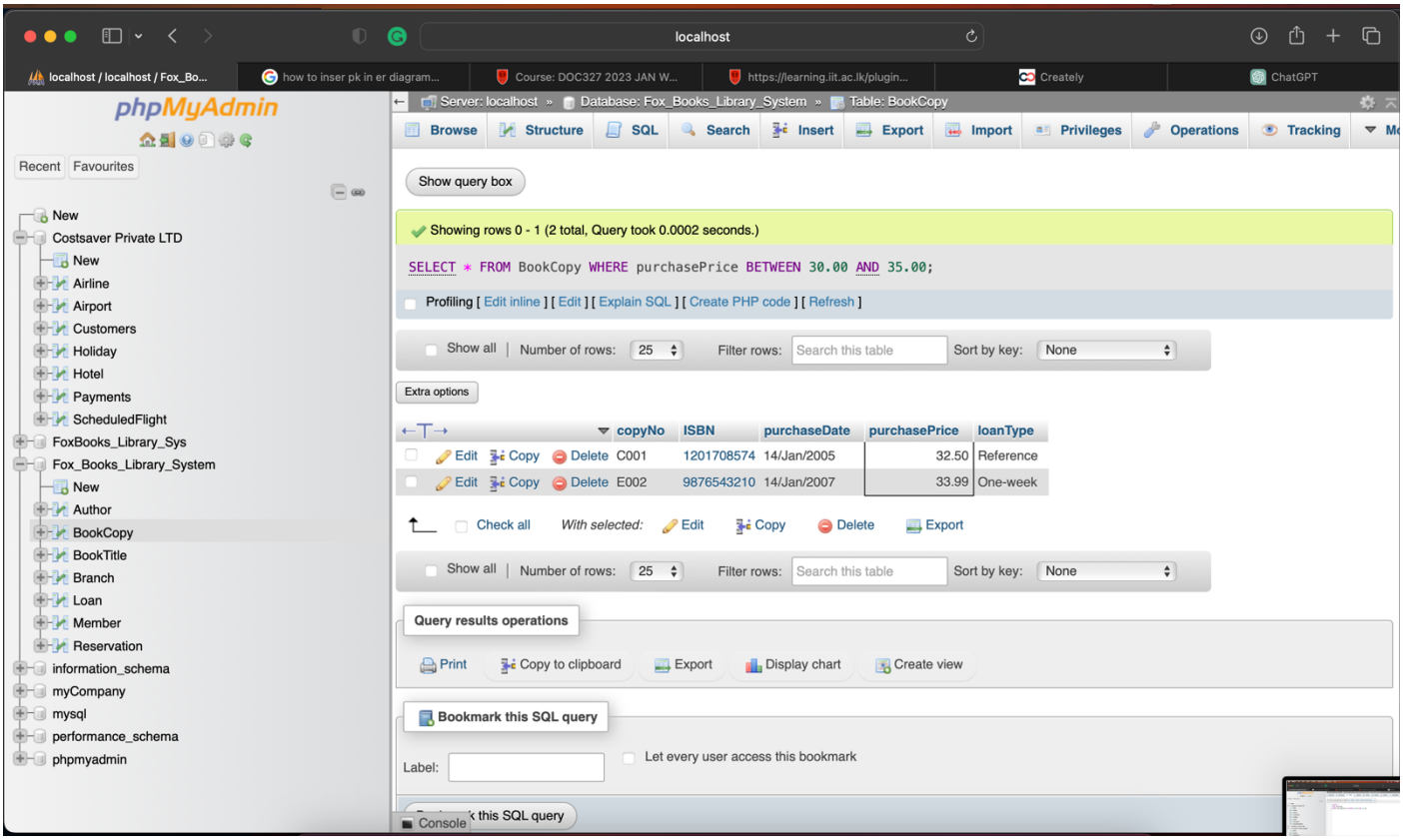

Figure 51 Q3

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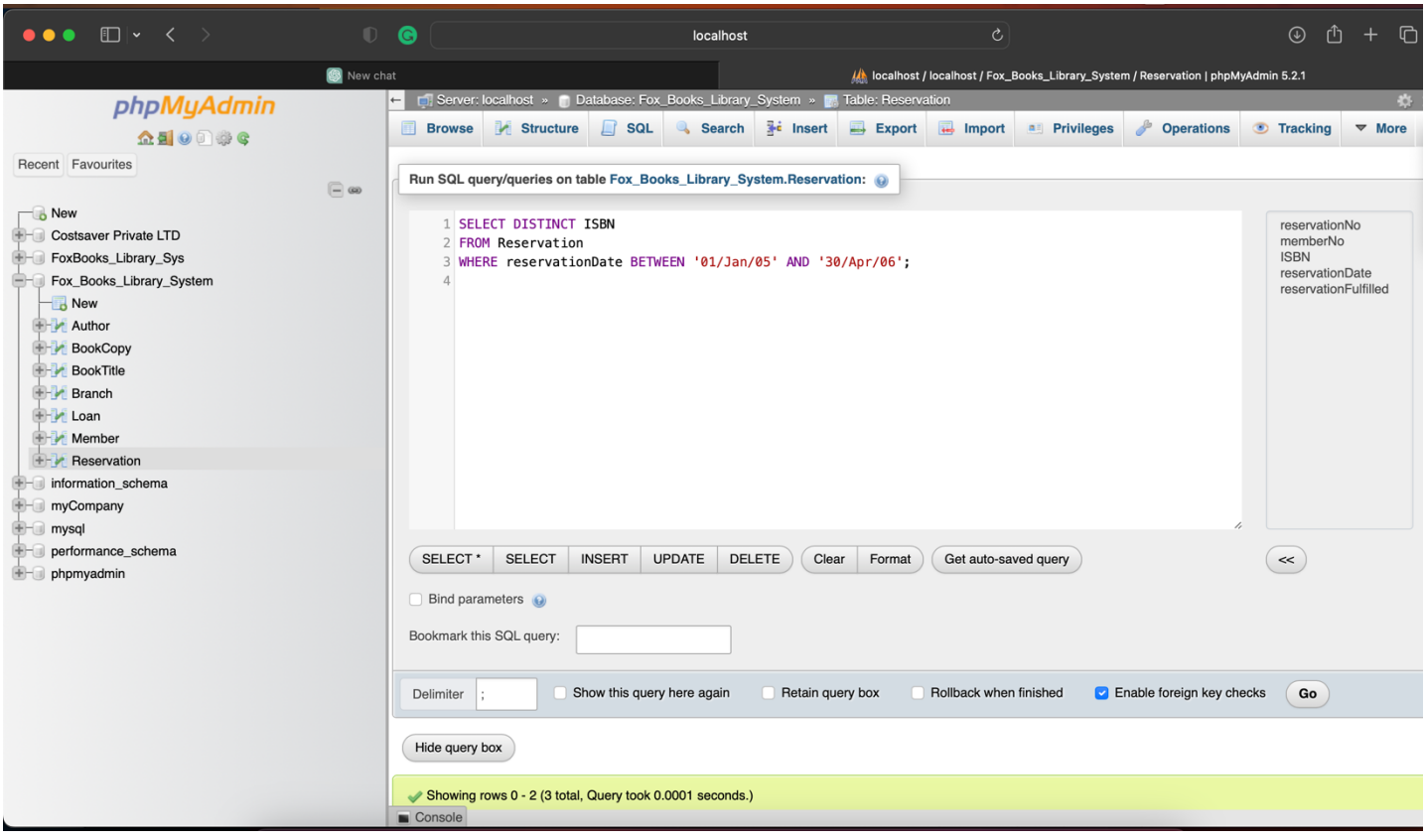

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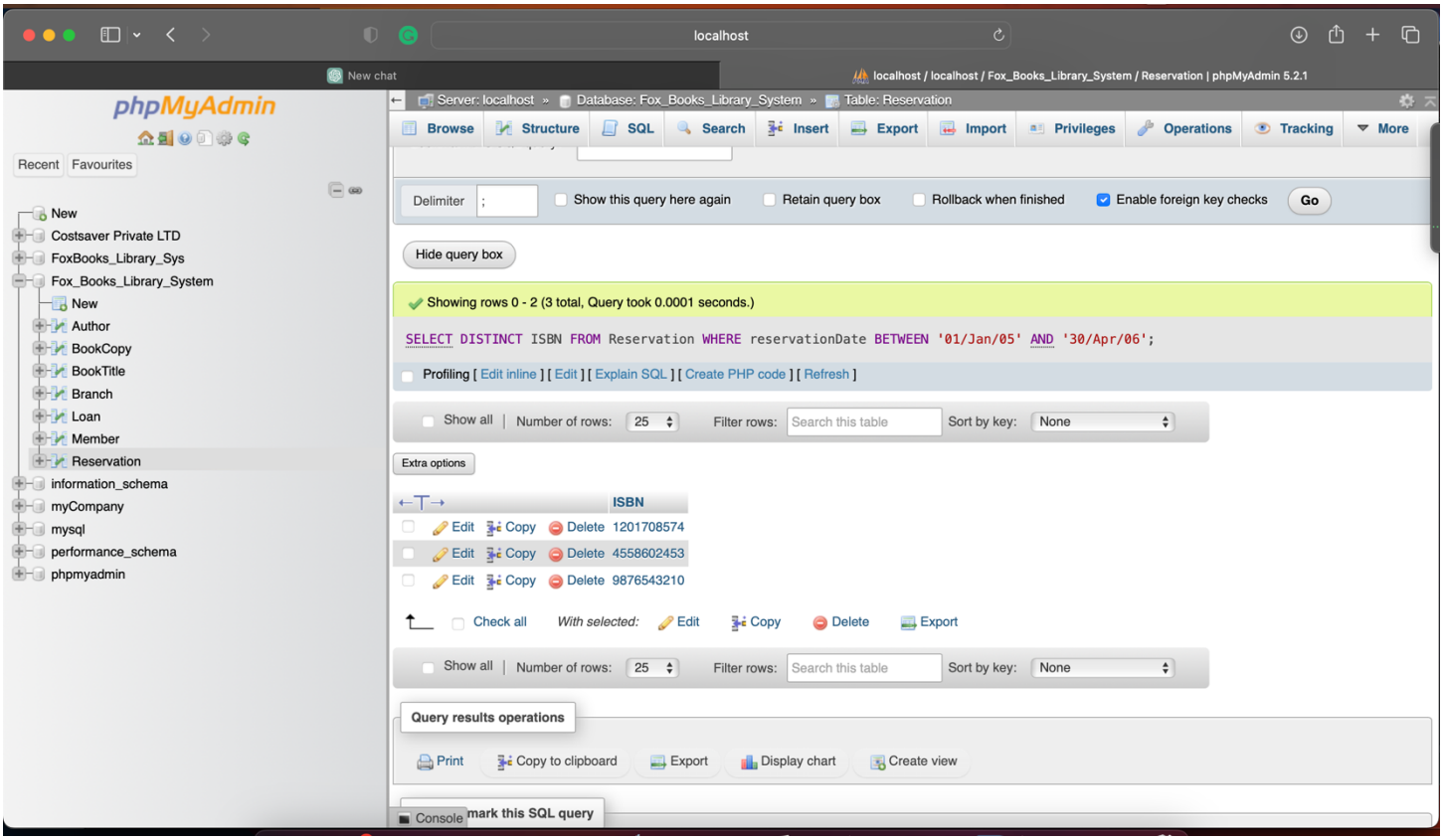

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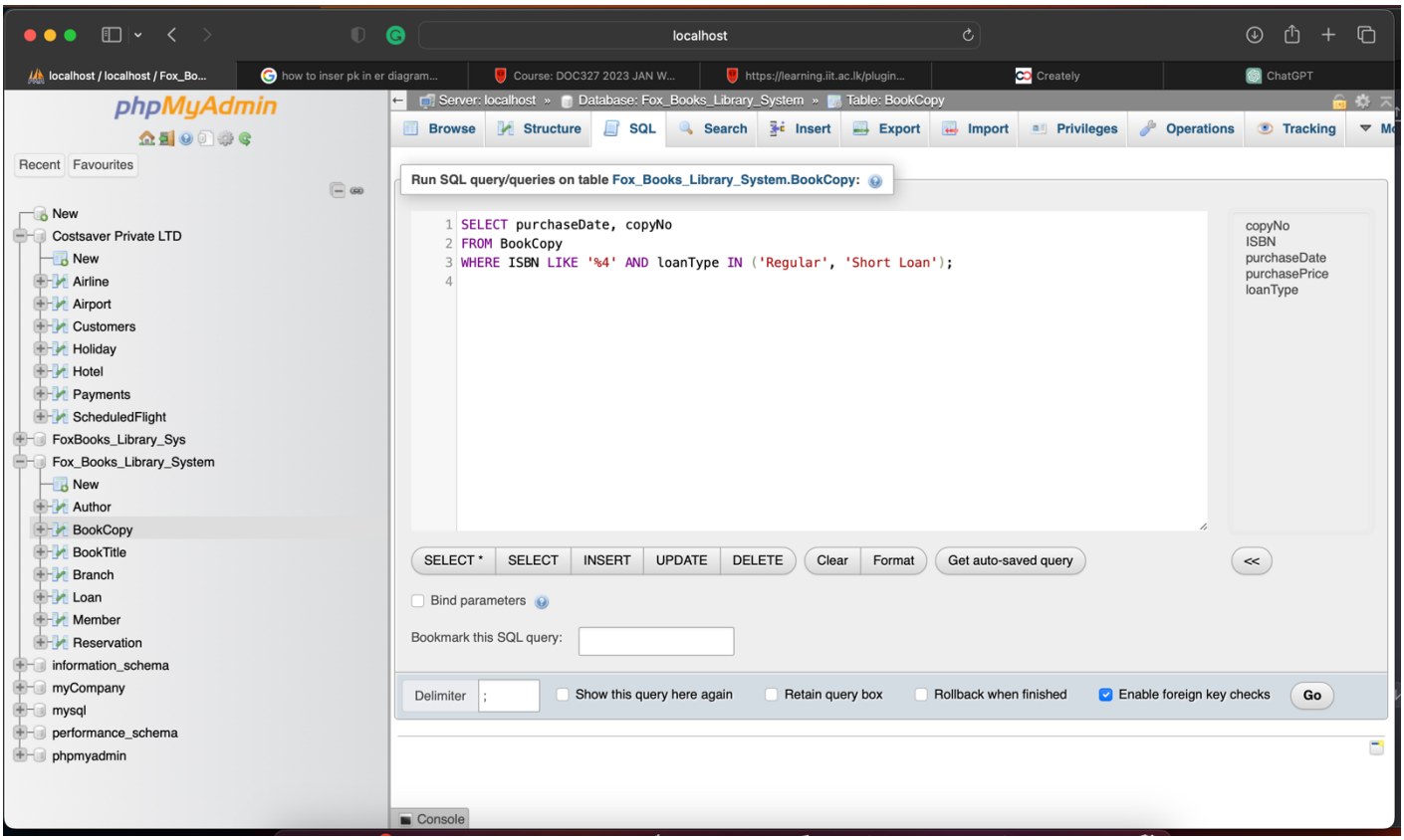

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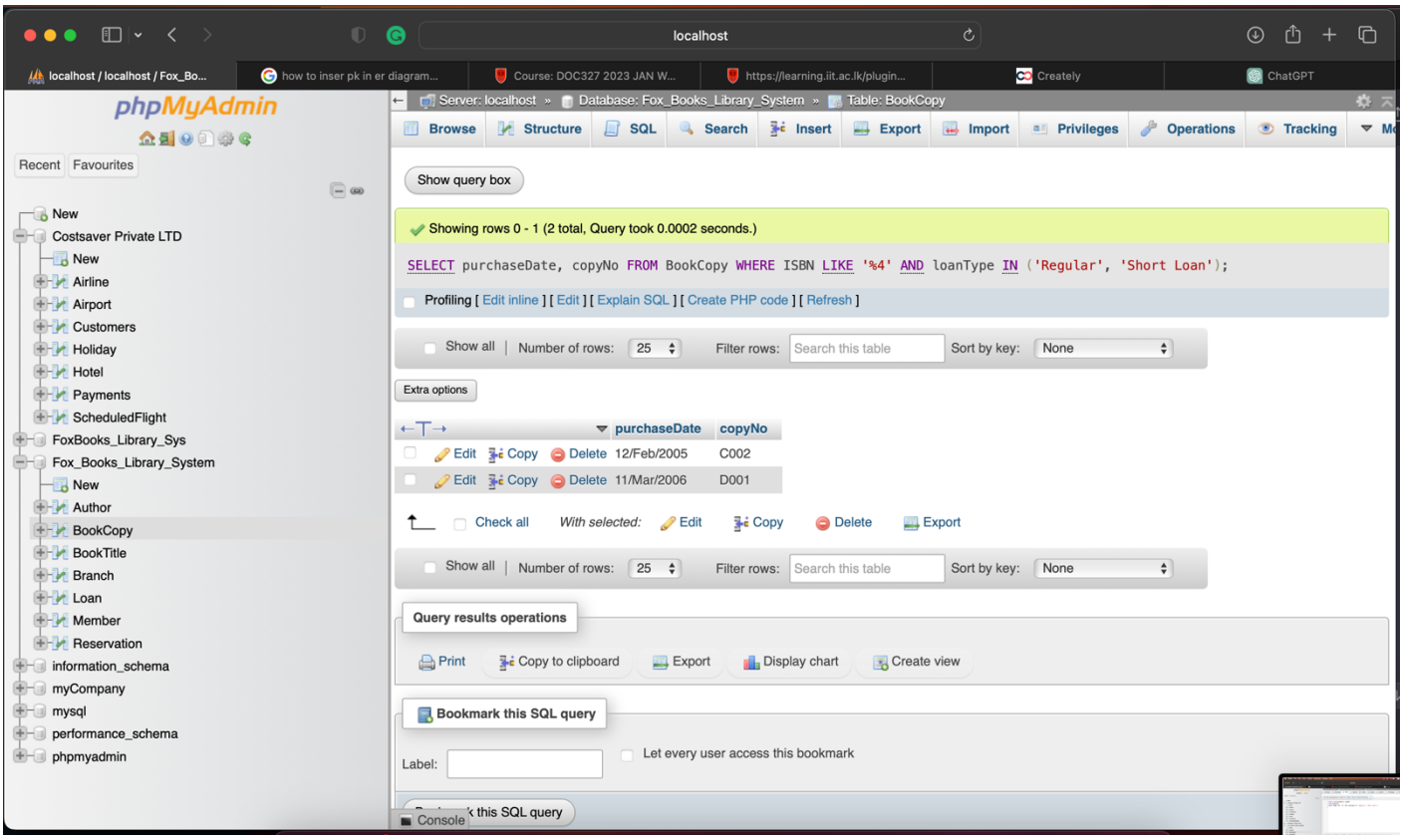

Figure 55 Q3

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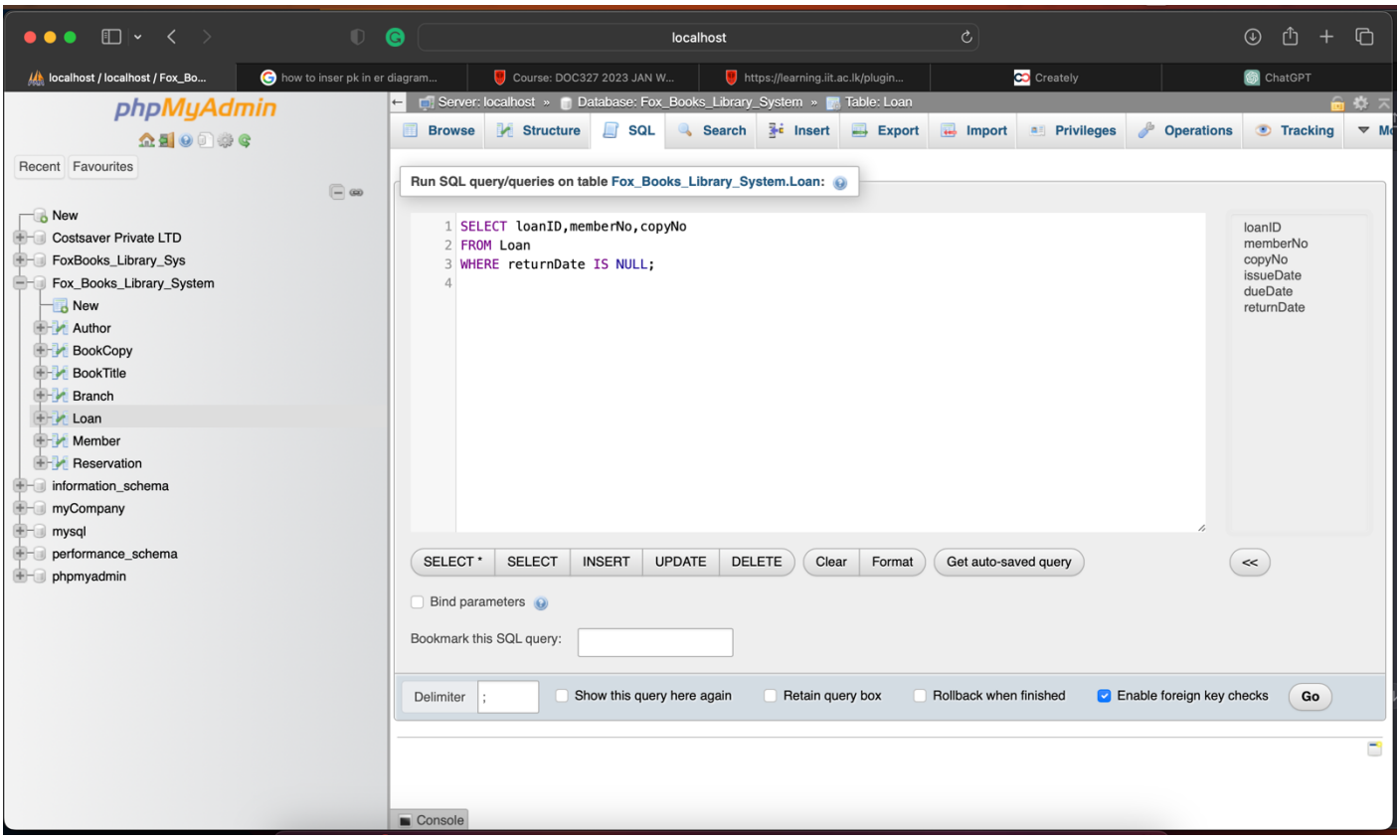

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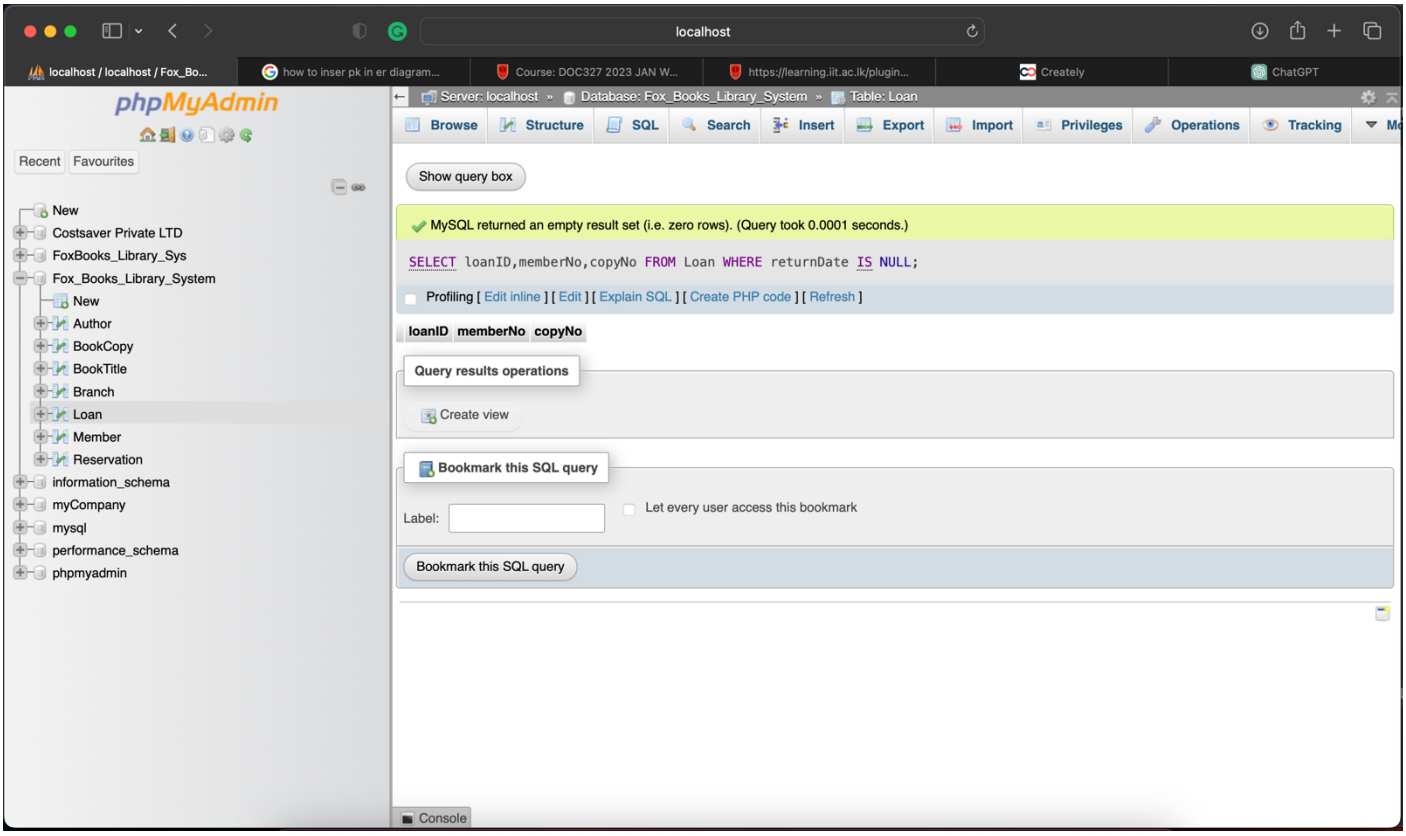


Figure 58 Q3

o)

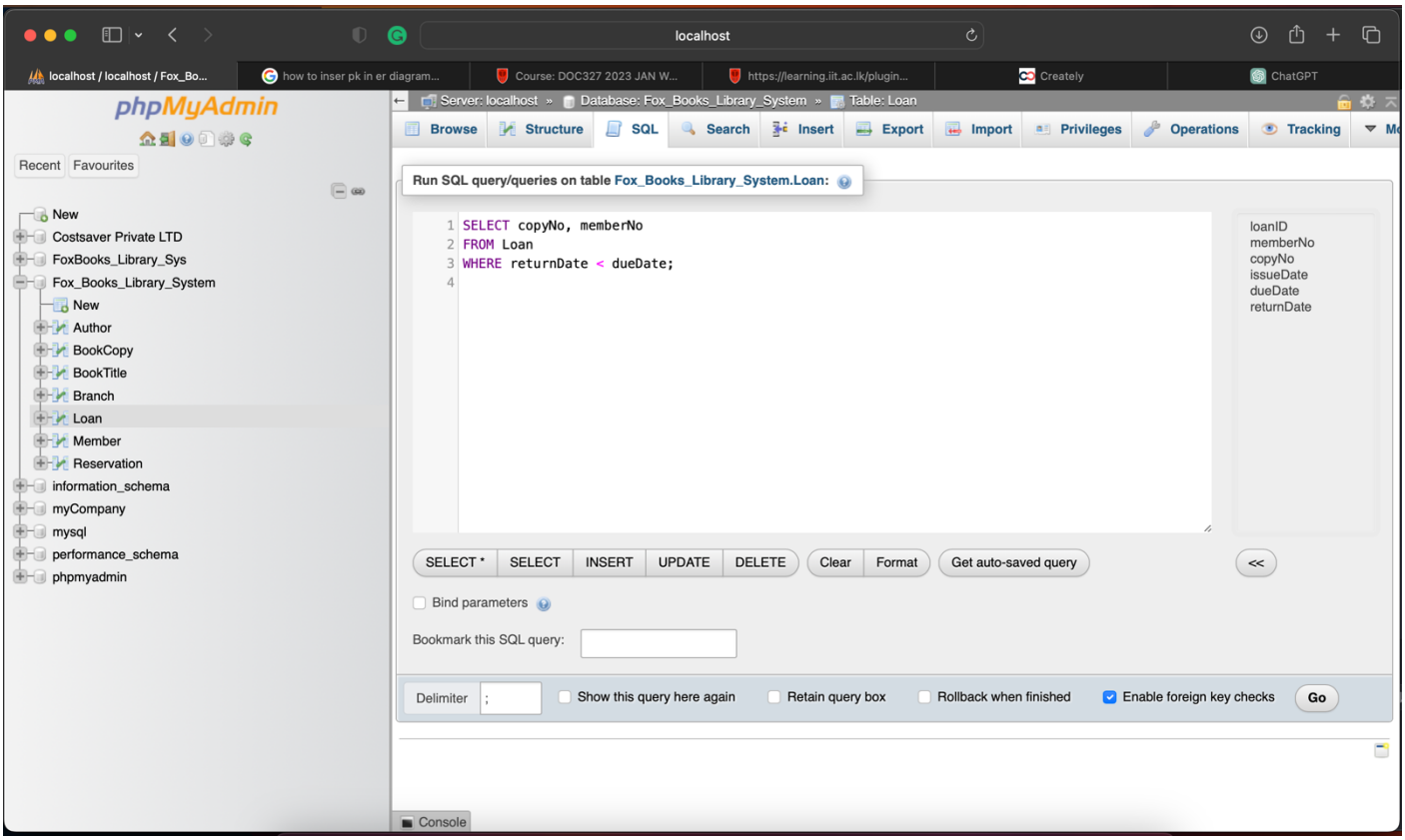
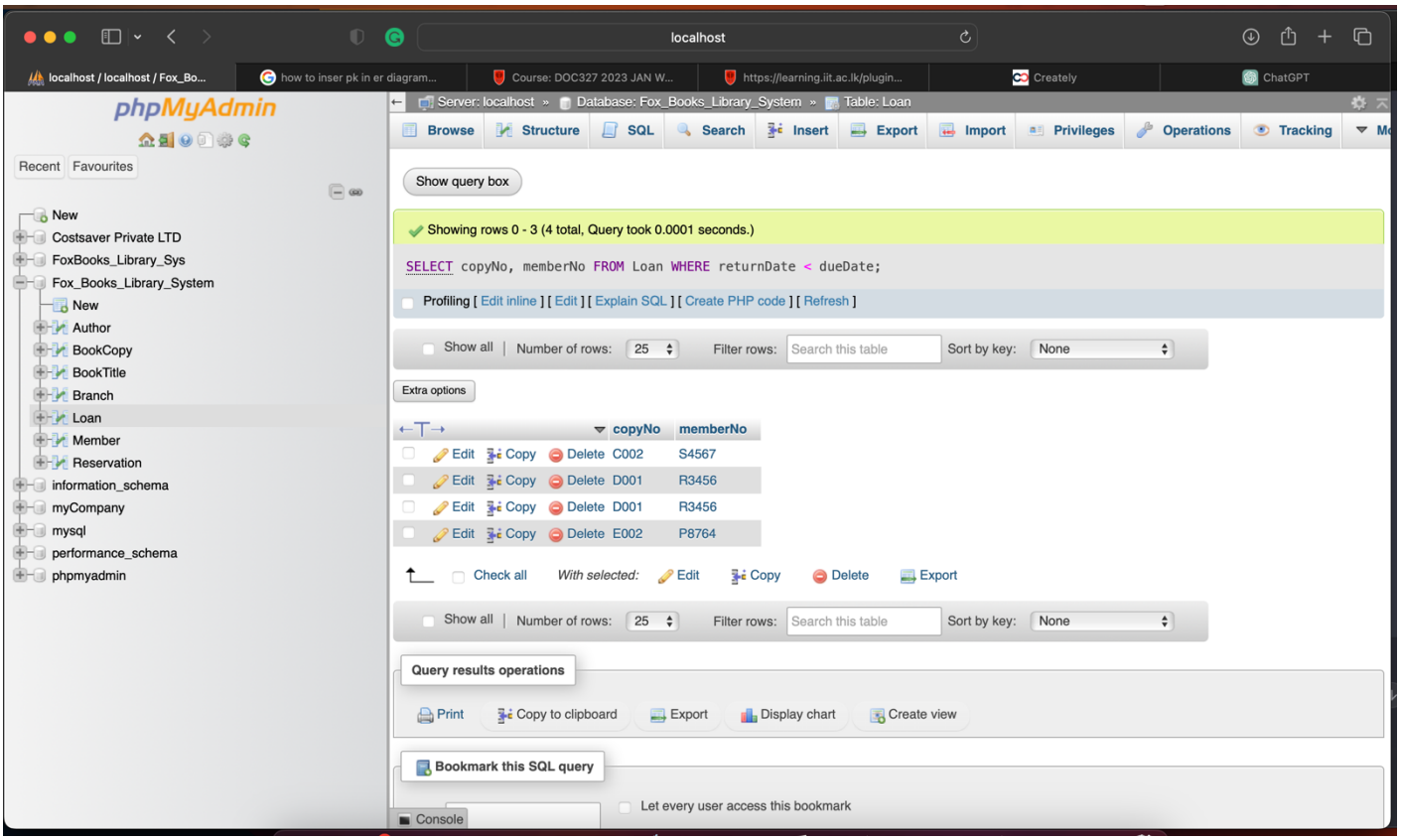


Figure 57 Q3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Recent
Favourites
New
Costsaver Private LTD
FoxBooks_Library_Sys
Fox_Books_Library_System
New
Author
BookCopy
BookTitle
Branch
Loan
Member
Reservation
information_schema
myCompany
mysql
performance_schema
phpmyadmin
Server: localhost » Database: Fox_Books_Library_System » Table: Loan
Browse
Structure
SQL
Search
Insert
Export
Import
Privileges
Operations
Tracking
Show query box
Showing rows 0 - 3 (4 total, Query took 0.0001 seconds.)
SELECT copyNo, memberNo FROM Loan WHERE returnDate < dueDate;
Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]
Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None
Extra options
copyNo memberNo
Edit Copy Delete C002 S4567
Edit Copy Delete D001 R3456
Edit Copy Delete D001 R3456
Edit Copy Delete E002 P8764
Check all With selected: Edit Copy Delete Export
Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None
Query results operations
Print
Copy to clipboard
Export
Display chart
Create view
Bookmark this SQL query
Let every user access this bookmark
Console

Figure 59 Q3

Self-Evaluations Sheets

Student Name:	Muhammed Saad Mazhar	Student Id:	20221804

State what tasks you carried out in the project.
Part A Question 1 Preparing the report
State what you enjoyed and did not enjoy about teamwork.
Enjoyed - Brainstorming meetings in which many ideas were discussed and original solutions were produced. Did not enjoyed - juggling personal obligations with team duties, which occasionally resulted in time difficulties.
State what you learnt about teamwork.
How to use the talents and abilities of each team member to further the group's objectives.
State what skills you gained/learnt from undertaking the project.
The value of attentive listening and being open-minded while having talks and addressing problems.
State any strengths about yourself that emerged while undertaking the project.
Problem-Solving Skills: I was exceptional at analysing complicated problems and locating workable solutions, which helped to successfully resolve a number of project-related concerns.
State any weaknesses about yourself that emerged while undertaking the project.
Communication: I had trouble articulating my thoughts effectively, which resulted in misunderstandings.
State how you would do things better if you were to undertake the project again.
To better manage your time, create more precise deadlines and routine progress check-ins.
Has undertaking this module made you reflect on/change your course?
Change in Career Path: The module broadened my understanding of the subject, which led me to reexamine my professional objectives and think about integrating them with this field of study.
Additional general or project specific comments:
Flexibility: I developed the ability to be adaptable while dealing with changes and changing plans as necessary.

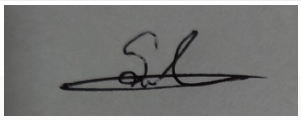
Student Signature:	Saad Mazhar 	Date:	2023-07-30

Table 2 Self Evaluation sheet (1)

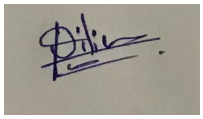
Student Name:	Dilina Perera	Student Id:	20221803
State what tasks you carried out in the project.			
Part A Question 2 Preparing the report			
State what you enjoyed and did not enjoy about teamwork.			
Enjoyed - Honouring everyone's efforts and sharing in team accomplishments. Did not enjoyed - resolving the infrequent controversies and disagreements that developed throughout the decision-making process.			
State what you learnt about teamwork.			
The necessity of clear communication in a group context in order to prevent misunderstandings and disputes.			
State what skills you gained/learnt from undertaking the project.			
Improved interpersonal and communication abilities as a result of frequent team encounters.			
State any strengths about yourself that emerged while undertaking the project.			
Strong Leadership: I showed that I could lead and manage the team during key project phases, keeping everyone on track and inspired to reach our objectives.			
State any weaknesses about yourself that emerged while undertaking the project.			
Time Management: I frequently missed deadlines due to my inability to adequately manage my time.			
State how you would do things better if you were to undertake the project again.			
To guarantee a fair distribution of labour, implement a more organised job delegation mechanism.			
Has undertaking this module made you reflect on/change your course?			
Increased Interest: Completing this programme generated a renewed interest in the topic, prompting me to think about pursuing a minor or specialisation in the area.			
Additional general or project specific comments:			
Task Management: I improved my task management and project progress tracking skills.			
Student Signature:	Dilina Perera 	Date:	2023-07-30

Table 3 Self Evaluation sheet (2)

Student Name:	Rafeek Mohammed Alfar	Student Id:	20221802

State what tasks you carried out in the project.
Part A Question 3 Abstract
State what you enjoyed and did not enjoy about teamwork.
Enjoyed - The teamwork and support among members, which helps to create a healthy work atmosphere. Did not enjoyed - addressing problems with the team's uneven workload distribution. ddressing problems with the team's uneven workload distribution.
State what you learnt about teamwork.
collaborating well with others in the team, delegating tasks, and achieving shared objectives.
State what skills you gained/learnt from undertaking the project.
Developed project management abilities to efficiently plan, arrange, and carry out activities.
State any strengths about yourself that emerged while undertaking the project.
Flexibility: I was very adaptable in responding to shifting project specifications and unanticipated roadblocks, which helped the team continue to make progress and fulfil deadlines.
State any weaknesses about yourself that emerged while undertaking the project.
Assignment delegation: I frequently tried to do every assignment on my own since I found it difficult to delegate.
State how you would do things better if you were to undertake the project again.
Improve my communication abilities so that I can explain thoughts and instructions more clearly.
Has undertaking this module made you reflect on/change your course?
Academic Excellence: Doing well in this module gave me greater confidence in the field and led me to consider putting more of my academic efforts into it.
Additional general or project specific comments:
Positivity: I overcame challenges and motivated the team by keeping a positive attitude.

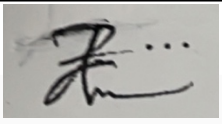
Student Signature:	Rafeek Alfar 	Date:	2023-07-30

Table 4 Self Evaluation sheet (3)

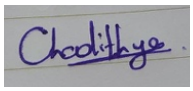
Student Name:	Chadithya Dinuwan Udayantha	Student Id:	20221798
State what tasks you carried out in the project.			
Part B Question 1 Question 2 (a)			
State what you enjoyed and did not enjoy about teamwork.			
Enjoyed - Gaining new perspectives on various facets of the project by utilising one another's experience. Did not enjoyed - managing communication difficulties like misconceptions and misinterpretations.			
State what you learnt about teamwork.			
How to give supportive criticism and encouragement to improve team performance.			
State what skills you gained/learnt from undertaking the project.			
By overcoming project hurdles, participants' problem-solving and critical thinking skills were improved			
State any strengths about yourself that emerged while undertaking the project.			
Attention to Detail: Our deliverables were highly accurate because of my excellent eye for detail, which helped spot possible mistakes and anomalies in project components.			
State any weaknesses about yourself that emerged while undertaking the project.			
Making decisions: Especially in unsure circumstances, I dreaded making choices.			
State how you would do things better if you were to undertake the project again.			
Encourage a more welcoming work atmosphere to promote fair participation from all team members.			
Has undertaking this module made you reflect on/change your course?			
Networking Opportunities: I made connections with like-minded peers and seasoned professionals through the module, which motivated me to consider career possibilities that entail working with subject matter experts in this area.			
Additional general or project specific comments:			
Initiative: When extra tasks were required, I took the initiative to step up and provide my suggestions.			
Student Signature:	Chadithya Dinuwan 	Date:	2023-07-28

Table 5 Self Evaluation sheet (4)

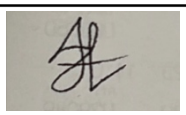
Student Name:	Mohamed Zabrin Mohamed Juzail	Student Id:	20222336
State what tasks you carried out in the project.			
Part B Question 2 (b)			
State what you enjoyed and did not enjoy about teamwork.			
Enjoyed - Fostering a culture of trust and solid professional connections among the team. Did not enjoyed - managing unforeseen project scope changes and adjusting to new needs.			
State what you learnt about teamwork.			
The significance of teamwork and mutual decision-making in building a strong workplace culture.			
State what skills you gained/learnt from undertaking the project.			
Gained expertise working with people who have a variety of personalities and viewpoints.			
State any strengths about yourself that emerged while undertaking the project.			
Communication: I successfully informed team members of project changes, concepts, and requirements, promoting understanding and productive cooperation.			
State any weaknesses about yourself that emerged while undertaking the project.			
Conflict Resolution: I had trouble settling disagreements inside the team.			
State how you would do things better if you were to undertake the project again.			
Create contingency plans in advance to account for unexpected project scope adjustments.			
Has undertaking this module made you reflect on/change your course?			
Research Possibilities: The module made me aware of fascinating research opportunities, which I think may have influenced my decision to look into future research-oriented projects or courses.			
Additional general or project specific comments:			
Resilience: I improved my ability to handle failures and overcome obstacles.			
Student Signature:	Zabrin Juzail 	Date:	2023-07-28

Table 6 Self Evaluation sheet (5)

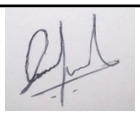
Student Name:	Akash Sathiyanathan	Student Id:	20222330
State what tasks you carried out in the project.			
Part B Question 3 Finalize the report			
State what you enjoyed and did not enjoy about teamwork.			
Enjoyed - Being able to rely on teammates for aid and support while confronting difficulties. Did not enjoyed - ensuring that everyone on the team remained enthusiastic and involved throughout the project.			
State what you learnt about teamwork.			
How to effectively manage time and strike a balance between individual and team obligations.			
State what skills you gained/learnt from undertaking the project.			
Improved public speaking and presentation abilities via project displays.			
State any strengths about yourself that emerged while undertaking the project.			
Time Management: I effectively balanced my personal obligations with team duties by managing my time and tasks.			
State any weaknesses about yourself that emerged while undertaking the project.			
Flexibility: I had trouble adjusting to unforeseen changes in the project.			
State how you would do things better if you were to undertake the project again.			
Ask team members for input frequently to find areas that might want improvement.			
Has undertaking this module made you reflect on/change your course?			
Personal Development: Completing this module pushed me academically and promoted personal development, which inspired me to think about additional research on subjects that would broaden my perspective and improve my educational experience. contemplate the flexibility of my current course and whether it aligns with the dynamic job market.			
Additional general or project specific comments:			
Paying Attention to input: To enhance my performance and the project's outcomes, I actively sought out and used input.			
Student Signature:	Akash Sathiyanathan 	Date:	2023-07-30

Table 7 Self Evaluation sheet (6)